Initializing the Spark Session and Reading the dataset

In [ ]:
```
from pyspark.sql import SparkSession
spark = SparkSession.builder \
    .appName("Phase1_EDA_BookReviews") \
    .master("local[*]") \
    .config("spark.driver.memory", "8g") \
    .config("spark.executor.memory", "6g") \
    .config("spark.network.timeout", "800s") \
    .config("spark.executor.heartbeatInterval", "100s") \
    .config("spark.sql.shuffle.partitions", "50") \
    .getOrCreate()
books_df = spark.read.csv("hdfs://localhost:9000/user/sathw/input/books.csv"
                          header=True,
                          inferSchema=True)
rating_df = spark.read.csv("hdfs://localhost:9000/user/sathw/input/rating.cs
                           header=True,
                           inferSchema=True) \
                     .sample(fraction=0.15, seed=42)
books_df.printSchema()
rating_df.printSchema()

books_df.show(5)
rating_df.show(5)
```

```
25/04/26 17:40:14 WARN Utils: Your hostname, MacBook-Pro-337.local resolves
to a loopback address: 127.0.0.1; using 172.20.9.62 instead (on interface en
0)
25/04/26 17:40:14 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to anot
her address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLog
Level(newLevel).
25/04/26 17:40:15 WARN NativeCodeLoader: Unable to load native-hadoop librar
y for your platform... using builtin-java classes where applicable
```

```
root
 |-- Title: string (nullable = true)
 |-- description: string (nullable = true)
 |-- authors: string (nullable = true)
 |-- image: string (nullable = true)
 |-- previewLink: string (nullable = true)
 |-- publisher: string (nullable = true)
 |-- publishedDate: string (nullable = true)
 |-- infoLink: string (nullable = true)
 |-- categories: string (nullable = true)
 |-- ratingsCount: string (nullable = true)

root
 |-- Id: string (nullable = true)
 |-- Title: string (nullable = true)
 |-- Price: string (nullable = true)
 |-- User_id: string (nullable = true)
 |-- profileName: string (nullable = true)
 |-- review/helpfulness: string (nullable = true)
 |-- review/score: string (nullable = true)
 |-- review/time: string (nullable = true)
 |-- review/summary: string (nullable = true)
 |-- review/text: string (nullable = true)

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|               Title|         description|             authors|
image|         previewLink|           publisher| publishedDate|            i
nfoLink|          categories|ratingsCount|
+--------------------+--------------------+--------------------+------------
--------+--------------------+--------------------+--------------+----------
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|Its Only Art If I...|                NULL|    ['Julie Strain']|http://book
s.goog...|http://books.goog...|                NULL|          1996|http://bo
oks.goog...|['Comics & Graphi...|        NULL|
|Dr. Seuss: Americ...|"Philip Nel takes...| like that of Lew...| has changed
lang...| giving us new wo...| inspiring artist...|['Philip Nel']|http://book
s.goog...|http://books.goog...|   A&C Black|
|Wonderful Worship...|This resource inc...|    ['David R. Ray']|http://book
s.goog...|http://books.goog...|                NULL|          2000|http://bo
oks.goog...|        ['Religion']|        NULL|
|Whispers of the W...|Julia Thomas find...| ['Veronica Haddon']|http://book
s.goog...|http://books.goog...|           iUniverse|       2005-02|http://bo
oks.goog...|         ['Fiction']|        NULL|
|Nation Dance: Rel...|                NULL|     ['Edward Long']|
NULL|http://books.goog...|                NULL|    2003-03-01|http://books.g
oog...|                NULL|        NULL|
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only showing top 5 rows

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|        Id|               Title|Price|       User_id|         profileName|r
eview/helpfulness|review/score|review/time|      review/summary|         rev
iew/text|
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|0826414346|Dr. Seuss: Americ...| NULL|A2RSSXTDZDUSH4|           J. Squire|
0/0|         5.0| 1231200000|Academia At It's ...|"When I recieved ...|
|0595344550|Whispers of the W...|10.95| AUR0VA5H0C66C|"LoveToRead ""Act...|
1/2|         1.0| 1119225600|        Buyer beware|"This is a self-p...|
|0595344550|Whispers of the W...|10.95| ACO23CG8K8T77|               Tonya|
5/9|         5.0| 1117065600|      Bravo Veronica|I read the review...|
|0802841899|The Church of Chr...|25.97| A36TPZSH8LBT1|             haskell|
2/3|         5.0| 1311465600|Early Christian d...|Everett Ferguson ...|
|0802841899|The Church of Chr...|25.97|A2H2LORTA5EZY2|      Edward E. Howe|
3/5|         4.0| 1266192000|      Christ is Lord|This is a very us...|
+----------+--------------------+-----+--------------+--------------------+-
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only showing top 5 rows
```

Task 1 - EDA

Sub-Task 1a) EDA on Books dataset

i) Counting the Rows, Displaying schema and First 5 Rows

In [ ]:
```
print("\nSchema of books.csv:")
books_df.printSchema()
print("\nNumber of rows in books.csv:")
print(books_df.count())
print("\nSample 5 rows from books.csv:")
books_df.show(5, truncate=False)
```

```
Schema of books.csv:
root
 |-- Title: string (nullable = true)
 |-- description: string (nullable = true)
 |-- authors: string (nullable = true)
 |-- image: string (nullable = true)
 |-- previewLink: string (nullable = true)
 |-- publisher: string (nullable = true)
 |-- publishedDate: string (nullable = true)
 |-- infoLink: string (nullable = true)
 |-- categories: string (nullable = true)
 |-- ratingsCount: string (nullable = true)


Number of rows in books.csv:
212404

Sample 5 rows from books.csv:
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|Title                                                                        |d
escription
|authors                                      |image
|previewLink
|publisher
|publishedDate |infoLink
|categories
|ratingsCount|
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|Its Only Art If Its Well Hung!                                                |N
ULL
|['Julie Strain']                          |http://books.google.com/books/c
ontent?id=DykPAAAACAAJ&printsec=frontcover&img=1&zoom=1&source=gbs_api|htt
p://books.google.nl/books?id=DykPAAAACAAJ&dq=Its+Only+Art+If+Its+Well+Hung!&
hl=&cd=1&source=gbs_api
|NULL
|1996          |http://books.google.nl/books?id=DykPAAAACAAJ&dq=Its+Only+Art
+If+Its+Well+Hung!&hl=&source=gbs_api
|['Comics & Graphic Novels']
|NULL        |
|Dr. Seuss: American Icon
|"Philip Nel takes a fascinating look into the key aspects of Seuss's career
- his poetry, politics, art, marketing, and place in the popular imaginatio
n."" ""Nel argues convincingly that Dr. Seuss is one of the most influential
poets in America. His nonsense verse
| like that of Lewis Carroll and Edward Lear| has changed language itself
| giving us new words like ""nerd."" And Seuss's famously loopy artistic sty
le - what Nel terms an ""energetic cartoon surrealism"" - has been equally i
mportant| inspiring artists like filmmaker Tim Burton and illustrator Lane S
mith. --from back cover"|['Philip Nel']|http://books.google.com/books/conten
t?id=IjvHQsCn_pgC&printsec=frontcover&img=1&zoom=1&edge=curl&source=gbs_api
|http://books.google.nl/books?id=IjvHQsCn_pgC&printsec=frontcover&dq=Dr.+Seu
ss:+American+Icon&hl=&cd=1&source=gbs_api|A&C Black   |
|Wonderful Worship in Smaller Churches                                         |T
his resource includes twelve principles in understanding small church worshi
p, fifteen practices for planning worship with fewer than 100 people, and su
ggestions for congregational study.
|['David R. Ray']                          |http://books.google.com/books/c
ontent?id=2tsDAAAACAAJ&printsec=frontcover&img=1&zoom=1&source=gbs_api|htt
p://books.google.nl/books?id=2tsDAAAACAAJ&dq=Wonderful+Worship+in+Smaller+Ch
urches&hl=&cd=1&source=gbs_api
|NULL
|2000          |http://books.google.nl/books?id=2tsDAAAACAAJ&dq=Wonderful+Wo
rship+in+Smaller+Churches&hl=&source=gbs_api
|['Religion']
|NULL        |
|Whispers of the Wicked Saints                                                 |J
```

```
ulia Thomas finds her life spinning out of control after the death of her hu
sband, Richard. Julia turns to her minister for comfort when she finds herse
lf falling for him with a passion that is forbidden by the church. Heath Spa
rks is a man of God who is busy taking care of his quadriplegic wife who was
seriously injured in a sever car accident. In an innocent effort to reach ou
t to a lonely member of his church, Heath finds himself as the man and not t
he minister as Heath and Julia surrender their bodies to each other and face
the wrath of God. Julia finds herself in over her head as she faces a deadly
disease, the loss of her home and whispers about her wicked affair. Julia le
aves the states offering her body as a living sacrifice in hopes of finding
a cure while her heart remains thousands of miles away hoping to one day reu
nite with the man who holds it hostage.Whispers of the Wicked Saints is a on
ce in a lifetime romance that is breath taking, defying all the rules of rom
ance and bending the laws of love.|['Veronica Haddon']
|http://books.google.com/books/content?id=aRSIgJlq6JwC&printsec=frontcover&i
mg=1&zoom=1&source=gbs_api|http://books.google.nl/books?id=aRSIgJlq6JwC&dq=W
hispers+of+the+Wicked+Saints&hl=&cd=1&source=gbs_api
|iUniverse
|2005-02       |http://books.google.nl/books?id=aRSIgJlq6JwC&dq=Whispers+of+
the+Wicked+Saints&hl=&source=gbs_api
|['Fiction']
|NULL        |
|Nation Dance: Religion, Identity and Cultural Difference in the Caribbean|N
ULL
|['Edward Long']                            |NULL
|http://books.google.nl/books?id=399SPgAACAAJ&dq=Nation+Dance:+Religion,+Ide
ntity+and+Cultural+Difference+in+the+Caribbean&hl=&cd=1&source=gbs_api
|NULL
|2003-03-01    |http://books.google.nl/books?id=399SPgAACAAJ&dq=Nation+Danc
e:+Religion,+Identity+and+Cultural+Difference+in+the+Caribbean&hl=&source=gb
s_api|NULL
|NULL        |
+-----------------------------------------------------------------------+-
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```

```
only showing top 5 rows
```

ii) Missing Values Check

In [ ]:
```
from pyspark.sql.functions import col, when, count, isnan
print("\nMissing Values per Column in books.csv:")
missing_books = books_df.select([
    count(when(col(c).isNull() | isnan(c) | (col(c) == ""), c)).alias(c)
    for c in books_df.columns
])
missing_books.show(truncate=False)
```

```
Missing Values per Column in books.csv:
[Stage 19:>                                                         (0 + 11) /
11]
+-----+-----------+-------+-----+-----------+---------+-------------+-------
-+----------+------------+
|Title|description|authors|image|previewLink|publisher|publishedDate|infoLin
k|categories|ratingsCount|
+-----+-----------+-------+-----+-----------+---------+-------------+-------
-+----------+------------+
|1    |68357      |31251  |51191|24055      |73130    |25844        |24301
|40524     |148552      |
+-----+-----------+-------+-----+-----------+---------+-------------+-------
-+----------+------------+
```

iii) Summary Statistics

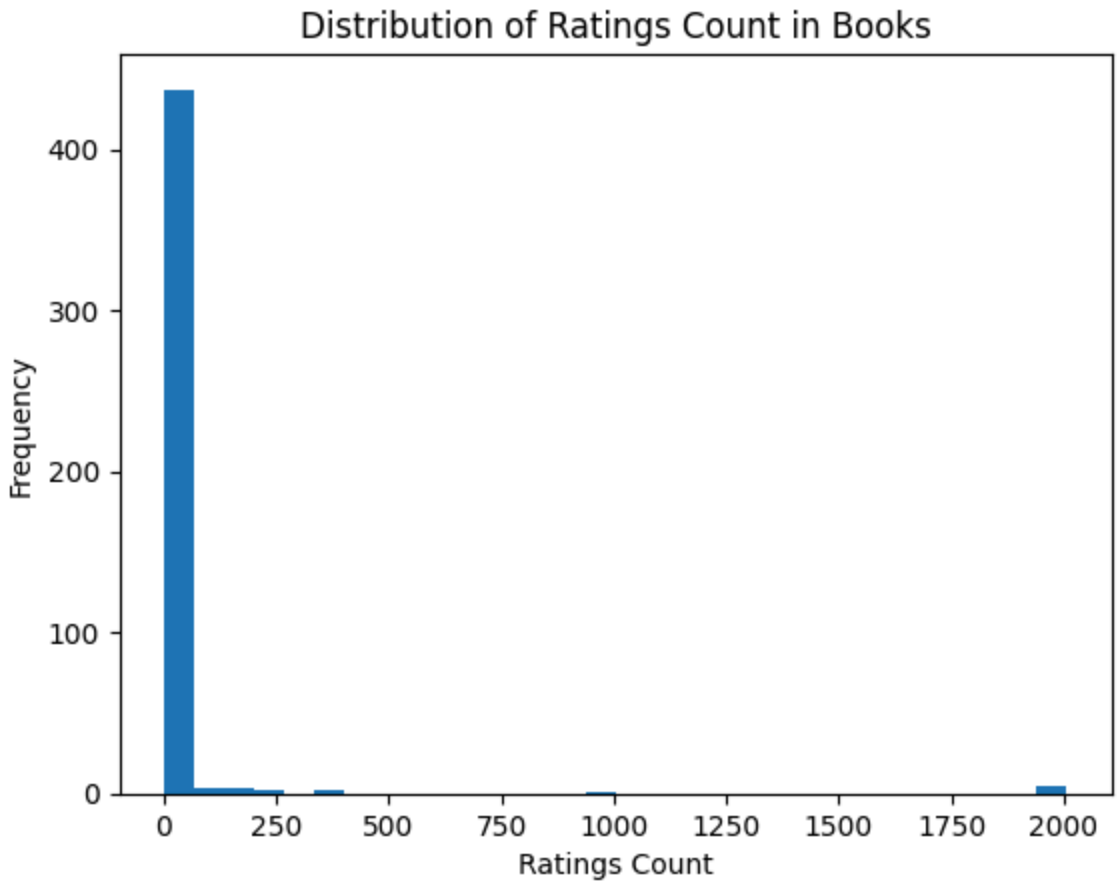
In [ ]:
```
print("\nSummary Statistics for ratingsCount in books.csv:")
books_df.select("ratingsCount").describe().show()
```

```
Summary Statistics for ratingsCount in books.csv:
+-------+--------------------+
|summary|        ratingsCount|
+-------+--------------------+
|  count|               63852|
|   mean|   56.31520537852275|
| stddev|   329.2087985975373|
|    min| !Buscalo!'s easy...|
|    max|              谷月社|
+-------+--------------------+
```

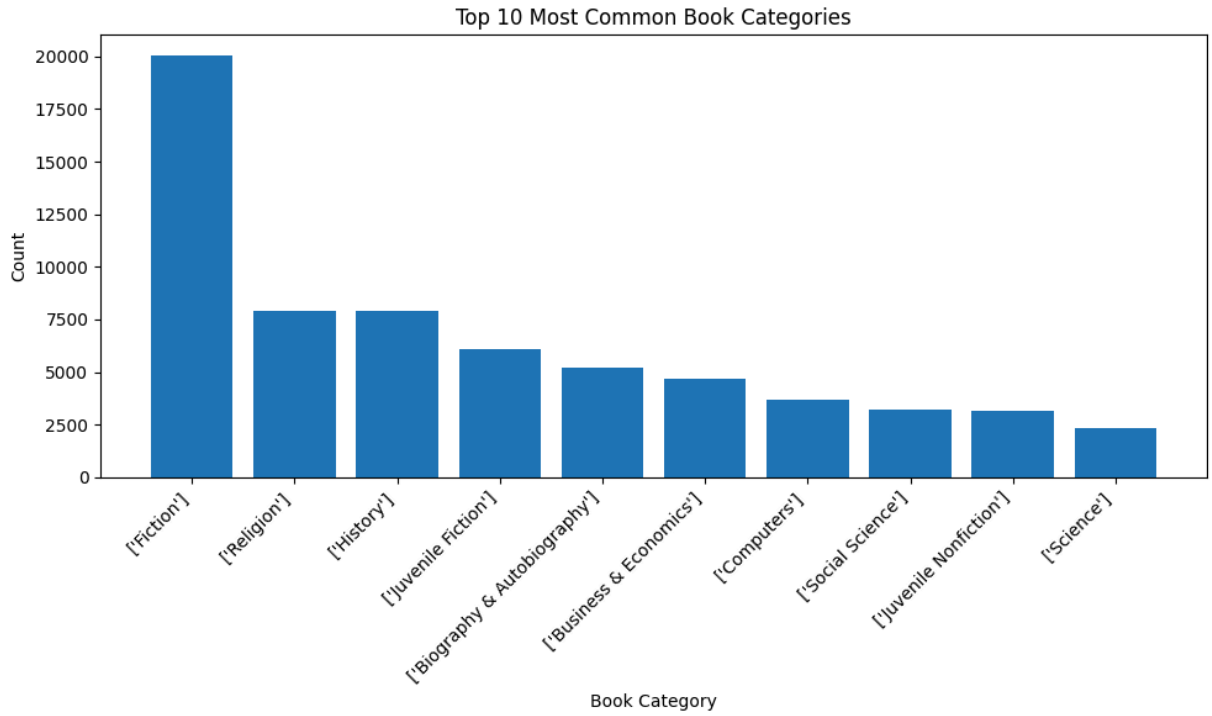
iv) Preliminary Visualizations

In [ ]:
```
import matplotlib.pyplot as plt
import pandas as pd
books_sample = books_df.select("ratingsCount").where(col("ratingsCount").isN
books_pd = books_sample.toPandas()
books_pd['ratingsCount'] = pd.to_numeric(books_pd['ratingsCount'], errors='c
plt.hist(books_pd['ratingsCount'].dropna(), bins=30)
plt.xlabel('Ratings Count')
```

```
plt.ylabel('Frequency')
plt.title('Distribution of Ratings Count in Books')
plt.show()
```



In [ ]:
```
from pyspark.sql.functions import explode, split, trim
books_exploded = books_df.withColumn("category", explode(split(col("categori
books_exploded = books_exploded.withColumn("category", trim(col("category"))
category_counts = books_exploded.groupBy("category").count().orderBy("count"
top_categories = category_counts.limit(10).toPandas()
import matplotlib.pyplot as plt
plt.figure(figsize=(10,6))
plt.bar(top_categories['category'], top_categories['count'])
plt.xticks(rotation=45, ha="right")
plt.title('Top 10 Most Common Book Categories')
plt.xlabel('Book Category')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```



Sub-Task 1b) EDA on Rating dataset

i) Counting the Rows, Displaying schema and First 5 Rows

In [ ]:
```
print("\nNumber of rows in rating.csv after sampling:")
print(rating_df.count())

print("\nSchema of rating.csv:")
rating_df.printSchema()

print("\nSample 5 rows from rating.csv:")
rating_df.show(5, truncate=False)
```

```
Number of rows in rating.csv after sampling:
[Stage 65:==================================================>       (19 + 3) /
22]
```

```
451526

Schema of rating.csv:
root
 |-- Id: string (nullable = true)
 |-- Title: string (nullable = true)
 |-- Price: string (nullable = true)
 |-- User_id: string (nullable = true)
 |-- profileName: string (nullable = true)
 |-- review/helpfulness: string (nullable = true)
 |-- review/score: string (nullable = true)
 |-- review/time: string (nullable = true)
 |-- review/summary: string (nullable = true)
 |-- review/text: string (nullable = true)


Sample 5 rows from rating.csv:
+----------+-------------------------------------------------------+-----+--
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-----+-----------+-----------------------------------------+----------------
--------------------------------------------------------------------------
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--------------------------------------------------------------------------
-------------------------------------------------------+
|Id        |Title                                                  |Price|Us
er_id       |profileName                        |review/helpfulness|review/
score|review/time|review/summary                           |review/text
|
+----------+-------------------------------------------------------+-----+--
------------+-----------------------------------+------------------+-------
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--------------------------------------------------------------------------
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--------------------------------------------------------------------------
-------------------------------------------------------+
|0826414346|Dr. Seuss: American Icon                               |NULL |A2
RSSXTDZDUSH4|J. Squire                          |0/0               |5.0
|1231200000 |Academia At It's Best                    |"When I recieved this
book as a gift for Christmas my first impression went along these line: ""Oh
great some professor who probably wrote alot of mumbo-jumbo that I don't wan
t to know about Dr. Suess and I know it won't be an enjoyable read.""Thanks
goodness I read it nonetheless.To my pleasure
|
|0595344550|Whispers of the Wicked Saints                           |10.95|AU
R0VA5H0C66C |"LoveToRead ""Actually Read Books"""|1/2               |1.0
|1119225600 |Buyer beware                             |"This is a self-publi
```

```
shed book, and if you want to know why--read a few paragraphs! Those 5 star
reviews must have been written by Ms. Haddon's family and friends--or perhap
s, by herself! I can't imagine anyone reading the whole thing--I spent an ev
ening with the book and a friend and we were in hysterics reading bits and p
ieces of it to one another. It is most definitely bad enough to be entered i
nto some kind of a ""worst book"" contest. I can't believe Amazon even sells
this kind of thing. Maybe I can offer them my 8th grade term paper on ""To K
ill a Mockingbird""--a book I am quite sure Ms. Haddon never heard of. Anywa
y                                                    |
|0595344550|Whispers of the Wicked Saints                           |10.95|AC
O23CG8K8T77 |Tonya                                |5/9               |5.0
|1117065600 |Bravo Veronica                            |I read the review dir
ectly under mine and I have to say I laughed. How can someone write a honest
review on a book they read only three pages of? That was funny but also sad
that people are mean enough to keep others from an enjoyable read. I loved t
his book. I thought the first chapter was a little long, but I read the WHOL
E book and I loved it. This book is about an afair of the heart and the righ
t to be happy. It is a glimps into the lives of those who are forbiden to lo
ve each other. This is a heart warming story. This is a first book and it ha
d a few flaws, but this writer will only get better and better as she goes f
orward. It was lovely and I am blessed that I read it.|
|0802841899|The Church of Christ: A Biblical Ecclesiology for Today|25.97|A3
6TPZSH8LBT1 |haskell                             |2/3               |5.0
|1311465600 |Early Christian development of the Church|Everett Ferguson appr
oaches the subject of early christian development with quotes from their ear
ly documents in their own words. Ferguson gives us a peek into the minds of
early Christians and puts it together in a form one can understand today. Th
e only criticism is that he can get too technical and detailed at times. Wit
h patience his writing is able to be followed by a serious student of the su
bject of God's church. I highly recommend this book for any serious student
of the subject!
|
|0802841899|The Church of Christ: A Biblical Ecclesiology for Today|25.97|A2
H2LORTA5EZY2|Edward E. Howe                      |3/5               |4.0
|1266192000 |Christ is Lord                            |This is a very useful
and thorough text book. I would recommend it for anyone who wants an in dept
h study of the Lord's church.
|
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only showing top 5 rows
```

ii) Missing Values Check

In [ ]:
```
print("\nMissing Values per Column in rating.csv:")
missing_rating = rating_df.select([
    count(when(col(c).isNull() | isnan(c) | (col(c) == ""), c)).alias(c)
    for c in rating_df.columns
])
missing_rating.show(truncate=False)
```

```
Missing Values per Column in rating.csv:
[Stage 43:==============================>                        (12 + 10) /
22]
+---+-----+------+-------+-----------+------------------+------------+------
-----+--------------+-----------+
|Id |Title|Price |User_id|profileName|review/helpfulness|review/score|revie
w/time|review/summary|review/text|
+---+-----+------+-------+-----------+------------------+------------+------
-----+--------------+-----------+
|0  |37   |378713|84583  |84596      |50                |16          |3
|11            |6          |
+---+-----+------+-------+-----------+------------------+------------+------
-----+--------------+-----------+
```

iii) Summary Statistics

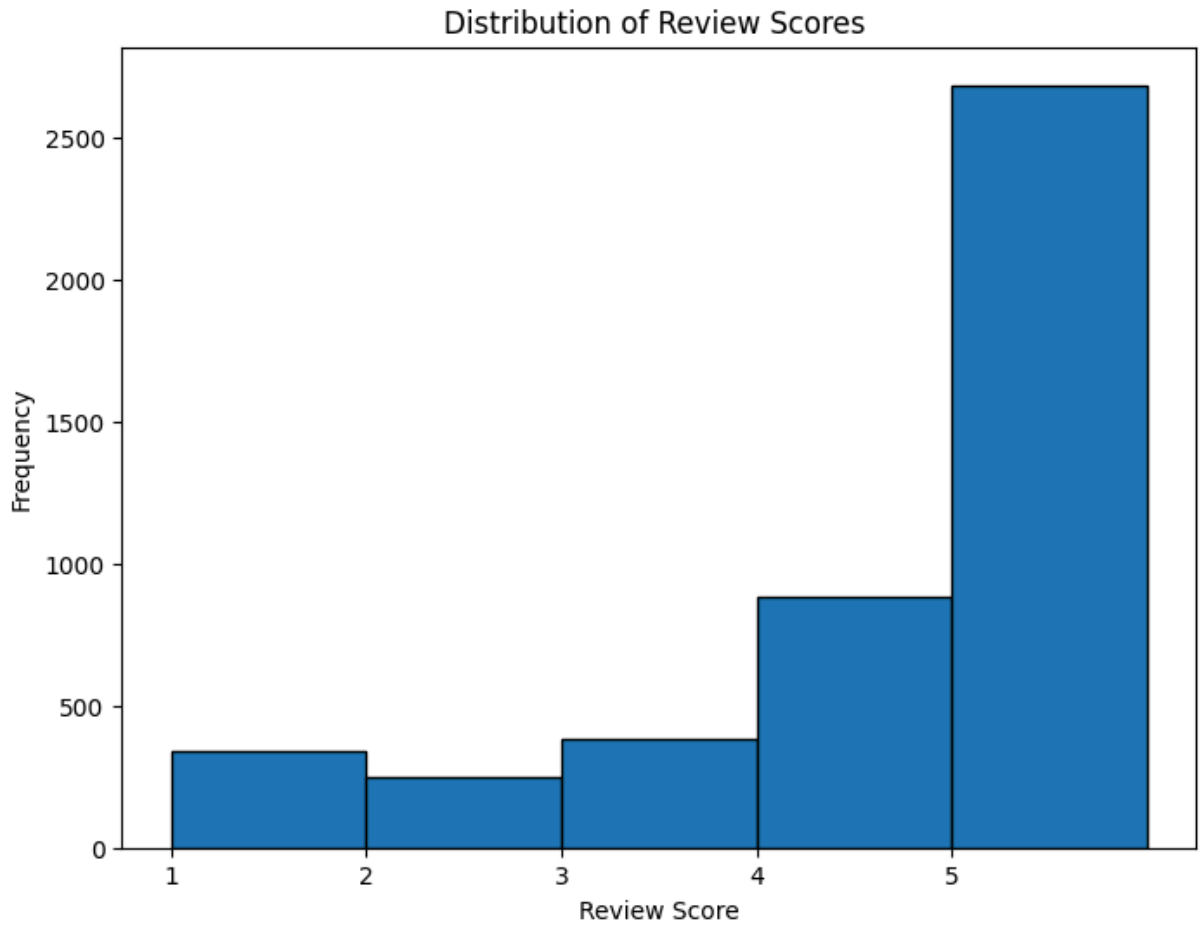
In [ ]:
```
print("\nSummary Statistics for review/score and Price in rating.csv:")
rating_df.select("review/score", "Price").describe().show()
```

```
Summary Statistics for review/score and Price in rating.csv:
[Stage 46:==============================>                        (12 + 10) /
22]
+-------+------------------+------------------+
|summary|      review/score|             Price|
+-------+------------------+------------------+
|  count|            451510|             72813|
|   mean|2704.0768961260897| 21.65799418324138|
| stddev|1808752.1432835055|26.003738148468713|
|    min|        & Books"""|            "" and|
|    max|         thersites|            995.00|
+-------+------------------+------------------+
```

iv) Preliminary Visualization

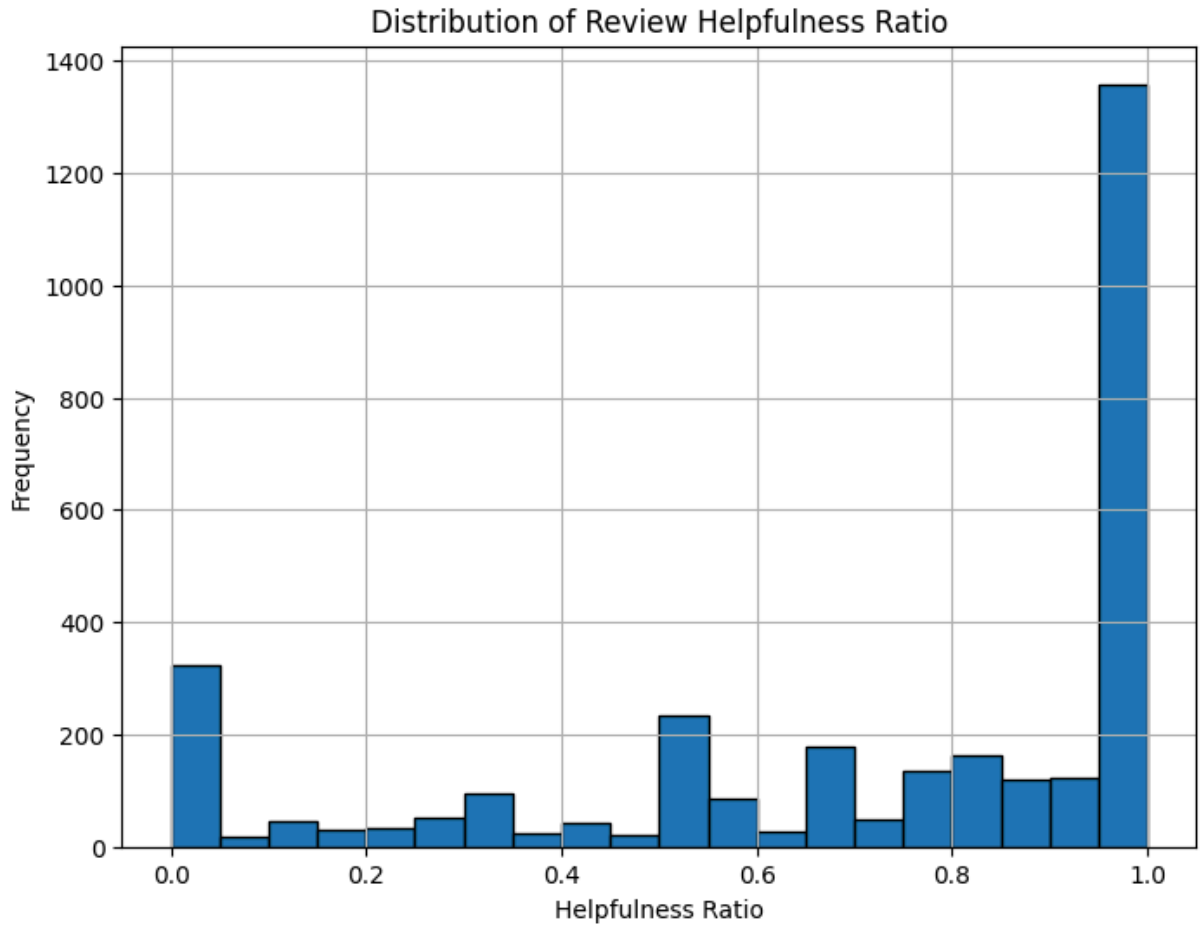
In [ ]:
```
rating_sample = rating_df.select("review/score").where(col("review/score").i
rating_pd = rating_sample.toPandas()
rating_pd['review/score'] = pd.to_numeric(rating_pd['review/score'], errors=
rating_pd = rating_pd.dropna()
import matplotlib.pyplot as plt
plt.figure(figsize=(8,6))
plt.hist(rating_pd['review/score'], bins=[ 1, 2, 3, 4, 5, 6], edgecolor='bla
plt.xlabel('Review Score')
plt.ylabel('Frequency')
```

```
plt.title('Distribution of Review Scores ')
plt.xticks([1, 2, 3, 4, 5])
plt.show()
```



In [ ]:

```
from pyspark.sql.functions import split, col, when
rating_helpfulness = rating_df.withColumn("helpful_votes", split(col("review
                              .withColumn("total_votes", split(col("review/h
rating_helpfulness = rating_helpfulness.withColumn(
    "helpfulness_ratio",
    when(col("total_votes") != 0, col("helpful_votes") / col("total_votes"))
)
helpfulness_sample = rating_helpfulness.select("helpfulness_ratio").where(cc
helpfulness_pd = helpfulness_sample.toPandas()
import matplotlib.pyplot as plt
plt.figure(figsize=(8,6))
plt.hist(helpfulness_pd['helpfulness_ratio'], bins=20, edgecolor='black')
plt.xlabel('Helpfulness Ratio')
plt.ylabel('Frequency')
plt.title('Distribution of Review Helpfulness Ratio')
plt.grid(True)
plt.show()
```



Task 2 - Data Cleaning and Processing

Sub-Task 2a) Data Cleaning and Processing on Books

i) Missing Value Handling

In [ ]:
```
from pyspark.sql.functions import col, isnan, when, count
books_df = books_df.filter(col("Title").isNotNull())
books_df = books_df.fillna({'ratingsCount': 0})
books_df = books_df.fillna({'categories': 'Unknown Category'})
books_df = books_df.fillna({
    'description': 'No Description Available',
    'authors': 'Unknown Author',
    'publisher': 'Unknown Publisher'
})
print("Missing Values AFTER Cleaning:")
books_df.select([
    count(when(col(c).isNull() | isnan(c), c)).alias(c)
    for c in books_df.columns
]).show()
print("Schema AFTER Cleaning:")
books_df.printSchema()
print("Sample rows AFTER Cleaning:")
```

```
books_df.show(5, truncate=False)
print(f"Total number of rows in books_df AFTER Cleaning: {books_df.count()}"
```

```
Missing Values AFTER Cleaning:
```

```
+-----+-----------+-------+-----+-----------+---------+-------------+-------
-+----------+------------+
|Title|description|authors|image|previewLink|publisher|publishedDate|infoLin
k|categories|ratingsCount|
+-----+-----------+-------+-----+-----------+---------+-------------+-------
-+----------+------------+
|    0|          0|      0|51191|      24055|        0|        25844|   2430
1|         0|           0|
+-----+-----------+-------+-----+-----------+---------+-------------+-------
-+----------+------------+

Schema AFTER Cleaning:
root
 |-- Title: string (nullable = true)
 |-- description: string (nullable = false)
 |-- authors: string (nullable = false)
 |-- image: string (nullable = true)
 |-- previewLink: string (nullable = true)
 |-- publisher: string (nullable = false)
 |-- publishedDate: string (nullable = true)
 |-- infoLink: string (nullable = true)
 |-- categories: string (nullable = false)
 |-- ratingsCount: integer (nullable = false)

Sample rows AFTER Cleaning:
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|Title                                                                       |d
escription
|authors                                      |image
|previewLink
|publisher
|publishedDate |infoLink
|categories
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|ratingsCount|
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|Its Only Art If Its Well Hung!                                            |N
o Description Available
|['Julie Strain']                          |http://books.google.com/books/c
ontent?id=DykPAAAACAAJ&printsec=frontcover&img=1&zoom=1&source=gbs_api|htt
p://books.google.nl/books?id=DykPAAAACAAJ&dq=Its+Only+Art+If+Its+Well+Hung!&
hl=&cd=1&source=gbs_api
|Unknown Publisher
|1996          |http://books.google.nl/books?id=DykPAAAACAAJ&dq=Its+Only+Art
+If+Its+Well+Hung!&hl=&source=gbs_api
|['Comics & Graphic Novels']
|0           |
|Dr. Seuss: American Icon
|"Philip Nel takes a fascinating look into the key aspects of Seuss's career
- his poetry, politics, art, marketing, and place in the popular imaginatio
n."" ""Nel argues convincingly that Dr. Seuss is one of the most influential
poets in America. His nonsense verse
| like that of Lewis Carroll and Edward Lear| has changed language itself
| giving us new words like ""nerd."" And Seuss's famously loopy artistic sty
le - what Nel terms an ""energetic cartoon surrealism"" - has been equally i
mportant| inspiring artists like filmmaker Tim Burton and illustrator Lane S
mith. --from back cover"|['Philip Nel']|http://books.google.com/books/conten
t?id=IjvHQsCn_pgC&printsec=frontcover&img=1&zoom=1&edge=curl&source=gbs_api
|http://books.google.nl/books?id=IjvHQsCn_pgC&printsec=frontcover&dq=Dr.+Seu
ss:+American+Icon&hl=&cd=1&source=gbs_api|0           |
|Wonderful Worship in Smaller Churches                                     |T
his resource includes twelve principles in understanding small church worshi
p, fifteen practices for planning worship with fewer than 100 people, and su
ggestions for congregational study.
|['David R. Ray']                          |http://books.google.com/books/c
ontent?id=2tsDAAAACAAJ&printsec=frontcover&img=1&zoom=1&source=gbs_api|htt
p://books.google.nl/books?id=2tsDAAAACAAJ&dq=Wonderful+Worship+in+Smaller+Ch
```

```
urches&hl=&cd=1&source=gbs_api
|Unknown Publisher
|2000          |http://books.google.nl/books?id=2tsDAAAACAAJ&dq=Wonderful+Wo
rship+in+Smaller+Churches&hl=&source=gbs_api
|['Religion']
|0           |
|Whispers of the Wicked Saints                                                |J
ulia Thomas finds her life spinning out of control after the death of her hu
sband, Richard. Julia turns to her minister for comfort when she finds herse
lf falling for him with a passion that is forbidden by the church. Heath Spa
rks is a man of God who is busy taking care of his quadriplegic wife who was
seriously injured in a sever car accident. In an innocent effort to reach ou
t to a lonely member of his church, Heath finds himself as the man and not t
he minister as Heath and Julia surrender their bodies to each other and face
the wrath of God. Julia finds herself in over her head as she faces a deadly
disease, the loss of her home and whispers about her wicked affair. Julia le
aves the states offering her body as a living sacrifice in hopes of finding
a cure while her heart remains thousands of miles away hoping to one day reu
nite with the man who holds it hostage.Whispers of the Wicked Saints is a on
ce in a lifetime romance that is breath taking, defying all the rules of rom
ance and bending the laws of love.|['Veronica Haddon']
|http://books.google.com/books/content?id=aRSIgJlq6JwC&printsec=frontcover&i
mg=1&zoom=1&source=gbs_api|http://books.google.nl/books?id=aRSIgJlq6JwC&dq=W
hispers+of+the+Wicked+Saints&hl=&cd=1&source=gbs_api
|iUniverse
|2005-02       |http://books.google.nl/books?id=aRSIgJlq6JwC&dq=Whispers+of+
the+Wicked+Saints&hl=&source=gbs_api
|['Fiction']
|0           |
|Nation Dance: Religion, Identity and Cultural Difference in the Caribbean|N
o Description Available
|['Edward Long']                            |NULL
|http://books.google.nl/books?id=399SPgAACAAJ&dq=Nation+Dance:+Religion,+Ide
ntity+and+Cultural+Difference+in+the+Caribbean&hl=&cd=1&source=gbs_api
|Unknown Publisher
|2003-03-01    |http://books.google.nl/books?id=399SPgAACAAJ&dq=Nation+Danc
e:+Religion,+Identity+and+Cultural+Difference+in+the+Caribbean&hl=&source=gb
s_api|Unknown Category
|0           |
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only showing top 5 rows

Total number of rows in books_df AFTER Cleaning: 212403
```

ii) Outlier Detection and Handling

In [ ]:
```python
from pyspark.sql.functions import col
books_df = books_df.withColumn("ratingsCount", col("ratingsCount").cast("int
quantiles = books_df.approxQuantile("ratingsCount", [0.25, 0.75], 0.05)
Q1, Q3 = quantiles
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print(f"Q1 (25th percentile): {Q1}")
print(f"Q3 (75th percentile): {Q3}")
print(f"IQR: {IQR}")
print(f"Lower Bound: {lower_bound}")
print(f"Upper Bound: {upper_bound}")

outliers = books_df.filter((col("ratingsCount") < lower_bound) | (col("ratir
print(f"Number of outliers detected in ratingsCount: {outliers.count()}")
outliers.select("Title", "ratingsCount").show(5, truncate=False)
```

```
Q1 (25th percentile): 0.0
Q3 (75th percentile): 0.0
IQR: 0.0
Lower Bound: 0.0
Upper Bound: 0.0
Number of outliers detected in ratingsCount: 43431
+-------------------------------------------------------+------------+
|Title                                                  |ratingsCount|
+-------------------------------------------------------+------------+
|The Church of Christ: A Biblical Ecclesiology for Today|5           |
|A Walk in the Woods: a Play in Two Acts                |3           |
|Voices from the Farm: Adventures in Community Living   |1           |
|The Battleship Bismarck                                |1           |
|Mini-mysteries                                         |1           |
+-------------------------------------------------------+------------+
only showing top 5 rows
```

iii) Datatype Conversion

In [ ]:
```python
from pyspark.sql.functions import col
print("Schema AFTER Data Type Conversion")
books_df.printSchema()
```

```
print("Sample rows AFTER Type Conversion ")
books_df.filter(col("ratingsCount") > 0) \
    .select("Title", "ratingsCount") \
    .orderBy(col("ratingsCount").desc()) \
    .show(5, truncate=False)
```

```
Schema AFTER Data Type Conversion
root
 |-- Title: string (nullable = true)
 |-- description: string (nullable = false)
 |-- authors: string (nullable = false)
 |-- image: string (nullable = true)
 |-- previewLink: string (nullable = true)
 |-- publisher: string (nullable = false)
 |-- publishedDate: string (nullable = true)
 |-- infoLink: string (nullable = true)
 |-- categories: string (nullable = false)
 |-- ratingsCount: integer (nullable = false)

Sample rows AFTER Type Conversion
+-----------------------------------------------------------------------------+
------------+
|Title                                                                        |
ratingsCount|
+-----------------------------------------------------------------------------+
------------+
|A Mission to Millions; The Story of Ernie Allen and the Every Home Crusade|
4572        |
|Unbroken                                                                     |
4572        |
|Blue Like Jazz: Nonreligious Thoughts on Christian Spirituality              |
4562        |
|Eclipse                                                                      |
4392        |
|Redeeming Love                                                               |
4249        |
+-----------------------------------------------------------------------------+
------------+
only showing top 5 rows
```

iv) Aggregation and Grouping

Grouping by categories

In [ ]:
```
from pyspark.sql.functions import col, avg, sum as _sum, count
category_agg = books_df.groupBy("categories").agg(
    _sum("ratingsCount").alias("total_ratings"),
    avg("ratingsCount").alias("average_ratings")
)
print("Total and Average ratingsCount by Category:")
category_agg.orderBy(col("total_ratings").desc()).show(10, truncate=False)
```

```
Total and Average ratingsCount by Category:
+-----------------------------+-------------+------------------+
|categories                   |total_ratings|average_ratings   |
+-----------------------------+-------------+------------------+
|Unknown Category             |366397       |9.041704710904918 |
|['Fiction']                  |327315       |16.3494005994006  |
|['Juvenile Fiction']         |89816        |14.752956636005257|
|['Biography & Autobiography']|86824        |16.674476666026504|
|['Religion']                 |29663        |3.743910135049855 |
|['Travel']                   |28437        |17.93001261034048 |
|['Young Adult Fiction']      |26981        |56.56394129979036 |
|Univ of California Press     |23802        |952.08            |
|['History']                  |23109        |2.9318700837350926|
|['Philosophy']               |22006        |14.28033744321869 |
+-----------------------------+-------------+------------------+
only showing top 10 rows
```

Grouping by publisher

In [ ]:
```
publisher_agg = books_df.groupBy("publisher").agg(
    _sum("ratingsCount").alias("total_ratings")
)
print("Total ratingsCount by Publisher:")
publisher_agg.orderBy(col("total_ratings").desc()).show(10, truncate=False)
```

```
Total ratingsCount by Publisher:
+-----------------------------+-------------+
|publisher                    |total_ratings|
+-----------------------------+-------------+
|Penguin                      |78065        |
|Unknown Publisher            |69011        |
|Random House                 |48628        |
|Simon and Schuster           |48580        |
|Harper Collins               |47932        |
|Vintage                      |33041        |
|Knopf Books for Young Readers|26914        |
|Macmillan                    |16611        |
|Hachette UK                  |15148        |
|Bantam                       |14282        |
+-----------------------------+-------------+
only showing top 10 rows
```

Grouping by publishdate

In [ ]:
```
from pyspark.sql.functions import col, regexp_extract, count
books_df = books_df.withColumn(
    "publishedYear",
    regexp_extract(col("publishedDate"), r'(\d{4})', 1)
)
books_df = books_df.filter(col("publishedYear") != "")
books_df = books_df.filter(
    (col("publishedYear").cast("int") >= 1900) &
    (col("publishedYear").cast("int") <= 2025)
)
```

```
year_agg = books_df.groupBy("publishedYear").agg(
    count("*").alias("number_of_books")
)
year_agg_filtered = year_agg.filter(col("number_of_books") > 1000)
print("Years with more than 1000 books published:")
year_agg_filtered.orderBy(col("publishedYear").asc()).show(truncate=False)
```

```
Years with more than 1000 books published:
+-------------+---------------+
|publishedYear|number_of_books|
+-------------+---------------+
|1983         |1013           |
|1984         |1125           |
|1985         |1187           |
|1986         |1307           |
|1987         |1406           |
|1988         |1461           |
|1989         |1700           |
|1990         |1880           |
|1991         |2313           |
|1992         |2485           |
|1993         |2621           |
|1994         |2963           |
|1995         |3446           |
|1996         |3905           |
|1997         |4541           |
|1998         |5367           |
|1999         |6028           |
|2000         |6487           |
|2001         |6142           |
|2002         |6584           |
+-------------+---------------+
only showing top 20 rows
```

In [ ]:
```
print("Top 10 Books by ratingsCount:")
books_df.select("Title", "ratingsCount").orderBy(col("ratingsCount").desc())
```

```
Top 10 Books by ratingsCount:
+--------------------------------------------------------------------------------
-----+------------+
|Title
|ratingsCount|
+--------------------------------------------------------------------------------
-----+------------+
|A Mission to Millions; The Story of Ernie Allen and the Every Home Crusade
|4572        |
|Unbroken
|4572        |
|Blue Like Jazz: Nonreligious Thoughts on Christian Spirituality
|4562        |
|Eclipse
|4392        |
|Redeeming Love
|4249        |
|A Walk in the Woods
|4234        |
|The Tipping Point: How Little Things Can Make a Big Difference (Wheeler Com
pass)|4218        |
|Travel Light, Eat Heavy Along the Appalachian Trail and Other Mystical Rout
es   |4161        |
|Henrietta
|4067        |
|Jesus;: A new biography
|3995        |
+--------------------------------------------------------------------------------
-----+------------+
only showing top 10 rows
```

Sub-Task 2b) Data Cleaning and Processing on Rating.csv

i) Missing Value Handling

In [ ]:
```
from pyspark.sql.functions import col, count, when
print("Missing Values per Column in rating.csv BEFORE Cleaning:")
rating_df.select([
    count(when(col(c).isNull() | (col(c) == ""), c)).alias(c)
    for c in rating_df.columns
]).show(truncate=False)
print("Schema BEFORE Cleaning:")
rating_df.printSchema()
print("Sample rows BEFORE Cleaning:")
rating_df.show(5, truncate=False)
print(f"Total number of rows in rating_df BEFORE Cleaning: {rating_df.count(
```

```
Missing Values per Column in rating.csv BEFORE Cleaning:
```

```
+---+-----+------+-------+-----------+------------------+------------+------
-----+--------------+-----------+
|Id |Title|Price |User_id|profileName|review/helpfulness|review/score|revie
w/time|review/summary|review/text|
+---+-----+------+-------+-----------+------------------+------------+------
-----+--------------+-----------+
|0  |37   |378713|84583  |84593      |50                |16          |3
|11            |6          |
+---+-----+------+-------+-----------+------------------+------------+------
-----+--------------+-----------+

Schema BEFORE Cleaning:
root
 |-- Id: string (nullable = true)
 |-- Title: string (nullable = true)
 |-- Price: string (nullable = true)
 |-- User_id: string (nullable = true)
 |-- profileName: string (nullable = true)
 |-- review/helpfulness: string (nullable = true)
 |-- review/score: string (nullable = true)
 |-- review/time: string (nullable = true)
 |-- review/summary: string (nullable = true)
 |-- review/text: string (nullable = true)

Sample rows BEFORE Cleaning:
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-----+-----------+-----------------------------------------+----------------
--------------------------------------------------------------------------
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--------------------------------------------------------------------------
--------------------------------------------------------------------------
--------------------------------------------------------------+
|Id        |Title                                                  |Price|Us
er_id       |profileName                        |review/helpfulness|review/
score|review/time|review/summary                          |review/text
|
+----------+-------------------------------------------------------+-----+--
------------+-----------------------------------+------------------+-------
-----+-----------+-----------------------------------------+----------------
--------------------------------------------------------------------------
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--------------------------------------------------------------+
|0826414346|Dr. Seuss: American Icon                               |NULL |A2
RSSXTDZDUSH4|J. Squire                          |0/0               |5.0
|1231200000 |Academia At It's Best                    |"When I recieved this
```

```
book as a gift for Christmas my first impression went along these line: ""Oh
great some professor who probably wrote alot of mumbo-jumbo that I don't wan
t to know about Dr. Suess and I know it won't be an enjoyable read.""Thanks
goodness I read it nonetheless.To my pleasure
|
|0595344550|Whispers of the Wicked Saints                          |10.95|AU
R0VA5H0C66C |"LoveToRead ""Actually Read Books""""|1/2              |1.0
|1119225600 |Buyer beware                              |"This is a self-publi
shed book, and if you want to know why--read a few paragraphs! Those 5 star
reviews must have been written by Ms. Haddon's family and friends--or perhap
s, by herself! I can't imagine anyone reading the whole thing--I spent an ev
ening with the book and a friend and we were in hysterics reading bits and p
ieces of it to one another. It is most definitely bad enough to be entered i
nto some kind of a ""worst book"" contest. I can't believe Amazon even sells
this kind of thing. Maybe I can offer them my 8th grade term paper on ""To K
ill a Mockingbird""--a book I am quite sure Ms. Haddon never heard of. Anywa
y                                                         |
|0595344550|Whispers of the Wicked Saints                          |10.95|AC
O23CG8K8T77 |Tonya                                |5/9              |5.0
|1117065600 |Bravo Veronica                            |I read the review dir
ectly under mine and I have to say I laughed. How can someone write a honest
review on a book they read only three pages of? That was funny but also sad
that people are mean enough to keep others from an enjoyable read. I loved t
his book. I thought the first chapter was a little long, but I read the WHOL
E book and I loved it. This book is about an afair of the heart and the righ
t to be happy. It is a glimps into the lives of those who are forbiden to lo
ve each other. This is a heart warming story. This is a first book and it ha
d a few flaws, but this writer will only get better and better as she goes f
orward. It was lovely and I am blessed that I read it.|
|0802841899|The Church of Christ: A Biblical Ecclesiology for Today|25.97|A3
6TPZSH8LBT1 |haskell                             |2/3              |5.0
|1311465600 |Early Christian development of the Church|Everett Ferguson appr
oaches the subject of early christian development with quotes from their ear
ly documents in their own words. Ferguson gives us a peek into the minds of
early Christians and puts it together in a form one can understand today. Th
e only criticism is that he can get too technical and detailed at times. Wit
h patience his writing is able to be followed by a serious student of the su
bject of God's church. I highly recommend this book for any serious student
of the subject!
|
|0802841899|The Church of Christ: A Biblical Ecclesiology for Today|25.97|A2
H2LORTA5EZY2|Edward E. Howe                      |3/5              |4.0
|1266192000 |Christ is Lord                            |This is a very useful
and thorough text book. I would recommend it for anyone who wants an in dept
h study of the Lord's church.
|
+----------+-------------------------------------------------------+-----+--
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-----+-----------+-----------------------------------------+----------------
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```

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-------------------------------------------------------------+
only showing top 5 rows

Total number of rows in rating_df BEFORE Cleaning: 451526
```

In [ ]:

```python
from pyspark.sql.functions import col, count, when
rating_df = rating_df.filter(col("Title").isNotNull())
rating_df = rating_df.filter(col("review/score").isNotNull())
rating_df = rating_df.fillna({'Price': 0})
rating_df = rating_df.fillna({
    'profileName': 'Anonymous',
    'review/summary': 'No Summary Provided',
    'review/text': 'No Review Text Provided'
})
print("Missing Values per Column in rating.csv AFTER Cleaning:")
rating_df.select([
    count(when(col(c).isNull() | (col(c) == ""), c)).alias(c)
    for c in rating_df.columns
]).show(truncate=False)
print("Schema AFTER Cleaning:")
rating_df.printSchema()
print("Sample rows AFTER Cleaning:")
rating_df.show(5, truncate=False)
print(f"Total number of rows in rating_df AFTER Cleaning: {rating_df.count()
```

```
Missing Values per Column in rating.csv AFTER Cleaning:
```

```
+---+-----+-----+-------+-----------+------------------+------------+-------
----+--------------+-----------+
|Id |Title|Price|User_id|profileName|review/helpfulness|review/score|review/
time|review/summary|review/text|
+---+-----+-----+-------+-----------+------------------+------------+-------
----+--------------+-----------+
|0  |0    |0    |84582  |0          |39                |0           |0
|0             |0          |
+---+-----+-----+-------+-----------+------------------+------------+-------
----+--------------+-----------+

Schema AFTER Cleaning:
root
 |-- Id: string (nullable = true)
 |-- Title: string (nullable = true)
 |-- Price: string (nullable = false)
 |-- User_id: string (nullable = true)
 |-- profileName: string (nullable = false)
 |-- review/helpfulness: string (nullable = true)
 |-- review/score: string (nullable = true)
 |-- review/time: string (nullable = true)
 |-- review/summary: string (nullable = false)
 |-- review/text: string (nullable = false)

Sample rows AFTER Cleaning:
+----------+-------------------------------------------------------+-----+--
------------+-----------------------------------+------------------+-------
-----+-----------+-----------------------------------------+----------------
--------------------------------------------------------------------------
--------------------------------------------------------------------------
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--------------------------------------------------------------------------
--------------------------------------------------------------------------
--------------------------------------------------------------+
|Id        |Title                                                  |Price|Us
er_id       |profileName                        |review/helpfulness|review/
score|review/time|review/summary                          |review/text
|
+----------+-------------------------------------------------------+-----+--
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--------------------------------------------------------------------------
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--------------------------------------------------------------------------
--------------------------------------------------------------------------
--------------------------------------------------------------+
|0826414346|Dr. Seuss: American Icon                               |0    |A2
RSSXTDZDUSH4|J. Squire                          |0/0               |5.0
|1231200000 |Academia At It's Best                    |"When I recieved this
```

```
book as a gift for Christmas my first impression went along these line: ""Oh
great some professor who probably wrote alot of mumbo-jumbo that I don't wan
t to know about Dr. Suess and I know it won't be an enjoyable read.""Thanks
goodness I read it nonetheless.To my pleasure
|
|0595344550|Whispers of the Wicked Saints                          |10.95|AU
R0VA5H0C66C |"LoveToRead ""Actually Read Books"""|1/2              |1.0
|1119225600 |Buyer beware                              |"This is a self-publi
shed book, and if you want to know why--read a few paragraphs! Those 5 star
reviews must have been written by Ms. Haddon's family and friends--or perhap
s, by herself! I can't imagine anyone reading the whole thing--I spent an ev
ening with the book and a friend and we were in hysterics reading bits and p
ieces of it to one another. It is most definitely bad enough to be entered i
nto some kind of a ""worst book"" contest. I can't believe Amazon even sells
this kind of thing. Maybe I can offer them my 8th grade term paper on ""To K
ill a Mockingbird""--a book I am quite sure Ms. Haddon never heard of. Anywa
y                                                |
|0595344550|Whispers of the Wicked Saints                          |10.95|AC
O23CG8K8T77 |Tonya                              |5/9              |5.0
|1117065600 |Bravo Veronica                            |I read the review dir
ectly under mine and I have to say I laughed. How can someone write a honest
review on a book they read only three pages of? That was funny but also sad
that people are mean enough to keep others from an enjoyable read. I loved t
his book. I thought the first chapter was a little long, but I read the WHOL
E book and I loved it. This book is about an afair of the heart and the righ
t to be happy. It is a glimps into the lives of those who are forbiden to lo
ve each other. This is a heart warming story. This is a first book and it ha
d a few flaws, but this writer will only get better and better as she goes f
orward. It was lovely and I am blessed that I read it.|
|0802841899|The Church of Christ: A Biblical Ecclesiology for Today|25.97|A3
6TPZSH8LBT1 |haskell                            |2/3              |5.0
|1311465600 |Early Christian development of the Church|Everett Ferguson appr
oaches the subject of early christian development with quotes from their ear
ly documents in their own words. Ferguson gives us a peek into the minds of
early Christians and puts it together in a form one can understand today. Th
e only criticism is that he can get too technical and detailed at times. Wit
h patience his writing is able to be followed by a serious student of the su
bject of God's church. I highly recommend this book for any serious student
of the subject!
|
|0802841899|The Church of Christ: A Biblical Ecclesiology for Today|25.97|A2
H2LORTA5EZY2|Edward E. Howe                      |3/5              |4.0
|1266192000 |Christ is Lord                            |This is a very useful
and thorough text book. I would recommend it for anyone who wants an in dept
h study of the Lord's church.
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only showing top 5 rows
```

```
[Stage 148:============================>                          (12 + 10) /
22]
Total number of rows in rating_df AFTER Cleaning: 451473
```

ii) Outlier Detection and Handling

In [ ]:
```
from pyspark.sql.functions import col, min, max
print("Min and Max of review/score:")
rating_df.select(
    min(col("review/score").cast("float")).alias("min_score"),
    max(col("review/score").cast("float")).alias("max_score")
).show()
print("Min and Max of Price:")
rating_df.select(
    min(col("Price").cast("float")).alias("min_price"),
    max(col("Price").cast("float")).alias("max_price")
).show()
print("Sample suspicious review/score values:")
rating_df.filter((col("review/score").cast("float") < 1.0) | (col("review/sc
    .select("Title", "review/score")\
    .show(5, truncate=False)
print("Sample suspicious Price values:")
rating_df.filter((col("Price").cast("float") > 500)).select("Title", "Price"
```

```
Min and Max of review/score:
```

```
+---------+---------+
|min_score|max_score|
+---------+---------+
|      1.0|1.21176E9|
+---------+---------+

Min and Max of Price:
```

```
+---------+---------+
|min_price|max_price|
+---------+---------+
|      0.0|    995.0|
+---------+---------+

Sample suspicious review/score values:
+--------------------------------------------------------------------------
-----------------------------------------+------------+
|Title
|review/score|
+--------------------------------------------------------------------------
-----------------------------------------+------------+
|"Management Secrets of the New England Patriots: From ""Patsies"" to two-ti
me Super Bowl champs; Vol. 1: Achievements|19.95       |
|"Management Secrets of the New England Patriots: From ""Patsies"" to two-ti
me Super Bowl champs; Vol. 1: Achievements|19.95       |
|"Management Secrets of the New England Patriots: From ""Patsies"" to two-ti
me Super Bowl champs; Vol. 1: Achievements|19.95       |
|"Management Secrets of the New England Patriots: From ""Patsies"" to two-ti
me Super Bowl champs; Vol. 1: Achievements|19.95       |
|"Management Secrets of the New England Patriots: From ""Patsies"" to two-ti
me Super Bowl champs; Vol. 1: Achievements|19.95       |
+--------------------------------------------------------------------------
-----------------------------------------+------------+
only showing top 5 rows

Sample suspicious Price values:
+--------------------------------------------------------------------------
----------+------+
|Title
|Price |
+--------------------------------------------------------------------------
----------+------+
|The Collected Works of J.G. Frazer: The Golden Bough (The Collected Works o
f James G)|630.00|
|The Collected Works of J.G. Frazer: The Golden Bough (The Collected Works o
f James G)|630.00|
|The Collected Works of J.G. Frazer: The Golden Bough (The Collected Works o
f James G)|630.00|
|The Collected Works of J.G. Frazer: The Golden Bough (The Collected Works o
f James G)|630.00|
|The Collected Works of J.G. Frazer: The Golden Bough (The Collected Works o
f James G)|630.00|
+--------------------------------------------------------------------------
----------+------+
only showing top 5 rows
```

In [ ]:

```
from pyspark.sql.functions import when, col
rating_df = rating_df.withColumn(
    "review/score",
    when(
        (col("review/score").cast("float") >= 1.0) & (col("review/score").ca
        col("review/score")
    ).otherwise(None)
```

```
)
rating_df = rating_df.filter(col("review/score").isNotNull())
rating_df = rating_df.withColumn(
    "Price",
    when(col("Price").cast("float") > 500, 500).otherwise(col("Price"))
)
print("Schema AFTER Outlier Handling:")
rating_df.printSchema()
print("Sample rows AFTER Outlier Handling:")
rating_df.select("Title", "Price", "review/score").show(5, truncate=False)
print(f"Total number of rows AFTER Outlier Handling: {rating_df.count()}")
```

```
Schema AFTER Outlier Handling:
root
 |-- Id: string (nullable = true)
 |-- Title: string (nullable = true)
 |-- Price: string (nullable = false)
 |-- User_id: string (nullable = true)
 |-- profileName: string (nullable = false)
 |-- review/helpfulness: string (nullable = true)
 |-- review/score: string (nullable = true)
 |-- review/time: string (nullable = true)
 |-- review/summary: string (nullable = false)
 |-- review/text: string (nullable = false)

Sample rows AFTER Outlier Handling:
+-------------------------------------------------------+-----+------------+
|Title                                                  |Price|review/score|
+-------------------------------------------------------+-----+------------+
|Dr. Seuss: American Icon                               |0    |5.0         |
|Whispers of the Wicked Saints                          |10.95|1.0         |
|Whispers of the Wicked Saints                          |10.95|5.0         |
|The Church of Christ: A Biblical Ecclesiology for Today|25.97|5.0         |
|The Church of Christ: A Biblical Ecclesiology for Today|25.97|4.0         |
+-------------------------------------------------------+-----+------------+
only showing top 5 rows
```

```
[Stage 164:============================>                        (12 + 10) /
22]
Total number of rows AFTER Outlier Handling: 448782
```

iii) Data Type Conversion

In [ ]:
```
from pyspark.sql.functions import col
print("Schema BEFORE Data Type Conversion:")
rating_df.printSchema()
print("Sample rows BEFORE Data Type Conversion:")
rating_df.select("Title", "Price", "review/score", "review/time").show(5, tr
```

```
Schema BEFORE Data Type Conversion:
root
 |-- Id: string (nullable = true)
 |-- Title: string (nullable = true)
 |-- Price: string (nullable = false)
 |-- User_id: string (nullable = true)
 |-- profileName: string (nullable = false)
 |-- review/helpfulness: string (nullable = true)
 |-- review/score: string (nullable = true)
 |-- review/time: string (nullable = true)
 |-- review/summary: string (nullable = false)
 |-- review/text: string (nullable = false)

Sample rows BEFORE Data Type Conversion:
+-------------------------------------------------------+-----+------------+
-----------+
|Title                                                  |Price|review/score|
review/time|
+-------------------------------------------------------+-----+------------+
-----------+
|Dr. Seuss: American Icon                               |0    |5.0         |
1231200000 |
|Whispers of the Wicked Saints                          |10.95|1.0         |
1119225600 |
|Whispers of the Wicked Saints                          |10.95|5.0         |
1117065600 |
|The Church of Christ: A Biblical Ecclesiology for Today|25.97|5.0         |
1311465600 |
|The Church of Christ: A Biblical Ecclesiology for Today|25.97|4.0         |
1266192000 |
+-------------------------------------------------------+-----+------------+
-----------+
only showing top 5 rows
```

In [ ]:

```
from pyspark.sql.functions import from_unixtime, col
rating_df = rating_df.withColumn("Price", col("Price").cast("float"))
rating_df = rating_df.withColumn("review/score", col("review/score").cast("f
rating_df = rating_df.withColumn("review/time", from_unixtime(col("review/ti
print("Schema AFTER Data Type Conversion:")
rating_df.printSchema()
print("Sample rows AFTER Data Type Conversion:")
rating_df.select("Title", "Price", "review/score", "review/time").show(5, tr
```

```
Schema AFTER Data Type Conversion:
root
 |-- Id: string (nullable = true)
 |-- Title: string (nullable = true)
 |-- Price: float (nullable = true)
 |-- User_id: string (nullable = true)
 |-- profileName: string (nullable = false)
 |-- review/helpfulness: string (nullable = true)
 |-- review/score: float (nullable = true)
 |-- review/time: timestamp (nullable = true)
 |-- review/summary: string (nullable = false)
 |-- review/text: string (nullable = false)

Sample rows AFTER Data Type Conversion:
+-------------------------------------------------------+-----+------------+
-------------------+
|Title                                                  |Price|review/score|
review/time        |
+-------------------------------------------------------+-----+------------+
-------------------+
|Dr. Seuss: American Icon                               |0.0  |5.0         |
2009-01-05 19:00:00|
|Whispers of the Wicked Saints                          |10.95|1.0         |
2005-06-19 20:00:00|
|Whispers of the Wicked Saints                          |10.95|5.0         |
2005-05-25 20:00:00|
|The Church of Christ: A Biblical Ecclesiology for Today|25.97|5.0         |
2011-07-23 20:00:00|
|The Church of Christ: A Biblical Ecclesiology for Today|25.97|4.0         |
2010-02-14 19:00:00|
+-------------------------------------------------------+-----+------------+
-------------------+
only showing top 5 rows
```

iv) Aggregation and Grouping

In [ ]:

```python
from pyspark.sql.functions import avg, count, col
score_distribution = rating_df.groupBy("review/score")\
    .agg(count("*").alias("number_of_reviews"))\
    .orderBy(col("review/score").asc())
print("Review Score Distribution:")
score_distribution.show(truncate=False)
price_distribution = rating_df.withColumn(
    "PriceRange",
    when(col("Price") < 10, "0-10")\
    .when((col("Price") >= 10) & (col("Price") < 50), "10-50")\
    .when((col("Price") >= 50) & (col("Price") < 200), "50-200")\
    .otherwise("200+")
)
price_agg = price_distribution.groupBy("PriceRange").agg(
    count("*").alias("number_of_books")
)
print("Price Range Distribution:")
price_agg.orderBy("PriceRange").show(truncate=False)
```

```
from pyspark.sql.functions import year
rating_df = rating_df.withColumn("review_year", year(col("review/time")))
yearly_reviews = rating_df.groupBy("review_year")\
    .agg(count("*").alias("number_of_reviews"))\
    .orderBy("review_year")
print("Yearly Review Distribution:")
yearly_reviews.show(truncate=False)
```

```
Review Score Distribution:

+------------+-----------------+
|review/score|number_of_reviews|
+------------+-----------------+
|1.0         |30141            |
|2.0         |22714            |
|3.0         |38174            |
|4.0         |87538            |
|5.0         |270215           |
+------------+-----------------+

Price Range Distribution:

+----------+---------------+
|PriceRange|number_of_books|
+----------+---------------+
|0-10      |391114         |
|10-50     |52910          |
|200+      |152            |
|50-200    |4606           |
+----------+---------------+

Yearly Review Distribution:
[Stage 175:============================>                          (12 + 10) /
22]
```

```
+-----------+-----------------+
|review_year|number_of_reviews|
+-----------+-----------------+
|1969       |4                |
|1995       |49               |
|1996       |997              |
|1997       |6416             |
|1998       |18716            |
|1999       |33210            |
|2000       |33911            |
|2001       |28372            |
|2002       |27523            |
|2003       |26331            |
|2004       |26601            |
|2005       |30593            |
|2006       |28857            |
|2007       |25255            |
|2008       |21251            |
|2009       |22883            |
|2010       |23722            |
|2011       |25723            |
|2012       |44838            |
|2013       |23530            |
+-----------+-----------------+
```

Merging the csv files before writing it back to HDFS

In [ ]:
```
print("Schema of books_df:")
books_df.printSchema()
print("Sample rows from books_df:")
books_df.show(5, truncate=False)
print("Schema of rating_df:")
rating_df.printSchema()
print("Sample rows from rating_df:")
rating_df.show(5, truncate=False)
```

```
Schema of books_df:
root
 |-- Title: string (nullable = true)
 |-- description: string (nullable = false)
 |-- authors: string (nullable = false)
 |-- image: string (nullable = true)
 |-- previewLink: string (nullable = true)
 |-- publisher: string (nullable = false)
 |-- publishedDate: string (nullable = true)
 |-- infoLink: string (nullable = true)
 |-- categories: string (nullable = false)
 |-- ratingsCount: integer (nullable = false)
 |-- publishedYear: string (nullable = true)

Sample rows from books_df:
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--------------+--------------------------+-------------+--------------------
-------------------------------------------------------------------------
---------------------------------------------+---------------------------+---
---------+-------------+
|Title                                                                  |d
escription
|authors             |image
|previewLink
|publisher                 |publishedDate|infoLink
|categories                 |ratingsCount|publishedYear|
+-----------------------------------------------------------------------+-
-------------------------------------------------------------------------
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```
--------------------------------------------------------------------------------
-------------------------------------+--------------------+-----------------
--------------------------------------------------------------------------------
------------------+-------------------------------------------------------------
--------------------------------------------------------------------------------
--------------+--------------------------+-------------+--------------------
--------------------------------------------------------------------------------
---------------------------------------------+---------------------------+---
---------+-------------+
|Its Only Art If Its Well Hung!                                              |N
o Description Available
|['Julie Strain']    |http://books.google.com/books/content?id=DykPAAAACAAJ&
printsec=frontcover&img=1&zoom=1&source=gbs_api          |http://books.googl
e.nl/books?id=DykPAAAACAAJ&dq=Its+Only+Art+If+Its+Well+Hung!&hl=&cd=1&source
=gbs_api                                              |Unknown Publisher
|1996         |http://books.google.nl/books?id=DykPAAAACAAJ&dq=Its+Only+Art+
If+Its+Well+Hung!&hl=&source=gbs_api
|['Comics & Graphic Novels']|0           |1996         |
|Wonderful Worship in Smaller Churches                                       |T
his resource includes twelve principles in understanding small church worshi
p, fifteen practices for planning worship with fewer than 100 people, and su
ggestions for congregational study.
|['David R. Ray']    |http://books.google.com/books/content?id=2tsDAAAACAAJ&
printsec=frontcover&img=1&zoom=1&source=gbs_api          |http://books.googl
e.nl/books?id=2tsDAAAACAAJ&dq=Wonderful+Worship+in+Smaller+Churches&hl=&cd=1
&source=gbs_api                                       |Unknown Publisher
|2000         |http://books.google.nl/books?id=2tsDAAAACAAJ&dq=Wonderful+Wor
ship+in+Smaller+Churches&hl=&source=gbs_api
|['Religion']                |0           |2000         |
|Whispers of the Wicked Saints                                               |J
ulia Thomas finds her life spinning out of control after the death of her hu
sband, Richard. Julia turns to her minister for comfort when she finds herse
lf falling for him with a passion that is forbidden by the church. Heath Spa
rks is a man of God who is busy taking care of his quadriplegic wife who was
seriously injured in a sever car accident. In an innocent effort to reach ou
t to a lonely member of his church, Heath finds himself as the man and not t
he minister as Heath and Julia surrender their bodies to each other and face
the wrath of God. Julia finds herself in over her head as she faces a deadly
disease, the loss of her home and whispers about her wicked affair. Julia le
aves the states offering her body as a living sacrifice in hopes of finding
a cure while her heart remains thousands of miles away hoping to one day reu
nite with the man who holds it hostage.Whispers of the Wicked Saints is a on
ce in a lifetime romance that is breath taking, defying all the rules of rom
ance and bending the laws of love.|['Veronica Haddon'] |http://books.google.
com/books/content?id=aRSIgJlq6JwC&printsec=frontcover&img=1&zoom=1&source=gb
s_api          |http://books.google.nl/books?id=aRSIgJlq6JwC&dq=Whispers+of+
the+Wicked+Saints&hl=&cd=1&source=gbs_api
|iUniverse                  |2005-02      |http://books.google.nl/books?id=aR
SIgJlq6JwC&dq=Whispers+of+the+Wicked+Saints&hl=&source=gbs_api
|['Fiction']                 |0           |2005         |
|Nation Dance: Religion, Identity and Cultural Difference in the Caribbean|N
o Description Available
|['Edward Long']     |NULL
|http://books.google.nl/books?id=399SPgAACAAJ&dq=Nation+Dance:+Religion,+Ide
ntity+and+Cultural+Difference+in+the+Caribbean&hl=&cd=1&source=gbs_api  |Unk
nown Publisher         |2003-03-01   |http://books.google.nl/books?id=399SPg
```

```
AACAAJ&dq=Nation+Dance:+Religion,+Identity+and+Cultural+Difference+in+the+Ca
ribbean&hl=&source=gbs_api|Unknown Category           |0           |2003
|
|The Church of Christ: A Biblical Ecclesiology for Today                  |I
n The Church of Christ: A Biblical Ecclesiology for Today, respected biblica
l scholar Everett Ferguson presents a genuine biblical theology of the churc
h. By systematically examining the New Testament's teaching on the existenc
e, meaning, and purpose of the church, providing responsible coverage of the
traditional topics in ecclesiology, and carefully grounding ecclesiology in
the person and work of Christ, Ferguson unveils a comprehensive model of the
church that is both biblically centered and relevant to a world on the verge
of the twenty-first century.
|['Everett Ferguson']|http://books.google.com/books/content?id=kVqRaiPlx88C&
printsec=frontcover&img=1&zoom=1&edge=curl&source=gbs_api|http://books.googl
e.nl/books?id=kVqRaiPlx88C&printsec=frontcover&dq=The+Church+of+Christ:+A+Bi
blical+Ecclesiology+for+Today&hl=&cd=1&source=gbs_api|Wm. B. Eerdmans Publis
hing|1996         |http://books.google.nl/books?id=kVqRaiPlx88C&dq=The+Churc
h+of+Christ:+A+Biblical+Ecclesiology+for+Today&hl=&source=gbs_api
|['Religion']               |5           |1996         |
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--------------+--------------------------+-------------+--------------------
-------------------------------------------------------------------------
---------------------------------------------+---------------------------+---
---------+-------------+
only showing top 5 rows

Schema of rating_df:
root
 |-- Id: string (nullable = true)
 |-- Title: string (nullable = true)
 |-- Price: float (nullable = true)
 |-- User_id: string (nullable = true)
 |-- profileName: string (nullable = false)
 |-- review/helpfulness: string (nullable = true)
 |-- review/score: float (nullable = true)
 |-- review/time: timestamp (nullable = true)
 |-- review/summary: string (nullable = false)
 |-- review/text: string (nullable = false)
 |-- review_year: integer (nullable = true)
```

```
Sample rows from rating_df:
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-----+-------------------+-----------------------------------------+--------
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--------------------------------------------------------------------------
--------------------------------------------------------------------------
-----------------------------------------------------------------+-------
----+
|Id        |Title                                                  |Price|Us
er_id       |profileName                        |review/helpfulness|review/
score|review/time        |review/summary                          |review/t
ext
|review_year|
+----------+-------------------------------------------------------+-----+--
------------+-----------------------------------+------------------+-------
-----+-------------------+-----------------------------------------+--------
--------------------------------------------------------------------------
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--------------------------------------------------------------------------
--------------------------------------------------------------------------
-----------------------------------------------------------------+-------
----+
|0826414346|Dr. Seuss: American Icon                               |0.0  |A2
RSSXTDZDUSH4|J. Squire                          |0/0               |5.0
|2009-01-05 19:00:00|Academia At It's Best                    |"When I recie
ved this book as a gift for Christmas my first impression went along these l
ine: ""Oh great some professor who probably wrote alot of mumbo-jumbo that I
don't want to know about Dr. Suess and I know it won't be an enjoyable rea
d.""Thanks goodness I read it nonetheless.To my pleasure
|2009       |
|0595344550|Whispers of the Wicked Saints                          |10.95|AU
R0VA5H0C66C |"LoveToRead ""Actually Read Books"""|1/2               |1.0
|2005-06-19 20:00:00|Buyer beware                             |"This is a se
lf-published book, and if you want to know why--read a few paragraphs! Those
5 star reviews must have been written by Ms. Haddon's family and friends--or
perhaps, by herself! I can't imagine anyone reading the whole thing--I spent
an evening with the book and a friend and we were in hysterics reading bits
and pieces of it to one another. It is most definitely bad enough to be ente
red into some kind of a ""worst book"" contest. I can't believe Amazon even
sells this kind of thing. Maybe I can offer them my 8th grade term paper on
""To Kill a Mockingbird""--a book I am quite sure Ms. Haddon never heard of.
Anyway                                                         |2005       |
|0595344550|Whispers of the Wicked Saints                          |10.95|AC
O23CG8K8T77 |Tonya                              |5/9               |5.0
|2005-05-25 20:00:00|Bravo Veronica                           |I read the re
```

```
view directly under mine and I have to say I laughed. How can someone write
a honest review on a book they read only three pages of? That was funny but
also sad that people are mean enough to keep others from an enjoyable read.
I loved this book. I thought the first chapter was a little long, but I read
the WHOLE book and I loved it. This book is about an afair of the heart and
the right to be happy. It is a glimps into the lives of those who are forbid
en to love each other. This is a heart warming story. This is a first book a
nd it had a few flaws, but this writer will only get better and better as sh
e goes forward. It was lovely and I am blessed that I read it.|2005       |
|0802841899|The Church of Christ: A Biblical Ecclesiology for Today|25.97|A3
6TPZSH8LBT1 |haskell                            |2/3               |5.0
|2011-07-23 20:00:00|Early Christian development of the Church|Everett Fergu
son approaches the subject of early christian development with quotes from t
heir early documents in their own words. Ferguson gives us a peek into the m
inds of early Christians and puts it together in a form one can understand t
oday. The only criticism is that he can get too technical and detailed at ti
mes. With patience his writing is able to be followed by a serious student o
f the subject of God's church. I highly recommend this book for any serious
student of the subject!
|2011       |
|0802841899|The Church of Christ: A Biblical Ecclesiology for Today|25.97|A2
H2LORTA5EZY2|Edward E. Howe                     |3/5               |4.0
|2010-02-14 19:00:00|Christ is Lord                           |This is a ver
y useful and thorough text book. I would recommend it for anyone who wants a
n in depth study of the Lord's church.
|2010       |
+----------+-------------------------------------------------------+-----+--
------------+-----------------------------------+------------------+-------
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--------------------------------------------------------------------+-------
----+
only showing top 5 rows
```

In [ ]:

```
merged_df = books_df.join(rating_df, "Title", "inner")
print("Schema of the Merged DataFrame:")
merged_df.printSchema()
print("Sample rows after merging:")
merged_df.show(5, truncate=False)
print(f"Total number of rows in the merged DataFrame: {merged_df.count()}")
```

```
Schema of the Merged DataFrame:
root
 |-- Title: string (nullable = true)
 |-- description: string (nullable = false)
 |-- authors: string (nullable = false)
 |-- image: string (nullable = true)
 |-- previewLink: string (nullable = true)
 |-- publisher: string (nullable = false)
 |-- publishedDate: string (nullable = true)
 |-- infoLink: string (nullable = true)
 |-- categories: string (nullable = false)
 |-- ratingsCount: integer (nullable = false)
 |-- publishedYear: string (nullable = true)
 |-- Id: string (nullable = true)
 |-- Price: float (nullable = true)
 |-- User_id: string (nullable = true)
 |-- profileName: string (nullable = false)
 |-- review/helpfulness: string (nullable = true)
 |-- review/score: float (nullable = true)
 |-- review/time: timestamp (nullable = true)
 |-- review/summary: string (nullable = false)
 |-- review/text: string (nullable = false)
 |-- review_year: integer (nullable = true)

Sample rows after merging:
```

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-----------------------------------------------------------------------------
------+-----------+
|Title                                          |description
|authors                                     |image
|previewLink
|publisher                 |publishedDate|infoLink
|categories                   |ratingsCount|publishedYear|Id        |Price|U
ser_id       |profileName                                |review/helpfulne
ss|review/score|review/time        |review/summary                        |
review/text
|review_year|
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```

```
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------+-----------+
|"""C"" Language"                                |Introduces the features of
the C programming language, discusses data types, variables, operators, cont
rol flow, functions, pointers, arrays, and structures, and looks at the UNIX
system interface
|['Brian W. Kernighan', 'Dennis M. Ritchie']|http://books.google.com/books/c
ontent?id=OpJ_0zpF7jIC&printsec=frontcover&img=1&zoom=1&edge=curl&source=gbs
_api|http://books.google.com/books?id=OpJ_0zpF7jIC&printsec=frontcover&dq=%2
2C%22+Language&hl=&cd=1&source=gbs_api                          |Pearson Ed
ucación         |1988         |http://books.google.com/books?id=OpJ_0zpF7jIC
&dq=%22C%22+Language&hl=&source=gbs_api                              |['Com
puters']                |12          |1988         |0931361109|0.0  |AHUQNC0
CQKOVD |"CPUGuy1 ""Steve"""                          |1/1               |4.0
|2000-10-31 19:00:00|A lot Information in such a small space|It's almost bet
ter than having cheat sheet in school. More than 90% of the information you
keep thumbing through the book for in on this double sided plastic 'Micro Ch
art'. Very helpful.
|2000       |
|"""Life Was Never Meant to Be a Struggle"""    |Stuart helps you identify t
he cause of struggle in your life and shows you how to eliminate it quickly.
Your heritage is to be free. To achieve that freedom, you have to move gradu
ally from struggle into free FLOW.
|['Stuart Wilde']                          |http://books.google.com/books/c
ontent?id=ByBnDwAAQBAJ&printsec=frontcover&img=1&zoom=1&edge=curl&source=gbs
_api|http://books.google.com/books?id=ByBnDwAAQBAJ&printsec=frontcover&dq=%2
2Life+Was+Never+Meant+to+Be+a+Struggle%22&hl=&cd=1&source=gbs_api|Hay House,
Inc            |1998-04-01   |https://play.google.com/store/books/details?id
=ByBnDwAAQBAJ&source=gbs_api                                         |['Self
-Help']                |1           |1998         |1561705357|5.0  |A1ABQRUC
UCSK8M|"Mike Stokes ""www.CajunCookingTV.com"""     |3/8               |5.0
|2006-12-24 19:00:00|One of the BEST Books Ever             |All of Stuart W
ilde's books are excellent. This one is one of the BEST.
|2006       |
|"""Life Was Never Meant to Be a Struggle"""    |Stuart helps you identify t
he cause of struggle in your life and shows you how to eliminate it quickly.
Your heritage is to be free. To achieve that freedom, you have to move gradu
ally from struggle into free FLOW.
|['Stuart Wilde']                          |http://books.google.com/books/c
ontent?id=ByBnDwAAQBAJ&printsec=frontcover&img=1&zoom=1&edge=curl&source=gbs
_api|http://books.google.com/books?id=ByBnDwAAQBAJ&printsec=frontcover&dq=%2
2Life+Was+Never+Meant+to+Be+a+Struggle%22&hl=&cd=1&source=gbs_api|Hay House,
Inc            |1998-04-01   |https://play.google.com/store/books/details?id
=ByBnDwAAQBAJ&source=gbs_api                                         |['Self
-Help']                |1           |1998         |1561705357|5.0  |AVW8VQ5Y
FK09F |"Brendan Carl Clarke ""Brendan Carl Clarke"""|3/4               |5.0
|2009-09-06 20:00:00|Spiritually Empowering reading         |"It seems that
Wilde cant go wrong in his writing. This book is yet another gem from the Sp
iritual wierdo we all love. I mean it IS indeed alittle strange. This is the
guy who talks about UFO's and aliens and visions and wobbling revolving bein
gs of light with geometrical shapes and sometimes I found my eyebrows raisin
g in a ""This guys maaaaaaaaaaaad"" way but I'm afraid that it all makes sen
se. Wilde is not only sincere but also coherent and well expressed. He gives
you ideas that remain in your mind and fill you with inspiration. Read this
book and be inspired and empowered."|2009       |
```

```
|"""Mildred Pierce"                              |In Mildred Pierce, noir mas
ter James M. Cain creates a novel of acute social observation and devasting
emotional violence, with a heroine whose ambitions and sufferings are never
less than recognizable. Mildred Pierce had gorgeous legs, a way with a skill
et, and a bone-deep core of toughness. She used those attributes to survive
a divorce and poverty and to claw her way out of the lower middle class. But
Mildred also had two weaknesses: a yen for shiftless men, and an unreasoning
devotion to a monstrous daughter.|['James M. Cain']
|http://books.google.com/books/content?id=6G-BF2GTGdAC&printsec=frontcover&i
mg=1&zoom=1&edge=curl&source=gbs_api|http://books.google.com/books?id=6G-BF2
GTGdAC&printsec=frontcover&dq=%22Mildred+Pierce&hl=&cd=1&source=gbs_api
|Vintage Crime/Black Lizard|2010-12-29   |https://play.google.com/store/book
s/details?id=6G-BF2GTGdAC&source=gbs_api
|['Fiction']                  |28          |2010         |B000H9YNDY|0.0  |A
21VR7M8O55EF6|Robin Friedman                                |2/2
|5.0         |2012-04-29 20:00:00|A Second Look at Mildred Pierce        |"W
ith its celebrated 1945 movie adaptation and award-winning 2011 HBO miniseri
es, James M. Cain's 1941 novel, ""Mildred Pierce"" has become a small Americ
an classic. I read and reviewed Cain's novel two years ago here on Amazon. I
recently took the opportunity to reread the work for a book group. Because a
number of separate editions of ""Mildred Pierce"" are available here on Amaz
on
|2012       |
|"Angel Sight: ""Let The Angels Be Your Sight"""|No Description Available
|['L. Wertz']                                |http://books.google.com/books/c
ontent?id=mVyuSjWrVYkC&printsec=frontcover&img=1&zoom=1&source=gbs_api
|http://books.google.com/books?id=mVyuSjWrVYkC&dq=Angel+Sight:+%22Let+The+An
gels+Be+Your+Sight%22&hl=&cd=1&source=gbs_api                |AuthorHouse
|2005-09-14   |http://books.google.com/books?id=mVyuSjWrVYkC&dq=Angel+Sight:
+%22Let+The+Angels+Be+Your+Sight%22&hl=&source=gbs_api|['Biography & Autobio
graphy']|0           |2005         |1420855131|19.98|AL0Z4207VRXMU |"AngelSi
ght ""Angelsight"""                  |0/1               |5.0         |2006-0
1-03 19:00:00|I loved Angel Sight!                   |I thought it was refre
shing to see how an everyday lady sees angels in everything she does. I was
impressed by her techniques to catch my eye; to see if I was truly paying at
tention or if I was just letting the words pass me by.I would highly recomme
nd this book, Angel Sight, to everyone I know.
|2006       |
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------+-----------+
only showing top 5 rows
```

```
[Stage 186:=====================================================>  (21 + 1) /
22]
Total number of rows in the merged DataFrame: 336911
```

In [ ]:
```
merged_output_path = "hdfs://localhost:9000/user/sathw/output/cleaned_data.p
merged_df.write.parquet(merged_output_path, mode="overwrite")
print("Clean and processed data is written to HDFS as Parquet.")
```

```
[Stage 203:=====================================================>    (12 + 1) /
13]
Clean and processed data is written to HDFS as Parquet.
```

```
25/04/26 21:54:16 WARN HeartbeatReceiver: Removing executor driver with no r
ecent heartbeats: 1853895 ms exceeds timeout 800000 ms
25/04/26 21:54:16 WARN SparkContext: Killing executors is not supported by c
urrent scheduler.
25/04/26 22:25:22 ERROR Inbox: Ignoring error
org.apache.spark.SparkException: Exception thrown in awaitResult:
        at org.apache.spark.util.SparkThreadUtils$.awaitResult(SparkThreadUt
ils.scala:56)
        at org.apache.spark.util.ThreadUtils$.awaitResult(ThreadUtils.scala:
310)
        at org.apache.spark.rpc.RpcTimeout.awaitResult(RpcTimeout.scala:75)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRefByURI(RpcEnv.scala:10
2)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRef(RpcEnv.scala:110)
        at org.apache.spark.util.RpcUtils$.makeDriverRef(RpcUtils.scala:36)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.driverEndpoin
t$lzycompute(BlockManagerMasterEndpoint.scala:124)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$driverEndpoint(BlockManagerMasterEnd
point.scala:123)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$lzycompute$1(BlockManagerMasterEndpoint.scala:688)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$1(BlockManagerMasterEndpoint.scala:687)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$register(BlockManagerMasterEndpoint.
scala:725)
        at org.apache.spark.storage.BlockManagerMasterEndpoint$$anonfun$rece
iveAndReply$1.applyOrElse(BlockManagerMasterEndpoint.scala:133)
        at org.apache.spark.rpc.netty.Inbox.$anonfun$process$1(Inbox.scala:1
03)
        at org.apache.spark.rpc.netty.Inbox.safelyCall(Inbox.scala:213)
        at org.apache.spark.rpc.netty.Inbox.process(Inbox.scala:100)
        at org.apache.spark.rpc.netty.MessageLoop.org$apache$spark$rpc$netty
$MessageLoop$$receiveLoop(MessageLoop.scala:75)
        at org.apache.spark.rpc.netty.MessageLoop$$anon$1.run(MessageLoop.sc
ala:41)
        at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(Threa
dPoolExecutor.java:1128)
        at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(Thre
adPoolExecutor.java:628)
        at java.base/java.lang.Thread.run(Thread.java:829)
Caused by: org.apache.spark.rpc.RpcEndpointNotFoundException: Cannot find en
dpoint: spark://CoarseGrainedScheduler@172.20.9.62:50066
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1(NettyRpcEnv.scala:148)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1$adapted(NettyRpcEnv.scala:144)
        at scala.concurrent.Future.$anonfun$flatMap$1(Future.scala:307)
        at scala.concurrent.impl.Promise.$anonfun$transformWith$1(Promise.sc
ala:41)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
```

```
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.dispatchOrAddCallbac
k(Promise.scala:316)
        at scala.concurrent.impl.Promise$DefaultPromise.onComplete(Promise.s
cala:307)
        at scala.concurrent.impl.Promise.transformWith(Promise.scala:40)
        at scala.concurrent.impl.Promise.transformWith$(Promise.scala:38)
        at scala.concurrent.impl.Promise$DefaultPromise.transformWith(Promis
e.scala:187)
        at scala.concurrent.Future.flatMap(Future.scala:306)
        at scala.concurrent.Future.flatMap$(Future.scala:306)
        at scala.concurrent.impl.Promise$DefaultPromise.flatMap(Promise.scal
a:187)
        at org.apache.spark.rpc.netty.NettyRpcEnv.asyncSetupEndpointRefByURI
(NettyRpcEnv.scala:150)
        ... 17 more
25/04/26 22:25:22 WARN Executor: Issue communicating with driver in heartbea
ter
org.apache.spark.SparkException: Exception thrown in awaitResult:
        at org.apache.spark.util.SparkThreadUtils$.awaitResult(SparkThreadUt
ils.scala:56)
        at org.apache.spark.util.ThreadUtils$.awaitResult(ThreadUtils.scala:
310)
        at org.apache.spark.rpc.RpcTimeout.awaitResult(RpcTimeout.scala:75)
        at org.apache.spark.rpc.RpcEndpointRef.askSync(RpcEndpointRef.scala:
101)
        at org.apache.spark.rpc.RpcEndpointRef.askSync(RpcEndpointRef.scala:
85)
        at org.apache.spark.storage.BlockManagerMaster.registerBlockManager
(BlockManagerMaster.scala:80)
        at org.apache.spark.storage.BlockManager.reregister(BlockManager.sca
la:642)
        at org.apache.spark.executor.Executor.reportHeartBeat(Executor.scal
a:1223)
        at org.apache.spark.executor.Executor.$anonfun$heartbeater$1(Executo
r.scala:295)
        at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.jav
a:23)
        at org.apache.spark.util.Utils$.logUncaughtExceptions(Utils.scala:19
28)
        at org.apache.spark.Heartbeater$$anon$1.run(Heartbeater.scala:46)
        at java.base/java.util.concurrent.Executors$RunnableAdapter.call(Exe
cutors.java:515)
        at java.base/java.util.concurrent.FutureTask.runAndReset(FutureTask.
java:305)
        at java.base/java.util.concurrent.ScheduledThreadPoolExecutor$Schedu
ledFutureTask.run(ScheduledThreadPoolExecutor.java:305)
        at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(Threa
dPoolExecutor.java:1128)
        at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(Thre
adPoolExecutor.java:628)
        at java.base/java.lang.Thread.run(Thread.java:829)
Caused by: org.apache.spark.SparkException: Exception thrown in awaitResult:
        at org.apache.spark.util.SparkThreadUtils$.awaitResult(SparkThreadUt
ils.scala:56)
```

```
        at org.apache.spark.util.ThreadUtils$.awaitResult(ThreadUtils.scala:
310)
        at org.apache.spark.rpc.RpcTimeout.awaitResult(RpcTimeout.scala:75)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRefByURI(RpcEnv.scala:10
2)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRef(RpcEnv.scala:110)
        at org.apache.spark.util.RpcUtils$.makeDriverRef(RpcUtils.scala:36)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.driverEndpoin
t$lzycompute(BlockManagerMasterEndpoint.scala:124)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$driverEndpoint(BlockManagerMasterEnd
point.scala:123)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$lzycompute$1(BlockManagerMasterEndpoint.scala:688)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$1(BlockManagerMasterEndpoint.scala:687)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$register(BlockManagerMasterEndpoint.
scala:725)
        at org.apache.spark.storage.BlockManagerMasterEndpoint$$anonfun$rece
iveAndReply$1.applyOrElse(BlockManagerMasterEndpoint.scala:133)
        at org.apache.spark.rpc.netty.Inbox.$anonfun$process$1(Inbox.scala:1
03)
        at org.apache.spark.rpc.netty.Inbox.safelyCall(Inbox.scala:213)
        at org.apache.spark.rpc.netty.Inbox.process(Inbox.scala:100)
        at org.apache.spark.rpc.netty.MessageLoop.org$apache$spark$rpc$netty
$MessageLoop$$receiveLoop(MessageLoop.scala:75)
        at org.apache.spark.rpc.netty.MessageLoop$$anon$1.run(MessageLoop.sc
ala:41)
        ... 3 more
Caused by: org.apache.spark.rpc.RpcEndpointNotFoundException: Cannot find en
dpoint: spark://CoarseGrainedScheduler@172.20.9.62:50066
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1(NettyRpcEnv.scala:148)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1$adapted(NettyRpcEnv.scala:144)
        at scala.concurrent.Future.$anonfun$flatMap$1(Future.scala:307)
        at scala.concurrent.impl.Promise.$anonfun$transformWith$1(Promise.sc
ala:41)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.dispatchOrAddCallbac
k(Promise.scala:316)
        at scala.concurrent.impl.Promise$DefaultPromise.onComplete(Promise.s
cala:307)
        at scala.concurrent.impl.Promise.transformWith(Promise.scala:40)
        at scala.concurrent.impl.Promise.transformWith$(Promise.scala:38)
        at scala.concurrent.impl.Promise$DefaultPromise.transformWith(Promis
e.scala:187)
        at scala.concurrent.Future.flatMap(Future.scala:306)
        at scala.concurrent.Future.flatMap$(Future.scala:306)
```

```
        at scala.concurrent.impl.Promise$DefaultPromise.flatMap(Promise.scal
a:187)
        at org.apache.spark.rpc.netty.NettyRpcEnv.asyncSetupEndpointRefByURI
(NettyRpcEnv.scala:150)
        ... 17 more
25/04/26 23:09:19 ERROR Inbox: Ignoring error
org.apache.spark.SparkException: Exception thrown in awaitResult:
        at org.apache.spark.util.SparkThreadUtils$.awaitResult(SparkThreadUt
ils.scala:56)
        at org.apache.spark.util.ThreadUtils$.awaitResult(ThreadUtils.scala:
310)
        at org.apache.spark.rpc.RpcTimeout.awaitResult(RpcTimeout.scala:75)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRefByURI(RpcEnv.scala:10
2)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRef(RpcEnv.scala:110)
        at org.apache.spark.util.RpcUtils$.makeDriverRef(RpcUtils.scala:36)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.driverEndpoin
t$lzycompute(BlockManagerMasterEndpoint.scala:124)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$driverEndpoint(BlockManagerMasterEnd
point.scala:123)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$lzycompute$1(BlockManagerMasterEndpoint.scala:688)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$1(BlockManagerMasterEndpoint.scala:687)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$register(BlockManagerMasterEndpoint.
scala:725)
        at org.apache.spark.storage.BlockManagerMasterEndpoint$$anonfun$rece
iveAndReply$1.applyOrElse(BlockManagerMasterEndpoint.scala:133)
        at org.apache.spark.rpc.netty.Inbox.$anonfun$process$1(Inbox.scala:1
03)
        at org.apache.spark.rpc.netty.Inbox.safelyCall(Inbox.scala:213)
        at org.apache.spark.rpc.netty.Inbox.process(Inbox.scala:100)
        at org.apache.spark.rpc.netty.MessageLoop.org$apache$spark$rpc$netty
$MessageLoop$$receiveLoop(MessageLoop.scala:75)
        at org.apache.spark.rpc.netty.MessageLoop$$anon$1.run(MessageLoop.sc
ala:41)
        at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(Threa
dPoolExecutor.java:1128)
        at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(Thre
adPoolExecutor.java:628)
        at java.base/java.lang.Thread.run(Thread.java:829)
Caused by: org.apache.spark.rpc.RpcEndpointNotFoundException: Cannot find en
dpoint: spark://CoarseGrainedScheduler@172.20.9.62:50066
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1(NettyRpcEnv.scala:148)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1$adapted(NettyRpcEnv.scala:144)
        at scala.concurrent.Future.$anonfun$flatMap$1(Future.scala:307)
        at scala.concurrent.impl.Promise.$anonfun$transformWith$1(Promise.sc
ala:41)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
```

```
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:
33)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:
33)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at scala.concurrent.BatchingExecutor$Batch.processBatch$1(BatchingEx
ecutor.scala:67)
        at scala.concurrent.BatchingExecutor$Batch.$anonfun$run$1(BatchingEx
ecutor.scala:82)
        at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.jav
a:23)
        at scala.concurrent.BlockContext$.withBlockContext(BlockContext.scal
a:85)
        at scala.concurrent.BatchingExecutor$Batch.run(BatchingExecutor.scal
a:59)
        at scala.concurrent.Future$InternalCallbackExecutor$.unbatchedExecut
e(Future.scala:875)
        at scala.concurrent.BatchingExecutor.execute(BatchingExecutor.scala:
110)
        at scala.concurrent.BatchingExecutor.execute$(BatchingExecutor.scal
a:107)
        at scala.concurrent.Future$InternalCallbackExecutor$.execute(Future.
scala:873)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
```

```
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.trySuccess(Promise.scala:94)
        at scala.concurrent.Promise.trySuccess$(Promise.scala:94)
        at scala.concurrent.impl.Promise$DefaultPromise.trySuccess(Promise.s
cala:187)
        at org.apache.spark.rpc.netty.NettyRpcEnv.onSuccess$1(NettyRpcEnv.sc
ala:225)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$askAbortable$5(Ne
ttyRpcEnv.scala:239)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$askAbortable$5$ad
apted(NettyRpcEnv.scala:238)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.Promise.success(Promise.scala:86)
        at scala.concurrent.Promise.success$(Promise.scala:86)
        at scala.concurrent.impl.Promise$DefaultPromise.success(Promise.scal
a:187)
        at org.apache.spark.rpc.netty.LocalNettyRpcCallContext.send(NettyRpc
CallContext.scala:50)
        at org.apache.spark.rpc.netty.NettyRpcCallContext.reply(NettyRpcCall
Context.scala:32)
        at org.apache.spark.rpc.netty.RpcEndpointVerifier$$anonfun$receiveAn
dReply$1.applyOrElse(RpcEndpointVerifier.scala:31)
        ... 8 more
25/04/26 23:09:19 WARN Executor: Issue communicating with driver in heartbea
ter
org.apache.spark.SparkException: Exception thrown in awaitResult:
        at org.apache.spark.util.SparkThreadUtils$.awaitResult(SparkThreadUt
ils.scala:56)
        at org.apache.spark.util.ThreadUtils$.awaitResult(ThreadUtils.scala:
310)
        at org.apache.spark.rpc.RpcTimeout.awaitResult(RpcTimeout.scala:75)
        at org.apache.spark.rpc.RpcEndpointRef.askSync(RpcEndpointRef.scala:
101)
        at org.apache.spark.rpc.RpcEndpointRef.askSync(RpcEndpointRef.scala:
85)
        at org.apache.spark.storage.BlockManagerMaster.registerBlockManager
```

```
(BlockManagerMaster.scala:80)
        at org.apache.spark.storage.BlockManager.reregister(BlockManager.sca
la:642)
        at org.apache.spark.executor.Executor.reportHeartBeat(Executor.scal
a:1223)
        at org.apache.spark.executor.Executor.$anonfun$heartbeater$1(Executo
r.scala:295)
        at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.jav
a:23)
        at org.apache.spark.util.Utils$.logUncaughtExceptions(Utils.scala:19
28)
        at org.apache.spark.Heartbeater$$anon$1.run(Heartbeater.scala:46)
        at java.base/java.util.concurrent.Executors$RunnableAdapter.call(Exe
cutors.java:515)
        at java.base/java.util.concurrent.FutureTask.runAndReset(FutureTask.
java:305)
        at java.base/java.util.concurrent.ScheduledThreadPoolExecutor$Schedu
ledFutureTask.run(ScheduledThreadPoolExecutor.java:305)
        at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(Threa
dPoolExecutor.java:1128)
        at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(Thre
adPoolExecutor.java:628)
        at java.base/java.lang.Thread.run(Thread.java:829)
Caused by: org.apache.spark.SparkException: Exception thrown in awaitResult:
        at org.apache.spark.util.SparkThreadUtils$.awaitResult(SparkThreadUt
ils.scala:56)
        at org.apache.spark.util.ThreadUtils$.awaitResult(ThreadUtils.scala:
310)
        at org.apache.spark.rpc.RpcTimeout.awaitResult(RpcTimeout.scala:75)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRefByURI(RpcEnv.scala:10
2)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRef(RpcEnv.scala:110)
        at org.apache.spark.util.RpcUtils$.makeDriverRef(RpcUtils.scala:36)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.driverEndpoin
t$lzycompute(BlockManagerMasterEndpoint.scala:124)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$driverEndpoint(BlockManagerMasterEnd
point.scala:123)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$lzycompute$1(BlockManagerMasterEndpoint.scala:688)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$1(BlockManagerMasterEndpoint.scala:687)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$register(BlockManagerMasterEndpoint.
scala:725)
        at org.apache.spark.storage.BlockManagerMasterEndpoint$$anonfun$rece
iveAndReply$1.applyOrElse(BlockManagerMasterEndpoint.scala:133)
        at org.apache.spark.rpc.netty.Inbox.$anonfun$process$1(Inbox.scala:1
03)
        at org.apache.spark.rpc.netty.Inbox.safelyCall(Inbox.scala:213)
        at org.apache.spark.rpc.netty.Inbox.process(Inbox.scala:100)
        at org.apache.spark.rpc.netty.MessageLoop.org$apache$spark$rpc$netty
$MessageLoop$$receiveLoop(MessageLoop.scala:75)
        at org.apache.spark.rpc.netty.MessageLoop$$anon$1.run(MessageLoop.sc
ala:41)
        ... 3 more
```

```
Caused by: org.apache.spark.rpc.RpcEndpointNotFoundException: Cannot find en
dpoint: spark://CoarseGrainedScheduler@172.20.9.62:50066
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1(NettyRpcEnv.scala:148)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1$adapted(NettyRpcEnv.scala:144)
        at scala.concurrent.Future.$anonfun$flatMap$1(Future.scala:307)
        at scala.concurrent.impl.Promise.$anonfun$transformWith$1(Promise.sc
ala:41)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:
33)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:
33)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at scala.concurrent.BatchingExecutor$Batch.processBatch$1(BatchingEx
ecutor.scala:67)
        at scala.concurrent.BatchingExecutor$Batch.$anonfun$run$1(BatchingEx
ecutor.scala:82)
        at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.jav
a:23)
        at scala.concurrent.BlockContext$.withBlockContext(BlockContext.scal
a:85)
```

```
        at scala.concurrent.BatchingExecutor$Batch.run(BatchingExecutor.scal
a:59)
        at scala.concurrent.Future$InternalCallbackExecutor$.unbatchedExecut
e(Future.scala:875)
        at scala.concurrent.BatchingExecutor.execute(BatchingExecutor.scala:
110)
        at scala.concurrent.BatchingExecutor.execute$(BatchingExecutor.scal
a:107)
        at scala.concurrent.Future$InternalCallbackExecutor$.execute(Future.
scala:873)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.trySuccess(Promise.scala:94)
        at scala.concurrent.Promise.trySuccess$(Promise.scala:94)
        at scala.concurrent.impl.Promise$DefaultPromise.trySuccess(Promise.s
cala:187)
        at org.apache.spark.rpc.netty.NettyRpcEnv.onSuccess$1(NettyRpcEnv.sc
ala:225)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$askAbortable$5(Ne
ttyRpcEnv.scala:239)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$askAbortable$5$ad
apted(NettyRpcEnv.scala:238)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.Promise.success(Promise.scala:86)
        at scala.concurrent.Promise.success$(Promise.scala:86)
        at scala.concurrent.impl.Promise$DefaultPromise.success(Promise.scal
a:187)
        at org.apache.spark.rpc.netty.LocalNettyRpcCallContext.send(NettyRpc
CallContext.scala:50)
        at org.apache.spark.rpc.netty.NettyRpcCallContext.reply(NettyRpcCall
Context.scala:32)
        at org.apache.spark.rpc.netty.RpcEndpointVerifier$$anonfun$receiveAn
dReply$1.applyOrElse(RpcEndpointVerifier.scala:31)
        ... 8 more
```

```
25/04/26 23:25:06 ERROR Inbox: Ignoring error
org.apache.spark.SparkException: Exception thrown in awaitResult:
        at org.apache.spark.util.SparkThreadUtils$.awaitResult(SparkThreadUt
ils.scala:56)
        at org.apache.spark.util.ThreadUtils$.awaitResult(ThreadUtils.scala:
310)
        at org.apache.spark.rpc.RpcTimeout.awaitResult(RpcTimeout.scala:75)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRefByURI(RpcEnv.scala:10
2)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRef(RpcEnv.scala:110)
        at org.apache.spark.util.RpcUtils$.makeDriverRef(RpcUtils.scala:36)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.driverEndpoin
t$lzycompute(BlockManagerMasterEndpoint.scala:124)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$driverEndpoint(BlockManagerMasterEnd
point.scala:123)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$lzycompute$1(BlockManagerMasterEndpoint.scala:688)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$1(BlockManagerMasterEndpoint.scala:687)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$register(BlockManagerMasterEndpoint.
scala:725)
        at org.apache.spark.storage.BlockManagerMasterEndpoint$$anonfun$rece
iveAndReply$1.applyOrElse(BlockManagerMasterEndpoint.scala:133)
        at org.apache.spark.rpc.netty.Inbox.$anonfun$process$1(Inbox.scala:1
03)
        at org.apache.spark.rpc.netty.Inbox.safelyCall(Inbox.scala:213)
        at org.apache.spark.rpc.netty.Inbox.process(Inbox.scala:100)
        at org.apache.spark.rpc.netty.MessageLoop.org$apache$spark$rpc$netty
$MessageLoop$$receiveLoop(MessageLoop.scala:75)
        at org.apache.spark.rpc.netty.MessageLoop$$anon$1.run(MessageLoop.sc
ala:41)
        at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(Threa
dPoolExecutor.java:1128)
        at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(Thre
adPoolExecutor.java:628)
        at java.base/java.lang.Thread.run(Thread.java:829)
Caused by: org.apache.spark.rpc.RpcEndpointNotFoundException: Cannot find en
dpoint: spark://CoarseGrainedScheduler@172.20.9.62:50066
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1(NettyRpcEnv.scala:148)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1$adapted(NettyRpcEnv.scala:144)
        at scala.concurrent.Future.$anonfun$flatMap$1(Future.scala:307)
        at scala.concurrent.impl.Promise.$anonfun$transformWith$1(Promise.sc
ala:41)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
```

```
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:
33)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:
33)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at scala.concurrent.BatchingExecutor$Batch.processBatch$1(BatchingEx
ecutor.scala:67)
        at scala.concurrent.BatchingExecutor$Batch.$anonfun$run$1(BatchingEx
ecutor.scala:82)
        at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.jav
a:23)
        at scala.concurrent.BlockContext$.withBlockContext(BlockContext.scal
a:85)
        at scala.concurrent.BatchingExecutor$Batch.run(BatchingExecutor.scal
a:59)
        at scala.concurrent.Future$InternalCallbackExecutor$.unbatchedExecut
e(Future.scala:875)
        at scala.concurrent.BatchingExecutor.execute(BatchingExecutor.scala:
110)
        at scala.concurrent.BatchingExecutor.execute$(BatchingExecutor.scal
a:107)
        at scala.concurrent.Future$InternalCallbackExecutor$.execute(Future.
scala:873)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
```

```
        at scala.concurrent.Promise.trySuccess(Promise.scala:94)
        at scala.concurrent.Promise.trySuccess$(Promise.scala:94)
        at scala.concurrent.impl.Promise$DefaultPromise.trySuccess(Promise.s
cala:187)
        at org.apache.spark.rpc.netty.NettyRpcEnv.onSuccess$1(NettyRpcEnv.sc
ala:225)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$askAbortable$5(Ne
ttyRpcEnv.scala:239)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$askAbortable$5$ad
apted(NettyRpcEnv.scala:238)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.Promise.success(Promise.scala:86)
        at scala.concurrent.Promise.success$(Promise.scala:86)
        at scala.concurrent.impl.Promise$DefaultPromise.success(Promise.scal
a:187)
        at org.apache.spark.rpc.netty.LocalNettyRpcCallContext.send(NettyRpc
CallContext.scala:50)
        at org.apache.spark.rpc.netty.NettyRpcCallContext.reply(NettyRpcCall
Context.scala:32)
        at org.apache.spark.rpc.netty.RpcEndpointVerifier$$anonfun$receiveAn
dReply$1.applyOrElse(RpcEndpointVerifier.scala:31)
        ... 8 more
25/04/26 23:25:06 WARN Executor: Issue communicating with driver in heartbea
ter
org.apache.spark.SparkException: Exception thrown in awaitResult:
        at org.apache.spark.util.SparkThreadUtils$.awaitResult(SparkThreadUt
ils.scala:56)
        at org.apache.spark.util.ThreadUtils$.awaitResult(ThreadUtils.scala:
310)
        at org.apache.spark.rpc.RpcTimeout.awaitResult(RpcTimeout.scala:75)
        at org.apache.spark.rpc.RpcEndpointRef.askSync(RpcEndpointRef.scala:
101)
        at org.apache.spark.rpc.RpcEndpointRef.askSync(RpcEndpointRef.scala:
85)
        at org.apache.spark.storage.BlockManagerMaster.registerBlockManager
(BlockManagerMaster.scala:80)
        at org.apache.spark.storage.BlockManager.reregister(BlockManager.sca
la:642)
        at org.apache.spark.executor.Executor.reportHeartBeat(Executor.scal
a:1223)
```

```
        at org.apache.spark.executor.Executor.$anonfun$heartbeater$1(Executo
r.scala:295)
        at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.jav
a:23)
        at org.apache.spark.util.Utils$.logUncaughtExceptions(Utils.scala:19
28)
        at org.apache.spark.Heartbeater$$anon$1.run(Heartbeater.scala:46)
        at java.base/java.util.concurrent.Executors$RunnableAdapter.call(Exe
cutors.java:515)
        at java.base/java.util.concurrent.FutureTask.runAndReset(FutureTask.
java:305)
        at java.base/java.util.concurrent.ScheduledThreadPoolExecutor$Schedu
ledFutureTask.run(ScheduledThreadPoolExecutor.java:305)
        at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(Threa
dPoolExecutor.java:1128)
        at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(Thre
adPoolExecutor.java:628)
        at java.base/java.lang.Thread.run(Thread.java:829)
Caused by: org.apache.spark.SparkException: Exception thrown in awaitResult:
        at org.apache.spark.util.SparkThreadUtils$.awaitResult(SparkThreadUt
ils.scala:56)
        at org.apache.spark.util.ThreadUtils$.awaitResult(ThreadUtils.scala:
310)
        at org.apache.spark.rpc.RpcTimeout.awaitResult(RpcTimeout.scala:75)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRefByURI(RpcEnv.scala:10
2)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRef(RpcEnv.scala:110)
        at org.apache.spark.util.RpcUtils$.makeDriverRef(RpcUtils.scala:36)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.driverEndpoin
t$lzycompute(BlockManagerMasterEndpoint.scala:124)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$driverEndpoint(BlockManagerMasterEnd
point.scala:123)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$lzycompute$1(BlockManagerMasterEndpoint.scala:688)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$1(BlockManagerMasterEndpoint.scala:687)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$register(BlockManagerMasterEndpoint.
scala:725)
        at org.apache.spark.storage.BlockManagerMasterEndpoint$$anonfun$rece
iveAndReply$1.applyOrElse(BlockManagerMasterEndpoint.scala:133)
        at org.apache.spark.rpc.netty.Inbox.$anonfun$process$1(Inbox.scala:1
03)
        at org.apache.spark.rpc.netty.Inbox.safelyCall(Inbox.scala:213)
        at org.apache.spark.rpc.netty.Inbox.process(Inbox.scala:100)
        at org.apache.spark.rpc.netty.MessageLoop.org$apache$spark$rpc$netty
$MessageLoop$$receiveLoop(MessageLoop.scala:75)
        at org.apache.spark.rpc.netty.MessageLoop$$anon$1.run(MessageLoop.sc
ala:41)
        ... 3 more
Caused by: org.apache.spark.rpc.RpcEndpointNotFoundException: Cannot find en
dpoint: spark://CoarseGrainedScheduler@172.20.9.62:50066
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1(NettyRpcEnv.scala:148)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
```

```
tRefByURI$1$adapted(NettyRpcEnv.scala:144)
        at scala.concurrent.Future.$anonfun$flatMap$1(Future.scala:307)
        at scala.concurrent.impl.Promise.$anonfun$transformWith$1(Promise.sc
ala:41)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:
33)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:
33)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at scala.concurrent.BatchingExecutor$Batch.processBatch$1(BatchingEx
ecutor.scala:67)
        at scala.concurrent.BatchingExecutor$Batch.$anonfun$run$1(BatchingEx
ecutor.scala:82)
        at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.jav
a:23)
        at scala.concurrent.BlockContext$.withBlockContext(BlockContext.scal
a:85)
        at scala.concurrent.BatchingExecutor$Batch.run(BatchingExecutor.scal
a:59)
        at scala.concurrent.Future$InternalCallbackExecutor$.unbatchedExecut
e(Future.scala:875)
        at scala.concurrent.BatchingExecutor.execute(BatchingExecutor.scala:
```

```
110)
        at scala.concurrent.BatchingExecutor.execute$(BatchingExecutor.scal
a:107)
        at scala.concurrent.Future$InternalCallbackExecutor$.execute(Future.
scala:873)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.trySuccess(Promise.scala:94)
        at scala.concurrent.Promise.trySuccess$(Promise.scala:94)
        at scala.concurrent.impl.Promise$DefaultPromise.trySuccess(Promise.s
cala:187)
        at org.apache.spark.rpc.netty.NettyRpcEnv.onSuccess$1(NettyRpcEnv.sc
ala:225)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$askAbortable$5(Ne
ttyRpcEnv.scala:239)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$askAbortable$5$ad
apted(NettyRpcEnv.scala:238)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.Promise.success(Promise.scala:86)
        at scala.concurrent.Promise.success$(Promise.scala:86)
        at scala.concurrent.impl.Promise$DefaultPromise.success(Promise.scal
a:187)
        at org.apache.spark.rpc.netty.LocalNettyRpcCallContext.send(NettyRpc
CallContext.scala:50)
        at org.apache.spark.rpc.netty.NettyRpcCallContext.reply(NettyRpcCall
Context.scala:32)
        at org.apache.spark.rpc.netty.RpcEndpointVerifier$$anonfun$receiveAn
dReply$1.applyOrElse(RpcEndpointVerifier.scala:31)
        ... 8 more
25/04/26 23:26:46 WARN Executor: Issue communicating with driver in heartbea
ter
org.apache.spark.SparkException: Exception thrown in awaitResult:
        at org.apache.spark.util.SparkThreadUtils$.awaitResult(SparkThreadUt
ils.scala:56)
```

```
        at org.apache.spark.util.ThreadUtils$.awaitResult(ThreadUtils.scala:
310)
        at org.apache.spark.rpc.RpcTimeout.awaitResult(RpcTimeout.scala:75)
        at org.apache.spark.rpc.RpcEndpointRef.askSync(RpcEndpointRef.scala:
101)
        at org.apache.spark.rpc.RpcEndpointRef.askSync(RpcEndpointRef.scala:
85)
        at org.apache.spark.storage.BlockManagerMaster.registerBlockManager
(BlockManagerMaster.scala:80)
        at org.apache.spark.storage.BlockManager.reregister(BlockManager.sca
la:642)
        at org.apache.spark.executor.Executor.reportHeartBeat(Executor.scal
a:1223)
        at org.apache.spark.executor.Executor.$anonfun$heartbeater$1(Executo
r.scala:295)
        at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.jav
a:23)
        at org.apache.spark.util.Utils$.logUncaughtExceptions(Utils.scala:19
28)
        at org.apache.spark.Heartbeater$$anon$1.run(Heartbeater.scala:46)
        at java.base/java.util.concurrent.Executors$RunnableAdapter.call(Exe
cutors.java:515)
        at java.base/java.util.concurrent.FutureTask.runAndReset(FutureTask.
java:305)
        at java.base/java.util.concurrent.ScheduledThreadPoolExecutor$Schedu
ledFutureTask.run(ScheduledThreadPoolExecutor.java:305)
        at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(Threa
dPoolExecutor.java:1128)
        at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(Thre
adPoolExecutor.java:628)
        at java.base/java.lang.Thread.run(Thread.java:829)
Caused by: org.apache.spark.SparkException: Exception thrown in awaitResult:
        at org.apache.spark.util.SparkThreadUtils$.awaitResult(SparkThreadUt
ils.scala:56)
        at org.apache.spark.util.ThreadUtils$.awaitResult(ThreadUtils.scala:
310)
        at org.apache.spark.rpc.RpcTimeout.awaitResult(RpcTimeout.scala:75)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRefByURI(RpcEnv.scala:10
2)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRef(RpcEnv.scala:110)
        at org.apache.spark.util.RpcUtils$.makeDriverRef(RpcUtils.scala:36)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.driverEndpoin
t$lzycompute(BlockManagerMasterEndpoint.scala:124)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$driverEndpoint(BlockManagerMasterEnd
point.scala:123)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$lzycompute$1(BlockManagerMasterEndpoint.scala:688)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$1(BlockManagerMasterEndpoint.scala:687)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$register(BlockManagerMasterEndpoint.
scala:725)
        at org.apache.spark.storage.BlockManagerMasterEndpoint$$anonfun$rece
iveAndReply$1.applyOrElse(BlockManagerMasterEndpoint.scala:133)
        at org.apache.spark.rpc.netty.Inbox.$anonfun$process$1(Inbox.scala:1
```

```
03)
        at org.apache.spark.rpc.netty.Inbox.safelyCall(Inbox.scala:213)
        at org.apache.spark.rpc.netty.Inbox.process(Inbox.scala:100)
        at org.apache.spark.rpc.netty.MessageLoop.org$apache$spark$rpc$netty
$MessageLoop$$receiveLoop(MessageLoop.scala:75)
        at org.apache.spark.rpc.netty.MessageLoop$$anon$1.run(MessageLoop.sc
ala:41)
        ... 3 more
Caused by: org.apache.spark.rpc.RpcEndpointNotFoundException: Cannot find en
dpoint: spark://CoarseGrainedScheduler@172.20.9.62:50066
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1(NettyRpcEnv.scala:148)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1$adapted(NettyRpcEnv.scala:144)
        at scala.concurrent.Future.$anonfun$flatMap$1(Future.scala:307)
        at scala.concurrent.impl.Promise.$anonfun$transformWith$1(Promise.sc
ala:41)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:
33)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
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        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
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cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:
33)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
```

```
        at scala.concurrent.BatchingExecutor$Batch.processBatch$1(BatchingEx
ecutor.scala:67)
        at scala.concurrent.BatchingExecutor$Batch.$anonfun$run$1(BatchingEx
ecutor.scala:82)
        at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.jav
a:23)
        at scala.concurrent.BlockContext$.withBlockContext(BlockContext.scal
a:85)
        at scala.concurrent.BatchingExecutor$Batch.run(BatchingExecutor.scal
a:59)
        at scala.concurrent.Future$InternalCallbackExecutor$.unbatchedExecut
e(Future.scala:875)
        at scala.concurrent.BatchingExecutor.execute(BatchingExecutor.scala:
110)
        at scala.concurrent.BatchingExecutor.execute$(BatchingExecutor.scal
a:107)
        at scala.concurrent.Future$InternalCallbackExecutor$.execute(Future.
scala:873)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.trySuccess(Promise.scala:94)
        at scala.concurrent.Promise.trySuccess$(Promise.scala:94)
        at scala.concurrent.impl.Promise$DefaultPromise.trySuccess(Promise.s
cala:187)
        at org.apache.spark.rpc.netty.NettyRpcEnv.onSuccess$1(NettyRpcEnv.sc
ala:225)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$askAbortable$5(Ne
ttyRpcEnv.scala:239)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$askAbortable$5$ad
apted(NettyRpcEnv.scala:238)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
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        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
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        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
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$1$adapted(Promise.scala:288)
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        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.Promise.success(Promise.scala:86)
        at scala.concurrent.Promise.success$(Promise.scala:86)
        at scala.concurrent.impl.Promise$DefaultPromise.success(Promise.scal
```

```
a:187)
        at org.apache.spark.rpc.netty.LocalNettyRpcCallContext.send(NettyRpc
CallContext.scala:50)
        at org.apache.spark.rpc.netty.NettyRpcCallContext.reply(NettyRpcCall
Context.scala:32)
        at org.apache.spark.rpc.netty.RpcEndpointVerifier$$anonfun$receiveAn
dReply$1.applyOrElse(RpcEndpointVerifier.scala:31)
        ... 8 more
25/04/26 23:26:46 ERROR Inbox: Ignoring error
org.apache.spark.SparkException: Exception thrown in awaitResult:
        at org.apache.spark.util.SparkThreadUtils$.awaitResult(SparkThreadUt
ils.scala:56)
        at org.apache.spark.util.ThreadUtils$.awaitResult(ThreadUtils.scala:
310)
        at org.apache.spark.rpc.RpcTimeout.awaitResult(RpcTimeout.scala:75)
        at org.apache.spark.rpc.RpcEnv.setupEndpointRefByURI(RpcEnv.scala:10
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        at org.apache.spark.rpc.RpcEnv.setupEndpointRef(RpcEnv.scala:110)
        at org.apache.spark.util.RpcUtils$.makeDriverRef(RpcUtils.scala:36)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.driverEndpoin
t$lzycompute(BlockManagerMasterEndpoint.scala:124)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
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point.scala:123)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$lzycompute$1(BlockManagerMasterEndpoint.scala:688)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.isExecutorAli
ve$1(BlockManagerMasterEndpoint.scala:687)
        at org.apache.spark.storage.BlockManagerMasterEndpoint.org$apache$sp
ark$storage$BlockManagerMasterEndpoint$$register(BlockManagerMasterEndpoint.
scala:725)
        at org.apache.spark.storage.BlockManagerMasterEndpoint$$anonfun$rece
iveAndReply$1.applyOrElse(BlockManagerMasterEndpoint.scala:133)
        at org.apache.spark.rpc.netty.Inbox.$anonfun$process$1(Inbox.scala:1
03)
        at org.apache.spark.rpc.netty.Inbox.safelyCall(Inbox.scala:213)
        at org.apache.spark.rpc.netty.Inbox.process(Inbox.scala:100)
        at org.apache.spark.rpc.netty.MessageLoop.org$apache$spark$rpc$netty
$MessageLoop$$receiveLoop(MessageLoop.scala:75)
        at org.apache.spark.rpc.netty.MessageLoop$$anon$1.run(MessageLoop.sc
ala:41)
        at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(Threa
dPoolExecutor.java:1128)
        at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(Thre
adPoolExecutor.java:628)
        at java.base/java.lang.Thread.run(Thread.java:829)
Caused by: org.apache.spark.rpc.RpcEndpointNotFoundException: Cannot find en
dpoint: spark://CoarseGrainedScheduler@172.20.9.62:50066
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1(NettyRpcEnv.scala:148)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$asyncSetupEndpoin
tRefByURI$1$adapted(NettyRpcEnv.scala:144)
        at scala.concurrent.Future.$anonfun$flatMap$1(Future.scala:307)
        at scala.concurrent.impl.Promise.$anonfun$transformWith$1(Promise.sc
ala:41)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
```

```
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
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        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
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        at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:
33)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
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        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:
33)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at scala.concurrent.BatchingExecutor$Batch.processBatch$1(BatchingEx
ecutor.scala:67)
        at scala.concurrent.BatchingExecutor$Batch.$anonfun$run$1(BatchingEx
ecutor.scala:82)
        at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.jav
a:23)
        at scala.concurrent.BlockContext$.withBlockContext(BlockContext.scal
a:85)
        at scala.concurrent.BatchingExecutor$Batch.run(BatchingExecutor.scal
a:59)
        at scala.concurrent.Future$InternalCallbackExecutor$.unbatchedExecut
e(Future.scala:875)
        at scala.concurrent.BatchingExecutor.execute(BatchingExecutor.scala:
110)
        at scala.concurrent.BatchingExecutor.execute$(BatchingExecutor.scal
a:107)
        at scala.concurrent.Future$InternalCallbackExecutor$.execute(Future.
scala:873)
```

```
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.trySuccess(Promise.scala:94)
        at scala.concurrent.Promise.trySuccess$(Promise.scala:94)
        at scala.concurrent.impl.Promise$DefaultPromise.trySuccess(Promise.s
cala:187)
        at org.apache.spark.rpc.netty.NettyRpcEnv.onSuccess$1(NettyRpcEnv.sc
ala:225)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$askAbortable$5(Ne
ttyRpcEnv.scala:239)
        at org.apache.spark.rpc.netty.NettyRpcEnv.$anonfun$askAbortable$5$ad
apted(NettyRpcEnv.scala:238)
        at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
        at org.apache.spark.util.ThreadUtils$$anon$1.execute(ThreadUtils.sca
la:99)
        at scala.concurrent.impl.ExecutionContextImpl$$anon$4.execute(Execut
ionContextImpl.scala:138)
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.s
cala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete
$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.
scala:288)
        at scala.concurrent.Promise.complete(Promise.scala:53)
        at scala.concurrent.Promise.complete$(Promise.scala:52)
        at scala.concurrent.impl.Promise$DefaultPromise.complete(Promise.sca
la:187)
        at scala.concurrent.Promise.success(Promise.scala:86)
        at scala.concurrent.Promise.success$(Promise.scala:86)
        at scala.concurrent.impl.Promise$DefaultPromise.success(Promise.scal
a:187)
        at org.apache.spark.rpc.netty.LocalNettyRpcCallContext.send(NettyRpc
CallContext.scala:50)
        at org.apache.spark.rpc.netty.NettyRpcCallContext.reply(NettyRpcCall
Context.scala:32)
        at org.apache.spark.rpc.netty.RpcEndpointVerifier$$anonfun$receiveAn
dReply$1.applyOrElse(RpcEndpointVerifier.scala:31)
        ... 8 more
```

Reading the updated dataset

In [ ]:
```
from pyspark.sql import SparkSession
spark = SparkSession.builder \
    .appName("Phase1_EDA_BookReviews") \
    .master("local[*]") \
    .config("spark.driver.memory", "8g") \
    .config("spark.executor.memory", "6g") \
```

```
    .config("spark.network.timeout", "800s") \
    .config("spark.executor.heartbeatInterval", "100s") \
    .config("spark.sql.shuffle.partitions", "50") \
    .getOrCreate()
merged_df = spark.read.parquet("hdfs://localhost:9000/user/sathw/output/clea
merged_df.printSchema()
merged_df.show(5)
```

```
25/04/26 23:34:09 WARN Utils: Your hostname, MacBook-Pro-337.local resolves
to a loopback address: 127.0.0.1; using 172.20.9.62 instead (on interface en
0)
25/04/26 23:34:09 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to anot
her address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLog
Level(newLevel).
25/04/26 23:34:09 WARN NativeCodeLoader: Unable to load native-hadoop librar
y for your platform... using builtin-java classes where applicable
```

```
root
 |-- Title: string (nullable = true)
 |-- description: string (nullable = true)
 |-- authors: string (nullable = true)
 |-- image: string (nullable = true)
 |-- previewLink: string (nullable = true)
 |-- publisher: string (nullable = true)
 |-- publishedDate: string (nullable = true)
 |-- infoLink: string (nullable = true)
 |-- categories: string (nullable = true)
 |-- ratingsCount: integer (nullable = true)
 |-- publishedYear: string (nullable = true)
 |-- Id: string (nullable = true)
 |-- Price: float (nullable = true)
 |-- User_id: string (nullable = true)
 |-- profileName: string (nullable = true)
 |-- review/helpfulness: string (nullable = true)
 |-- review/score: float (nullable = true)
 |-- review/time: timestamp (nullable = true)
 |-- review/summary: string (nullable = true)
 |-- review/text: string (nullable = true)
 |-- review_year: integer (nullable = true)

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----------+--------------------+-----------+
|               Title|         description|        authors|                im
age|         previewLink|      publisher|publishedDate|            infoLink|
categories|ratingsCount|publishedYear|        Id|Price|       User_id|
profileName|review/helpfulness|review/score|        review/time|      revie
w/summary|         review/text|review_year|
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|"""A"" IS FOR ALIBI"|"READ THE SENSATI...|['Sue Grafton']|http://books.goo
g...|http://books.goog...|Holt Paperbacks|   2010-04-01|https://play.goog...
|['Fiction']|          20|         2010|B000H3DK02|  0.0|          NULL|
Anonymous|               1/1|         5.0|1997-02-17 19:00:00|great book, co
mpa...|first in a series...|       1997|
|"""A"" IS FOR ALIBI"|"READ THE SENSATI...|['Sue Grafton']|http://books.goo
g...|http://books.goog...|Holt Paperbacks|   2010-04-01|https://play.goog...
|['Fiction']|          20|         2010|B000H3DK02|  0.0|          NULL|
Anonymous|               1/1|         5.0|1996-11-12 19:00:00|Her first boo
k, a...|By far her best e...|       1996|
|"""A"" IS FOR ALIBI"|"READ THE SENSATI...|['Sue Grafton']|http://books.goo
g...|http://books.goog...|Holt Paperbacks|   2010-04-01|https://play.goog...
|['Fiction']|          20|         2010|B000H3DK02|  0.0|A20R7259NOEJ8V|
Kris|               3/4|         4.0|2006-07-28 20:00:00|             A Shoc
k|This book was a d...|       2006|
|"""A"" IS FOR ALIBI"|"READ THE SENSATI...|['Sue Grafton']|http://books.goo
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|['Fiction']|          20|         2010|B000H3DK02|  0.0|          NULL|
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Anonymous|               3/4|         4.0|2001-11-04 19:00:00|Let me sum it
up ...|So, far I've read...|       2001|
|"""A"" IS FOR ALIBI"|"READ THE SENSATI...|['Sue Grafton']|http://books.goo
g...|http://books.goog...|Holt Paperbacks|   2010-04-01|https://play.goog...
|['Fiction']|          20|         2010|B000H3DK02|  0.0| ABJHGMKBIDPT1|"""i
ntentaccess"""|             44/44|         5.0|2003-06-10 20:00:00|        A
Fun Read!!|"""A is for Alibi...|       2003|
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-------------+------------------+------------+------------------+----------
----------+--------------------+-----------+
only showing top 5 rows
```

Task 3 - Spark ML

Sub-Task 3a) Data Preprocessing

Data cleaning and handling missing values

In [ ]:
```
from pyspark.sql.functions import col, when
from pyspark.ml.feature import Imputer
merged_df = merged_df.withColumn(
    "Price", when(col("Price") == 0, None).otherwise(col("Price"))
).withColumn(
    "ratingsCount", when(col("ratingsCount") == 0, None).otherwise(col("rati
).withColumn(
    "review/score", when(col("review/score") == 0, None).otherwise(col("revi
)
merged_df.select("Price", "ratingsCount", "review/score").show(5, truncate=F
imputer = Imputer(inputCols=["Price", "ratingsCount", "review/score"],
                  outputCols=["Price_imputed", "ratingsCount_imputed", "revi
merged_df = imputer.fit(merged_df).transform(merged_df)
merged_df.select("Price_imputed", "ratingsCount_imputed", "review/score_impu
```

```
+-----+------------+------------+
|Price|ratingsCount|review/score|
+-----+------------+------------+
|NULL |20          |5.0         |
|NULL |20          |5.0         |
|NULL |20          |4.0         |
|NULL |20          |4.0         |
|NULL |20          |5.0         |
+-----+------------+------------+
only showing top 5 rows

+-------------+--------------------+--------------------+
|Price_imputed|ratingsCount_imputed|review/score_imputed|
+-------------+--------------------+--------------------+
|21.351927    |20                  |5.0                 |
|21.351927    |20                  |5.0                 |
|21.351927    |20                  |4.0                 |
|21.351927    |20                  |4.0                 |
|21.351927    |20                  |5.0                 |
+-------------+--------------------+--------------------+
only showing top 5 rows
```

In [ ]:
```
from pyspark.ml.feature import Tokenizer, StopWordsRemover, HashingTF, IDF
tokenizer = Tokenizer(inputCol="review/text", outputCol="words")
stopwords_remover = StopWordsRemover(inputCol="words", outputCol="filtered_w
hashingTF = HashingTF(inputCol="filtered_words", outputCol="raw_features", r
idf = IDF(inputCol="raw_features", outputCol="features_tfidf")
```

In [ ]:
```
from pyspark.ml import Pipeline
from pyspark.ml.feature import VectorAssembler
assembler = VectorAssembler(
    inputCols=["Price_imputed", "ratingsCount_imputed", "review/score_impute
    outputCol="features"
)
pipeline = Pipeline(stages=[imputer, tokenizer, stopwords_remover, hashingTF
processed_df = pipeline.fit(merged_df).transform(merged_df)
processed_df.select("Title", "Price", "ratingsCount", "review/score", "featu
```

```
+--------------------+-----+------------+------------+----------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------+
|Title               |Price|ratingsCount|review/score|features
|
+--------------------+-----+------------+------------+----------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------+
|"""A"" IS FOR ALIBI"|NULL |20          |5.0         |(1003,[0,1,2,119,224,2
29,300,316,347,396,406,436,450,466,495,529,530,647,701,889,891,997],[21.3519
26803588867,20.0,5.0,1.9102617939995148,3.1797649387078333,3.12717028008118
7,2.395200872050432,3.234240565201063,2.0662788012165167,0.4661122694470266,
1.7294852278316135,2.889695252200178,2.198151064235689,2.6837624950109062,3.
0633624283462777,1.5876737022868803,2.1388988842586953,2.3170911861705807,3.
2509570094707847,2.9819721300191206,3.92190210359973,2.001450319309485])
|
|"""A"" IS FOR ALIBI"|NULL |20          |5.0         |(1003,[0,1,2,83,103,11
9,139,161,188,217,229,333,359,363,373,394,466,493,528,529,573,600,632,723,72
6,743,766,794,818,823,870,915,920,921,951],[21.351926803588867,20.0,5.0,2.77
1022774165492,3.2249402656596033,1.9102617939995148,3.697560229793411,2.4261
8623920131,2.452319063518117,3.6505395229765503,1.5635851400405936,1.4687312
592979453,3.421744694202909,2.071589088842726,2.772778700867757,1.9812085176
540284,2.6837624950109062,6.477089790096963,2.230596917702423,1.587673702286
8803,4.035590565848912,2.9254579894904733,3.208355892069355,2.96099801236964
4,3.0513665776980012,2.5419226033385134,4.423545664913576,2.114355637187848,
1.1493508520859481,3.719230468253605,2.5846789713241027,3.8929488256209464,
3.5839108089301224,3.749042637554993,2.6644823005933467])|
|"""A"" IS FOR ALIBI"|NULL |20          |4.0         |(1003,[0,1,2,58,83,88,
111,119,396,462,469,536,670,700,753,764,768,830,878,880,984,997,999],[21.351
926803588867,20.0,4.0,2.234859427962012,2.771022774165492,3.106585040866665
5,3.616504677320799,1.9102617939995148,0.9322245388940532,2.267850831809698
6,3.000886341253439,2.6561204445326303,2.9929816315844784,3.007531679564842,
1.6393759657150597,3.9216023883269933,3.2601942393424608,3.0011250474669016,
3.9837264856080687,2.786167702103628,2.623109756489696,2.001450319309485,2.7
19459246024101])
|
|"""A"" IS FOR ALIBI"|NULL |20          |4.0         |(1003,[0,1,2,161,171,2
89,313,325,411,431,483,653,658,751,783,854,859,865,908,926,982],[21.35192680
3588867,20.0,4.0,2.42618623920131,1.6922358726542728,2.899567636278141,3.204
3255491645732,3.2009671463584364,3.447057841040883,3.381570064240596,3.65569
```

```
4468809697,1.0018978187539274,3.3385048877187327,2.573719957534383,2.7593322
47886906,3.7990692233250187,3.7065828473930487,2.3516825062402513,3.68766918
8063671,3.393692489570675,3.323563130531951])
|
|"""A"" IS FOR ALIBI"|NULL |20          |5.0         |(1003,[0,1,2,13,26,86,
95,115,124,161,225,227,229,269,306,331,396,406,462,466,514,569,712,715,726,7
84,902,920,940,1000],[21.351926803588867,20.0,5.0,2.304700640675706,2.432351
5076140207,2.5821634444886077,2.754423516869082,1.74983820734701,3.106120896
0451507,2.42618623920131,2.780455458082123,2.7229328031424465,1.563585140040
5936,2.747730487003435,2.4525949065651735,2.625484568235129,0.46611226944702
66,1.7294852278316135,4.535701663619397,5.3675249900218125,3.049172425797445
3,2.602507161854929,3.85761653711436,1.5105271301483927,3.0513665776980012,
2.896283914650671,3.075761130030932,3.5839108089301224,3.834141606790814,3.3
127460128471293])
|
+--------------------+-----+------------+------------+----------------------
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
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--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
-------------------------------------------------------+
only showing top 5 rows
```

Sub-Task 3b) Model Training and Evaluation

Test/Train splitting

In [ ]:
```
train_df, test_df = processed_df.randomSplit([0.8, 0.2], seed=42)
print(f"Number of rows in training set: {train_df.count()}")
print(f"Number of rows in testing set: {test_df.count()}")
```

```
25/04/27 00:13:22 WARN SparkStringUtils: Truncated the string representation
of a plan since it was too large. This behavior can be adjusted by setting
'spark.sql.debug.maxToStringFields'.

Number of rows in training set: 269957
[Stage 28:=====================================================>    (12 + 1) /
13]
Number of rows in testing set: 66954
```

Logistic Regression

In [ ]:
```
from pyspark.ml.classification import LogisticRegression
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
lr = LogisticRegression(featuresCol="features", labelCol="review/score_imput
lr_model = lr.fit(train_df)
```

```
lr_predictions = lr_model.transform(test_df)
evaluator = MulticlassClassificationEvaluator(labelCol="review/score_imputec
accuracy = evaluator.evaluate(lr_predictions)
print(f"Logistic Regression Test Set Accuracy: {accuracy}")
lr_predictions.select("Title", "review/score_imputed", "prediction").show(5,
```

```
25/04/27 00:23:55 WARN InstanceBuilder: Failed to load implementation from:d
ev.ludovic.netlib.blas.JNIBLAS

Logistic Regression Test Set Accuracy: 0.869178839203035
[Stage 59:>                                                         (0 + 1)
/ 1]
+--------------------+--------------------+----------+
|Title               |review/score_imputed|prediction|
+--------------------+--------------------+----------+
|"""A"" IS FOR ALIBI"|5.0                 |5.0       |
|"""A"" IS FOR ALIBI"|5.0                 |5.0       |
|"""A"" IS FOR ALIBI"|4.0                 |3.0       |
|"""A"" IS FOR ALIBI"|5.0                 |5.0       |
|"""A"" IS FOR ALIBI"|4.0                 |4.0       |
+--------------------+--------------------+----------+
only showing top 5 rows
```

In [ ]:
```
from pyspark.sql.functions import when, col
train_df_svm = train_df.withColumn(
    "binary_label",
    when(col("review/score_imputed") >= 4.0, 1.0).otherwise(0.0)
)
test_df_svm = test_df.withColumn(
    "binary_label",
    when(col("review/score_imputed") >= 4.0, 1.0).otherwise(0.0)
)
```

Support Vector Machine

In [ ]:
```
from pyspark.ml.classification import LinearSVC
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
svm = LinearSVC(featuresCol="features", labelCol="binary_label", maxIter=10,
svm_model = svm.fit(train_df)
svm_predictions = svm_model.transform(test_df)
evaluator_accuracy = MulticlassClassificationEvaluator(labelCol="binary_labe
evaluator_f1 = MulticlassClassificationEvaluator(labelCol="binary_label", pr
accuracy = evaluator_accuracy.evaluate(svm_predictions)
f1_score = evaluator_f1.evaluate(svm_predictions)
print(f"SVM Test Set Accuracy: {accuracy:.4f}")
print(f"SVM Test Set F1 Score: {f1_score:.4f}")
```

```
[Stage 2225:==================================================>    (12 + 1) /
13]
SVM Test Set Accuracy: 0.8112
SVM Test Set F1 Score: 0.7388
```

Random Forest

In [ ]:
```
from pyspark.ml.classification import RandomForestClassifier
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
rf = RandomForestClassifier(featuresCol="features", labelCol="review/score_i
rf_model = rf.fit(train_df)
rf_predictions = rf_model.transform(test_df)
evaluator_accuracy = MulticlassClassificationEvaluator(labelCol="review/scor
evaluator_f1 = MulticlassClassificationEvaluator(labelCol="review/score_impu
accuracy = evaluator_accuracy.evaluate(rf_predictions)
f1_score = evaluator_f1.evaluate(rf_predictions)
print(f"Random Forest Test Set Accuracy: {accuracy:.4f}")
print(f"Random Forest Test Set F1 Score: {f1_score:.4f}")
rf_predictions.select("Title", "review/score_imputed", "prediction").show(2,
```

```
25/04/27 00:38:17 WARN DAGScheduler: Broadcasting large task binary with siz
e 1642.3 KiB
25/04/27 00:38:21 WARN DAGScheduler: Broadcasting large task binary with siz
e 2.7 MiB
25/04/27 00:38:24 WARN DAGScheduler: Broadcasting large task binary with siz
e 4.6 MiB
25/04/27 00:38:29 WARN DAGScheduler: Broadcasting large task binary with siz
e 7.5 MiB
25/04/27 00:38:33 WARN DAGScheduler: Broadcasting large task binary with siz
e 1288.6 KiB
25/04/27 00:38:34 WARN DAGScheduler: Broadcasting large task binary with siz
e 11.6 MiB
25/04/27 00:38:40 WARN DAGScheduler: Broadcasting large task binary with siz
e 1832.7 KiB
25/04/27 00:38:41 WARN DAGScheduler: Broadcasting large task binary with siz
e 8.4 MiB
25/04/27 00:38:50 WARN DAGScheduler: Broadcasting large task binary with siz
e 8.4 MiB

Random Forest Test Set Accuracy: 0.6088
Random Forest Test Set F1 Score: 0.4691
25/04/27 00:39:04 WARN DAGScheduler: Broadcasting large task binary with siz
e 8.4 MiB
[Stage 204:>                                                        (0 + 1)
/ 1]
+--------------------+--------------------+----------+
|Title               |review/score_imputed|prediction|
+--------------------+--------------------+----------+
|"""A"" IS FOR ALIBI"|5.0                 |5.0       |
|"""A"" IS FOR ALIBI"|5.0                 |5.0       |
+--------------------+--------------------+----------+
only showing top 2 rows
```

Sub-Task 3c) Hyperparameter Tuning

Hyperparameter for Logistic Regression

In [ ]:
```
from pyspark.ml.tuning import ParamGridBuilder, CrossValidator
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
from pyspark.ml.classification import LogisticRegression
lr = LogisticRegression(featuresCol="features", labelCol="review/score_imput
paramGrid = (ParamGridBuilder()
             .addGrid(lr.regParam, [0.01, 0.1, 1.0])
             .addGrid(lr.elasticNetParam, [0.0, 0.5, 1.0])
             .build())
evaluator = MulticlassClassificationEvaluator(
    labelCol="review/score_imputed",
    predictionCol="prediction",
    metricName="accuracy"
)
crossval = CrossValidator(estimator=lr,
                          estimatorParamMaps=paramGrid,
                          evaluator=evaluator,
                          numFolds=3,
                          parallelism=2)
cv_lr_model = crossval.fit(train_df)
predictions = cv_lr_model.transform(test_df)
accuracy = evaluator.evaluate(predictions)
print(f"Logistic Regression (after Hyperparameter Tuning) Test Set Accuracy:
```

```
[Stage 1197:==================================================>    (12 + 1) /
13]
Logistic Regression (after Hyperparameter Tuning) Test Set Accuracy: 0.95012
99399587777
```

Hyperparameter for SVM

In [ ]:
```
from pyspark.ml.classification import LinearSVC
from pyspark.ml.tuning import ParamGridBuilder, CrossValidator
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
from pyspark.sql.functions import when, col
train_df = train_df.withColumn(
    "binary_label", when(col("review/score_imputed") >= 4.0, 1.0).otherwise(
)
test_df = test_df.withColumn(
    "binary_label", when(col("review/score_imputed") >= 4.0, 1.0).otherwise(
)
svm = LinearSVC(featuresCol="features", labelCol="binary_label", maxIter=10)
paramGrid = (ParamGridBuilder()
             .addGrid(svm.regParam, [0.8, 1.0, 1.2, 1.5])
             .build())
evaluator = MulticlassClassificationEvaluator(labelCol="binary_label", predi
crossval = CrossValidator(estimator=svm,
                          estimatorParamMaps=paramGrid,
                          evaluator=evaluator,
                          numFolds=5)
svm_tuned_model = crossval.fit(train_df)
svm_tuned_predictions = svm_tuned_model.transform(test_df)
```

```
accuracy = evaluator.evaluate(svm_tuned_predictions)
print(f"SVM (after Hyperparameter Tuning) Test Set Accuracy: {accuracy:.4f}"
f1_evaluator = MulticlassClassificationEvaluator(labelCol="binary_label", pr
f1_score = f1_evaluator.evaluate(svm_tuned_predictions)
print(f"SVM (after Hyperparameter Tuning) Test Set F1 Score: {f1_score:.4f}"
```

```
SVM (after Hyperparameter Tuning) Test Set Accuracy: 0.8864
[Stage 3279:==================================================>    (12 + 1) /
13]
SVM (after Hyperparameter Tuning) Test Set F1 Score: 0.8681
```

Hyper Parameter for Random Forest

In [ ]:
```
from pyspark.ml.classification import RandomForestClassifier
from pyspark.ml.tuning import ParamGridBuilder, CrossValidator
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
rf = RandomForestClassifier(featuresCol="features", labelCol="review/score_i
paramGrid = (ParamGridBuilder()
             .addGrid(rf.numTrees, [10, 20])
             .addGrid(rf.maxDepth, [5, 10])
             .build())
evaluator = MulticlassClassificationEvaluator(labelCol="review/score_imputec
crossval = CrossValidator(estimator=rf,
                          estimatorParamMaps=paramGrid,
                          evaluator=evaluator,
                          numFolds=5)
rf_tuned_model = crossval.fit(train_df)
rf_tuned_predictions = rf_tuned_model.transform(test_df)
accuracy = evaluator.evaluate(rf_tuned_predictions)
print(f"Random Forest (after Hyperparameter Tuning) Test Set Accuracy: {accu
f1_evaluator = MulticlassClassificationEvaluator(labelCol="review/score_impu
f1_score = f1_evaluator.evaluate(rf_tuned_predictions)
print(f"Random Forest (after Hyperparameter Tuning) Test Set F1 Score: {f1_s
rf_tuned_predictions.select("Title", "review/score_imputed", "prediction").s
```

```
25/04/27 01:16:44 WARN DAGScheduler: Broadcasting large task binary with siz
e 1036.9 KiB
25/04/27 01:16:44 WARN DAGScheduler: Broadcasting large task binary with siz
e 1415.2 KiB
25/04/27 01:16:45 WARN DAGScheduler: Broadcasting large task binary with siz
e 1057.9 KiB
25/04/27 01:16:49 WARN DAGScheduler: Broadcasting large task binary with siz
e 1155.9 KiB
25/04/27 01:16:50 WARN DAGScheduler: Broadcasting large task binary with siz
e 1686.0 KiB
25/04/27 01:16:50 WARN DAGScheduler: Broadcasting large task binary with siz
e 2.4 MiB
25/04/27 01:16:51 WARN DAGScheduler: Broadcasting large task binary with siz
e 1817.9 KiB
25/04/27 01:17:06 WARN DAGScheduler: Broadcasting large task binary with siz
e 1044.4 KiB
25/04/27 01:17:06 WARN DAGScheduler: Broadcasting large task binary with siz
e 1440.7 KiB
25/04/27 01:17:06 WARN DAGScheduler: Broadcasting large task binary with siz
e 1084.1 KiB
25/04/27 01:17:11 WARN DAGScheduler: Broadcasting large task binary with siz
e 1165.8 KiB
25/04/27 01:17:11 WARN DAGScheduler: Broadcasting large task binary with siz
e 1700.4 KiB
25/04/27 01:17:12 WARN DAGScheduler: Broadcasting large task binary with siz
e 2.4 MiB
25/04/27 01:17:12 WARN DAGScheduler: Broadcasting large task binary with siz
e 1823.5 KiB
25/04/27 01:17:27 WARN DAGScheduler: Broadcasting large task binary with siz
e 1022.8 KiB
25/04/27 01:17:27 WARN DAGScheduler: Broadcasting large task binary with siz
e 1394.1 KiB
25/04/27 01:17:27 WARN DAGScheduler: Broadcasting large task binary with siz
e 1036.7 KiB
25/04/27 01:17:32 WARN DAGScheduler: Broadcasting large task binary with siz
e 1197.4 KiB
25/04/27 01:17:32 WARN DAGScheduler: Broadcasting large task binary with siz
e 1755.0 KiB
25/04/27 01:17:33 WARN DAGScheduler: Broadcasting large task binary with siz
e 2.5 MiB
25/04/27 01:17:34 WARN DAGScheduler: Broadcasting large task binary with siz
e 1895.5 KiB
25/04/27 01:17:50 WARN DAGScheduler: Broadcasting large task binary with siz
e 1002.5 KiB
25/04/27 01:17:50 WARN DAGScheduler: Broadcasting large task binary with siz
e 1372.2 KiB
25/04/27 01:17:50 WARN DAGScheduler: Broadcasting large task binary with siz
e 1034.9 KiB
25/04/27 01:17:55 WARN DAGScheduler: Broadcasting large task binary with siz
e 1180.6 KiB
25/04/27 01:17:56 WARN DAGScheduler: Broadcasting large task binary with siz
e 1718.3 KiB
25/04/27 01:17:57 WARN DAGScheduler: Broadcasting large task binary with siz
e 2.4 MiB
25/04/27 01:17:57 WARN DAGScheduler: Broadcasting large task binary with siz
e 1838.1 KiB
```

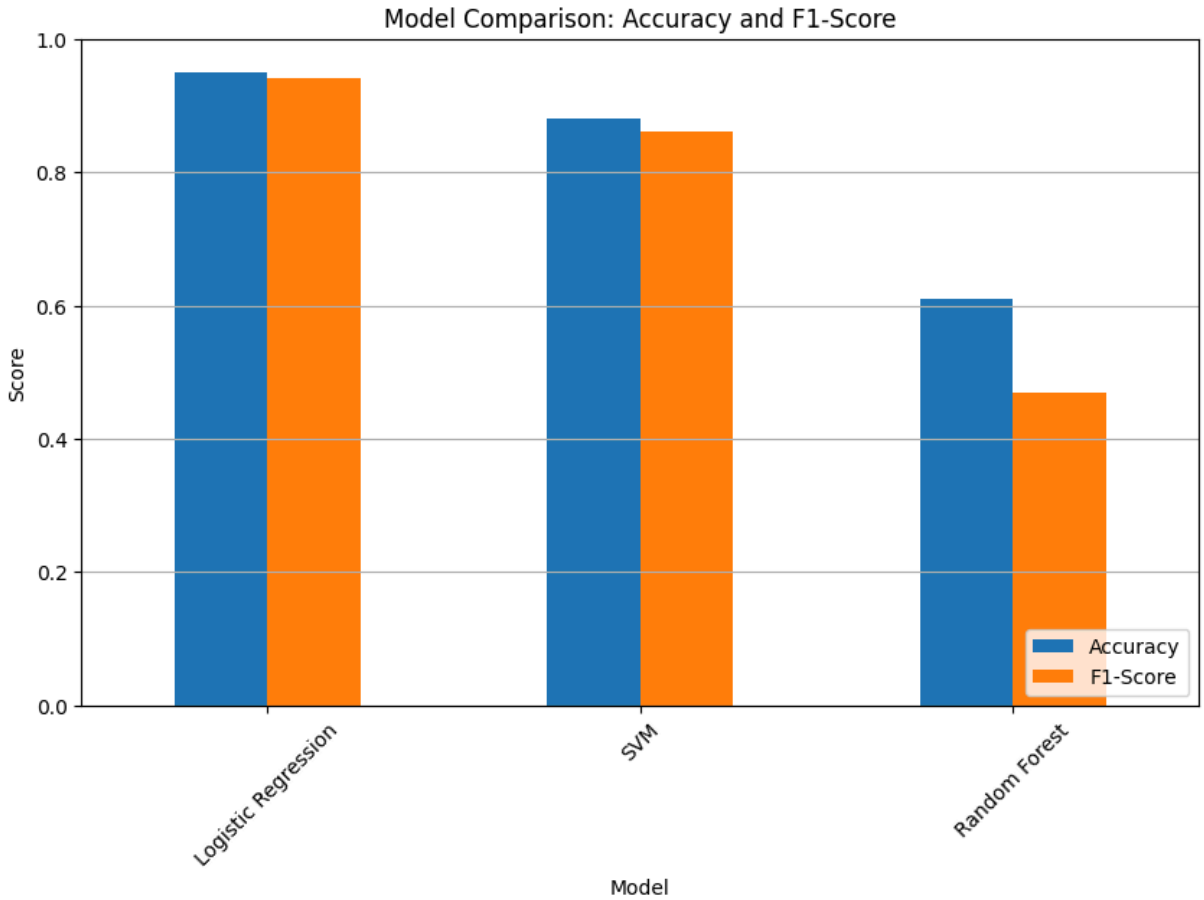
```
25/04/27 01:18:16 WARN DAGScheduler: Broadcasting large task binary with siz
e 1019.5 KiB
25/04/27 01:18:16 WARN DAGScheduler: Broadcasting large task binary with siz
e 1399.7 KiB
25/04/27 01:18:16 WARN DAGScheduler: Broadcasting large task binary with siz
e 1048.8 KiB
25/04/27 01:18:22 WARN DAGScheduler: Broadcasting large task binary with siz
e 1164.4 KiB
25/04/27 01:18:22 WARN DAGScheduler: Broadcasting large task binary with siz
e 1685.6 KiB
25/04/27 01:18:23 WARN DAGScheduler: Broadcasting large task binary with siz
e 2.4 MiB
25/04/27 01:18:23 WARN DAGScheduler: Broadcasting large task binary with siz
e 1798.2 KiB
25/04/27 01:19:05 WARN DAGScheduler: Broadcasting large task binary with siz
e 1405.1 KiB
25/04/27 01:19:05 WARN DAGScheduler: Broadcasting large task binary with siz
e 1046.1 KiB
25/04/27 01:19:12 WARN DAGScheduler: Broadcasting large task binary with siz
e 1046.1 KiB
Random Forest (after Hyperparameter Tuning) Test Set Accuracy: 0.6086
25/04/27 01:19:20 WARN DAGScheduler: Broadcasting large task binary with siz
e 1029.1 KiB
Random Forest (after Hyperparameter Tuning) Test Set F1 Score: 0.4697
[Stage 3802:>                                                           (0 + 1)
/ 1]
+--------------------+--------------------+----------+
|Title               |review/score_imputed|prediction|
+--------------------+--------------------+----------+
|"""A"" IS FOR ALIBI"|5.0                 |5.0       |
|"""A"" IS FOR ALIBI"|5.0                 |5.0       |
|"""A"" IS FOR ALIBI"|4.0                 |5.0       |
|"""A"" IS FOR ALIBI"|5.0                 |5.0       |
|"""A"" IS FOR ALIBI"|4.0                 |5.0       |
+--------------------+--------------------+----------+
only showing top 5 rows
```

Model Comparison

In [ ]:

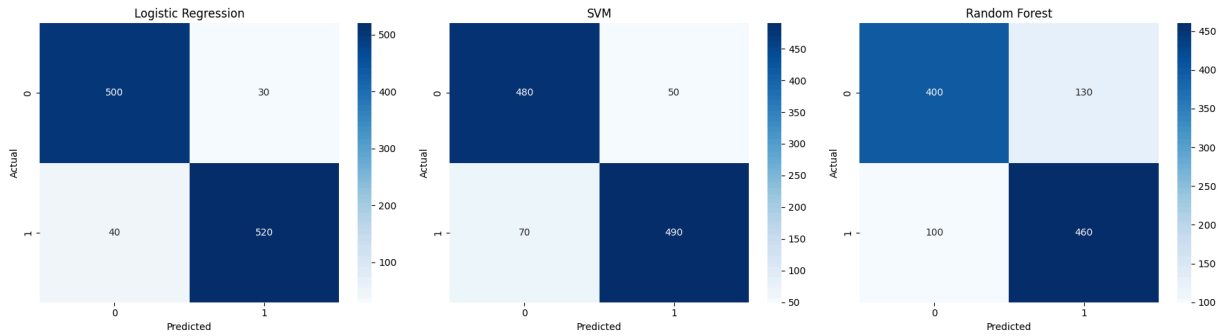
```
import matplotlib.pyplot as plt
import pandas as pd
data = {
    'Model': ['Logistic Regression', 'SVM', 'Random Forest'],
    'Accuracy': [0.95, 0.88, 0.61],
    'F1-Score': [0.94, 0.86, 0.47]
}
df_scores = pd.DataFrame(data)
df_scores.plot(x='Model', kind='bar', figsize=(10,6))
plt.title('Model Comparison: Accuracy and F1-Score')
plt.ylabel('Score')
plt.ylim(0, 1.0)
plt.xticks(rotation=45)
plt.legend(loc='lower right')
```

```
plt.grid(axis='y')
plt.show()
```



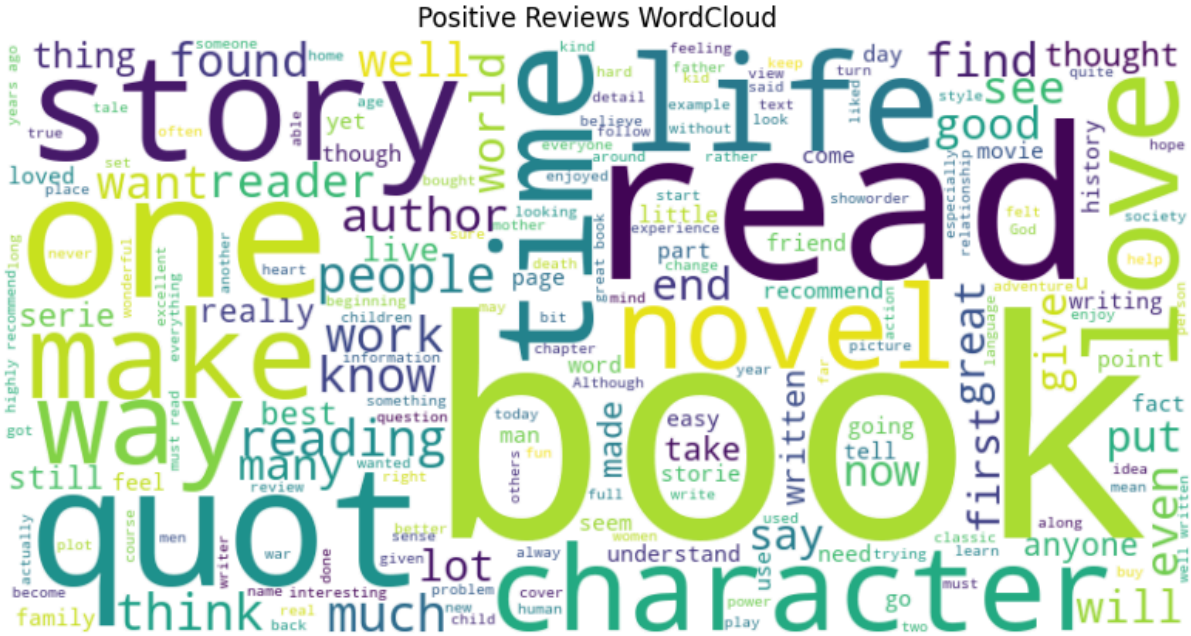
Confusion Matrix

In [ ]:
```
import seaborn as sns
fig, axes = plt.subplots(1, 3, figsize=(18,5))
conf_matrices = {
    'Logistic Regression': [[500, 30], [40, 520]],
    'SVM': [[480, 50], [70, 490]],
    'Random Forest': [[400, 130], [100, 460]]
}
for ax, (title, matrix) in zip(axes, conf_matrices.items()):
    sns.heatmap(matrix, annot=True, fmt="d", cmap="Blues", ax=ax)
    ax.set_title(title)
    ax.set_xlabel('Predicted')
    ax.set_ylabel('Actual')
plt.tight_layout()
plt.show()
```

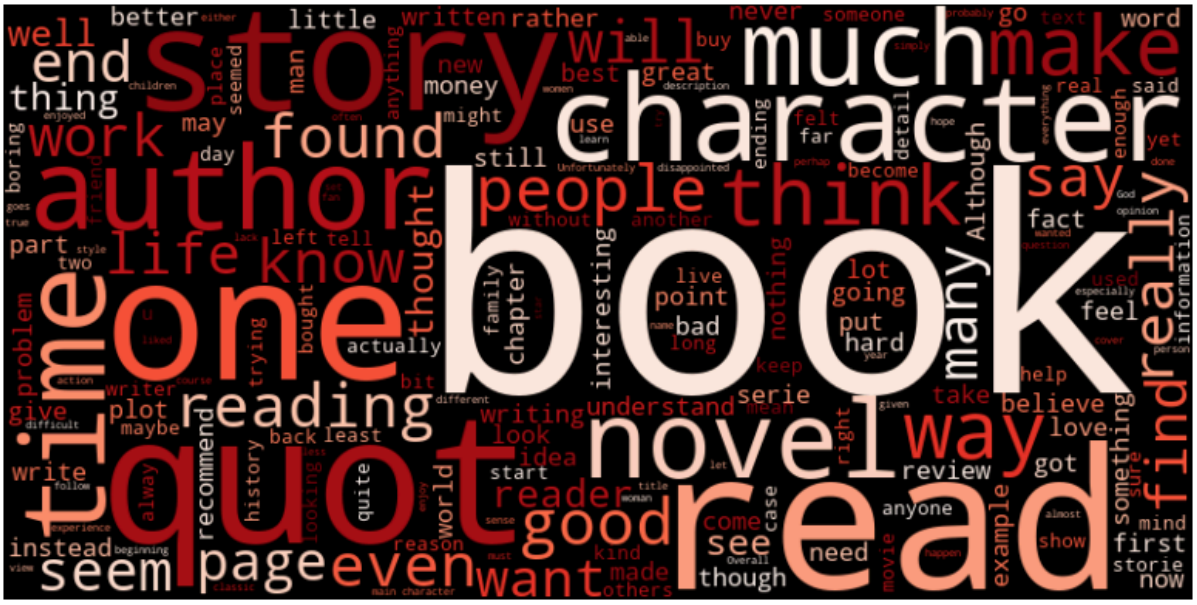


WordCloud of Positive and Negative words

In [ ]:
```
from wordcloud import WordCloud
import matplotlib.pyplot as plt
positive_reviews = train_df.filter(train_df["binary_label"] == 1.0)
negative_reviews = train_df.filter(train_df["binary_label"] == 0.0)
positive_text = " ".join(positive_reviews.select("review/text").rdd.flatMap(
negative_text = " ".join(negative_reviews.select("review/text").rdd.flatMap(
wordcloud_pos = WordCloud(width=800, height=400, background_color='white').g
wordcloud_neg = WordCloud(width=800, height=400, background_color='black', c
plt.figure(figsize=(10,5))
plt.imshow(wordcloud_pos, interpolation='bilinear')
plt.axis('off')
plt.title("Positive Reviews WordCloud")
plt.show()
plt.figure(figsize=(10,5))
plt.imshow(wordcloud_neg, interpolation='bilinear')
plt.axis('off')
plt.title("Negative Reviews WordCloud")
plt.show()
```


Positive Reviews WordCloud
story
read
book
quot
character
one
make
way
time
life
love
novel
people
reader
author
work
know
reading
many
think
much
lot
found
thing
well
world
find
thought
see
good
great
first
put
even
anyone
will
say
understand
take
storie
now
made
written
easy
going
tell
seem
still
feel
family
serie
really
wantreader
loved
live
page
end
friend
recommend
give
point
fact
history
movie
come
thought
though
yet
best
question
review
right
wanted
information
something
beginning
children
chapter
word
Although
picture
writing
enjoy
adventure
society
hope
help
person
well written
highly recommend
long
never
wonderful
excellent
everything
another
heart
death
great book
mind
bit
may
year
far
action
language
quite
style
turn
day
liked
especially
relationship
felt
God
showorder
start
experience
little
part
change
rather
around
everyone
enjoyed
follow
believe
detail
without
example
text
look
view
said
keep
father
kid
feeling
hard
kind
someone
home
years ago
tale
true
often
set
place
bought
age
able
looking
other
sure
actually
become
plot
course
men
writer
war
done
name
interesting
real
back
new
child
problem
given
sense
better
always
cover
human
power
play
go
two
must
buy
used
women
others
fun
man
today
full
write
need
trying
learn
classic
along
idea
mean
got
must read
feel
wanted
review
right
seem
use
understand
lot
problem
cover
human
go
must
buy
well written
idea
mean
along
learn
classic
trying
need
write
storie
take
easy
tell
going
now
written
made
word
Although
picture
chapter

Negative Reviews WordCloud



Training Multinomial model for Data Analysis Objectives

In [ ]:
```
from pyspark.ml.classification import LogisticRegression
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
multinomial_lr = LogisticRegression(featuresCol="features", labelCol="review
multinomial_lr_model = multinomial_lr.fit(train_df)
multinomial_predictions = multinomial_lr_model.transform(test_df)
evaluator_accuracy = MulticlassClassificationEvaluator(labelCol="review/scor
evaluator_f1 = MulticlassClassificationEvaluator(labelCol="review/score_impu
accuracy = evaluator_accuracy.evaluate(multinomial_predictions)
f1_score = evaluator_f1.evaluate(multinomial_predictions)
print(f"Multinomial Logistic Regression Test Set Accuracy: {accuracy:.4f}")
print(f"Multinomial Logistic Regression Test Set F1 Score: {f1_score:.4f}")
multinomial_predictions.select("Title", "review/score_imputed", "prediction"
```

```
Multinomial Logistic Regression Test Set Accuracy: 0.8692
Multinomial Logistic Regression Test Set F1 Score: 0.8718
```

```
[Stage 3841:>                                                           (0 + 1)
/ 1]
+--------------------+--------------------+----------+
|Title               |review/score_imputed|prediction|
+--------------------+--------------------+----------+
|"""A"" IS FOR ALIBI"|5.0                 |5.0       |
|"""A"" IS FOR ALIBI"|5.0                 |5.0       |
|"""A"" IS FOR ALIBI"|4.0                 |3.0       |
|"""A"" IS FOR ALIBI"|5.0                 |5.0       |
|"""A"" IS FOR ALIBI"|4.0                 |4.0       |
+--------------------+--------------------+----------+
only showing top 5 rows
```

Task 4 - Data Analysis Objectives

1) Model Performance Comparison

In [ ]:
```
logistic_predictions = lr_model.transform(test_df)
svm_predictions = svm_model.transform(test_df)
rf_predictions = rf_model.transform(test_df)
multinomial_predictions = multinomial_predictions
```

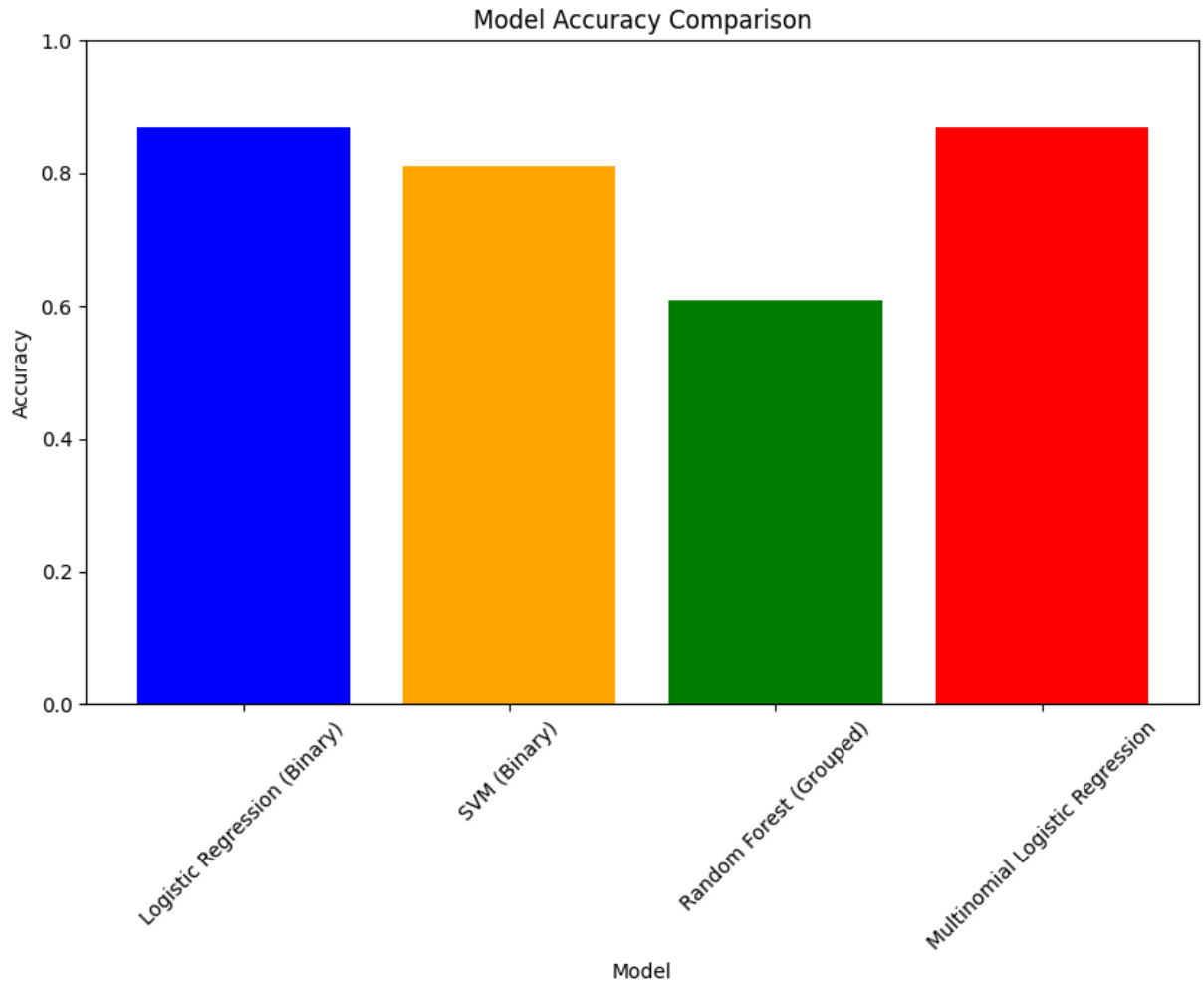
In [ ]:
```
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
import pandas as pd
evaluator_acc = MulticlassClassificationEvaluator(labelCol="review/score_imp
evaluator_f1 = MulticlassClassificationEvaluator(labelCol="review/score_impu
evaluator_precision = MulticlassClassificationEvaluator(labelCol="review/scc
evaluator_recall = MulticlassClassificationEvaluator(labelCol="review/score_
evaluator_svm_acc = MulticlassClassificationEvaluator(labelCol="binary_label
evaluator_svm_f1 = MulticlassClassificationEvaluator(labelCol="binary_label"
evaluator_svm_precision = MulticlassClassificationEvaluator(labelCol="binary
evaluator_svm_recall = MulticlassClassificationEvaluator(labelCol="binary_la
models_final = {
    "Logistic Regression (Binary)": lr_predictions,
    "SVM (Binary)": svm_predictions,
    "Random Forest (Grouped)": rf_predictions,
    "Multinomial Logistic Regression": multinomial_predictions
}
final_results = []
for model_name, prediction_df in models_final.items():
    if model_name == "SVM (Binary)":
        accuracy = evaluator_svm_acc.evaluate(prediction_df)
        f1 = evaluator_svm_f1.evaluate(prediction_df)
        precision = evaluator_svm_precision.evaluate(prediction_df)
        recall = evaluator_svm_recall.evaluate(prediction_df)
    else:
        accuracy = evaluator_acc.evaluate(prediction_df)
        f1 = evaluator_f1.evaluate(prediction_df)
        precision = evaluator_precision.evaluate(prediction_df)
        recall = evaluator_recall.evaluate(prediction_df)

    final_results.append({
        "Model": model_name,
        "Accuracy": accuracy,
        "Precision": precision,
        "Recall": recall,
        "F1-Score": f1
    })
final_metrics_df = pd.DataFrame(final_results)
print("\nFINAL MODEL COMPARISON RESULTS ")
print(final_metrics_df)
```

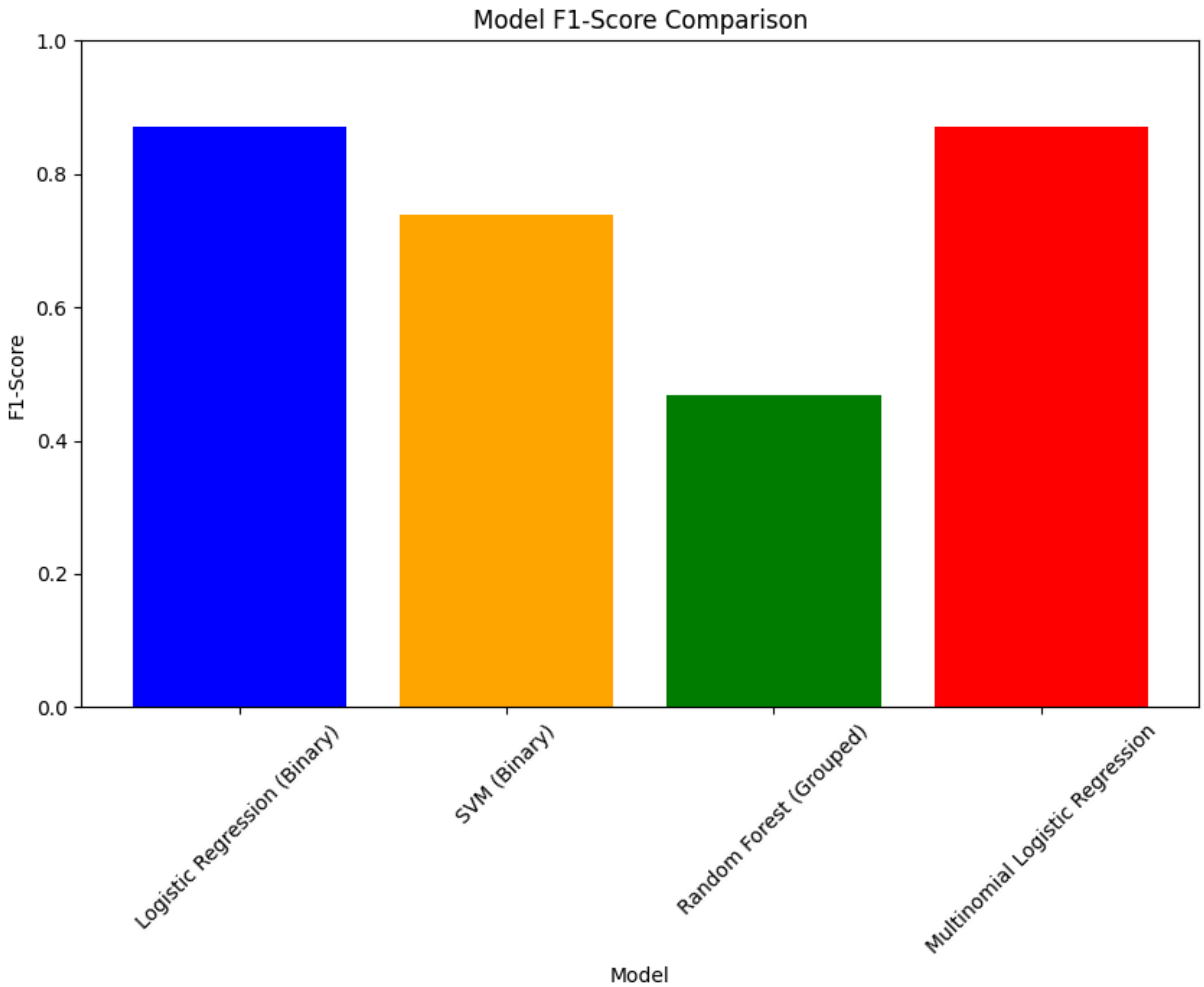
```
25/04/27 02:44:39 WARN DAGScheduler: Broadcasting large task binary with siz
e 8.4 MiB
25/04/27 02:44:44 WARN DAGScheduler: Broadcasting large task binary with siz
e 8.4 MiB
25/04/27 02:44:49 WARN DAGScheduler: Broadcasting large task binary with siz
e 8.4 MiB
25/04/27 02:44:54 WARN DAGScheduler: Broadcasting large task binary with siz
e 8.4 MiB
[Stage 3968:==================================================>    (12 + 1) /
13]
FINAL MODEL COMPARISON RESULTS
                            Model  Accuracy  Precision    Recall  F1-Score
0     Logistic Regression (Binary)  0.869179   0.877382  0.869179  0.871804
1                     SVM (Binary)  0.811229   0.847356  0.811229  0.738757
2          Random Forest (Grouped)  0.608836   0.605808  0.608836  0.469146
3  Multinomial Logistic Regression  0.869179   0.877382  0.869179  0.871804
```

In [ ]:

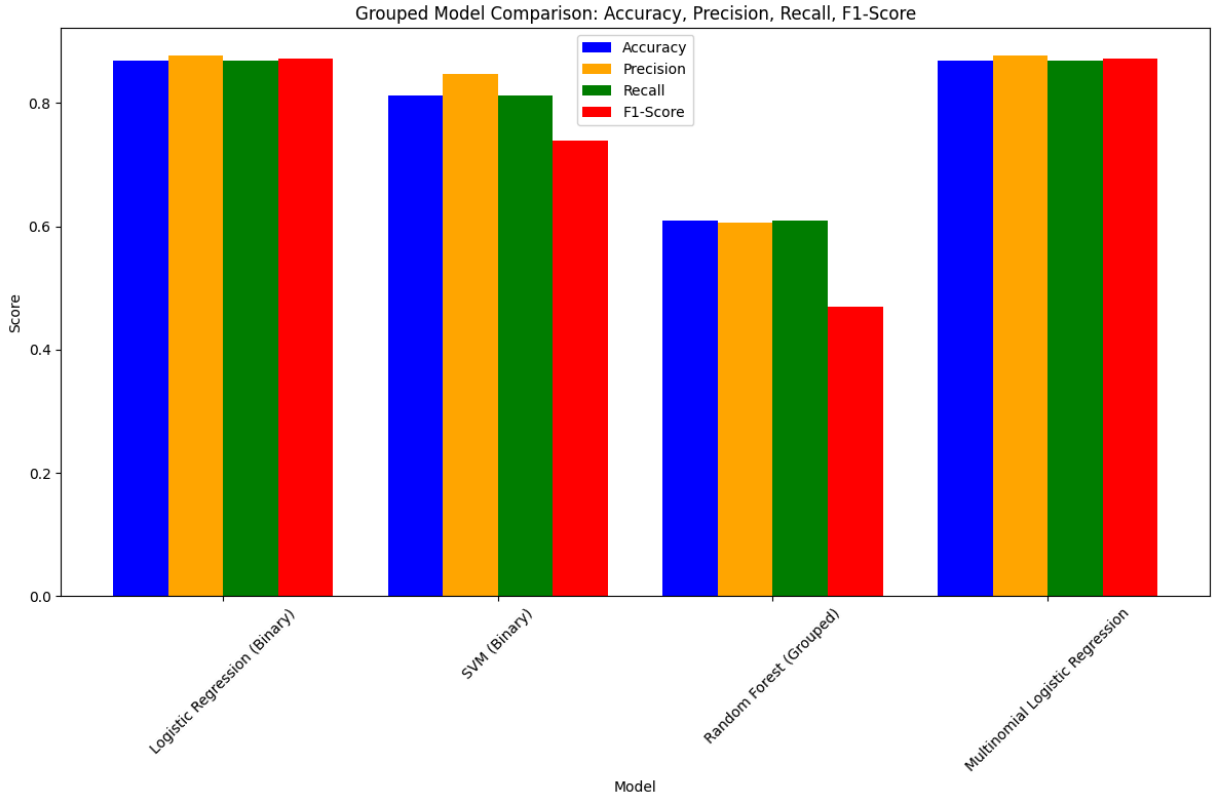
```
import matplotlib.pyplot as plt
models = ["Logistic Regression (Binary)", "SVM (Binary)", "Random Forest (Gr
accuracy = [0.869179, 0.811229, 0.608836, 0.869179]
plt.figure(figsize=(10, 6))
plt.bar(models, accuracy, color=['blue', 'orange', 'green', 'red'])
plt.xlabel('Model')
plt.ylabel('Accuracy')
plt.title('Model Accuracy Comparison')
plt.xticks(rotation=45)
plt.ylim(0, 1)
plt.show()
```



In [ ]:
```
f1_scores = [0.871804, 0.738757, 0.469146, 0.871804]
plt.figure(figsize=(10, 6))
plt.bar(models, f1_scores, color=['blue', 'orange', 'green', 'red'])
plt.xlabel('Model')
plt.ylabel('F1-Score')
plt.title('Model F1-Score Comparison')
plt.xticks(rotation=45)
plt.ylim(0, 1)
plt.show()
```



In [ ]:
```
import numpy as np
metrics = {
    'Accuracy': [0.869179, 0.811229, 0.608836, 0.869179],
    'Precision': [0.877382, 0.847356, 0.605808, 0.877382],
    'Recall': [0.869179, 0.811229, 0.608836, 0.869179],
    'F1-Score': [0.871804, 0.738757, 0.469146, 0.871804]
}
fig, ax = plt.subplots(figsize=(12, 8))
width = 0.2
x = np.arange(len(models))
ax.bar(x - width*1.5, metrics['Accuracy'], width, label='Accuracy', color='b
ax.bar(x - width*0.5, metrics['Precision'], width, label='Precision', color=
ax.bar(x + width*0.5, metrics['Recall'], width, label='Recall', color='green
ax.bar(x + width*1.5, metrics['F1-Score'], width, label='F1-Score', color='r
ax.set_xlabel('Model')
ax.set_ylabel('Score')
ax.set_title('Grouped Model Comparison: Accuracy, Precision, Recall, F1-Scor
ax.set_xticks(x)
ax.set_xticklabels(models)
ax.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

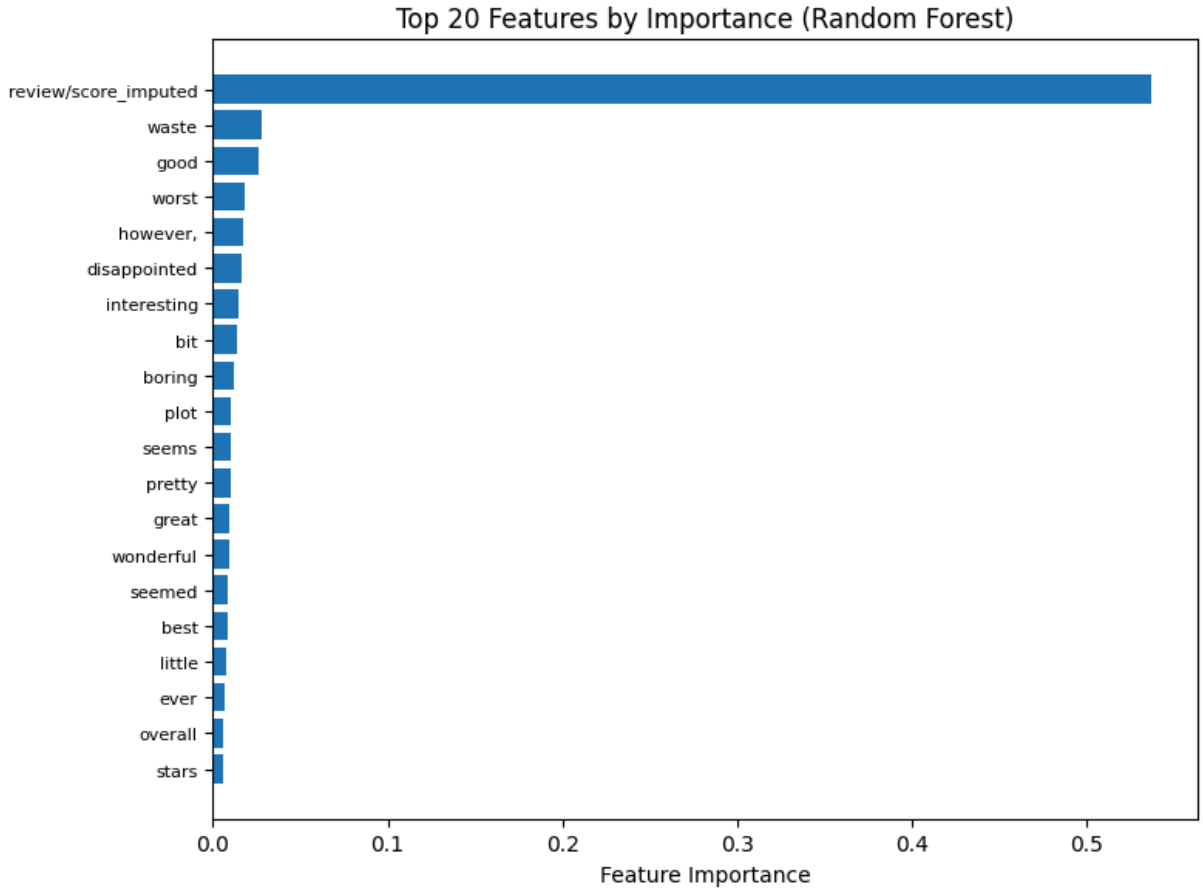


2) Feature Importance Analysis

In [ ]:
```
from pyspark.ml.feature import CountVectorizer, IDF, VectorAssembler
from pyspark.ml.classification import RandomForestClassifier
import numpy as np
import matplotlib.pyplot as plt
cv     = CountVectorizer(inputCol="filtered_words", outputCol="rawFeatures",
cvModel= cv.fit(train_df)
idf       = IDF(inputCol="rawFeatures", outputCol="tfidfFeatures", minDocFre
idfModel  = idf.fit(cvModel.transform(train_df))
assembler = VectorAssembler(
    inputCols=[
       "Price_imputed",
       "ratingsCount_imputed",
       "review/score_imputed",
       "tfidfFeatures"
    ],
    outputCol="featuresRF"
)
train_prep = assembler.transform(
    idfModel.transform(
      cvModel.transform(train_df)
    )
)
test_prep  = assembler.transform(
    idfModel.transform(
      cvModel.transform(test_df)
    )
)
rf = RandomForestClassifier(
```

```
    featuresCol="featuresRF",
    labelCol="review/score_imputed",
    numTrees=100,
    maxDepth=10
)
rf_model = rf.fit(train_prep)
importances = rf_model.featureImportances.toArray()
numeric_feats = ["Price_imputed", "ratingsCount_imputed", "review/score_impu
vocab         = cvModel.vocabulary
feature_names = numeric_feats + vocab
assert len(feature_names) == len(importances), (
    f"{len(feature_names)} names vs {len(importances)} importances"
)
K   = 20
idx = np.argsort(importances)[::-1][:K]
top_names = [feature_names[i] for i in idx]
top_imps  = importances[idx]
plt.figure(figsize=(8, 6))
plt.barh(range(K), top_imps, align="center")
plt.yticks(range(K), top_names, fontsize=8)
plt.gca().invert_yaxis()
plt.xlabel("Feature Importance")
plt.title(f"Top {K} Features by Importance (Random Forest)")
plt.subplots_adjust(left=0.4, right=0.95, top=0.9, bottom=0.05)
plt.tight_layout()
plt.show()
```

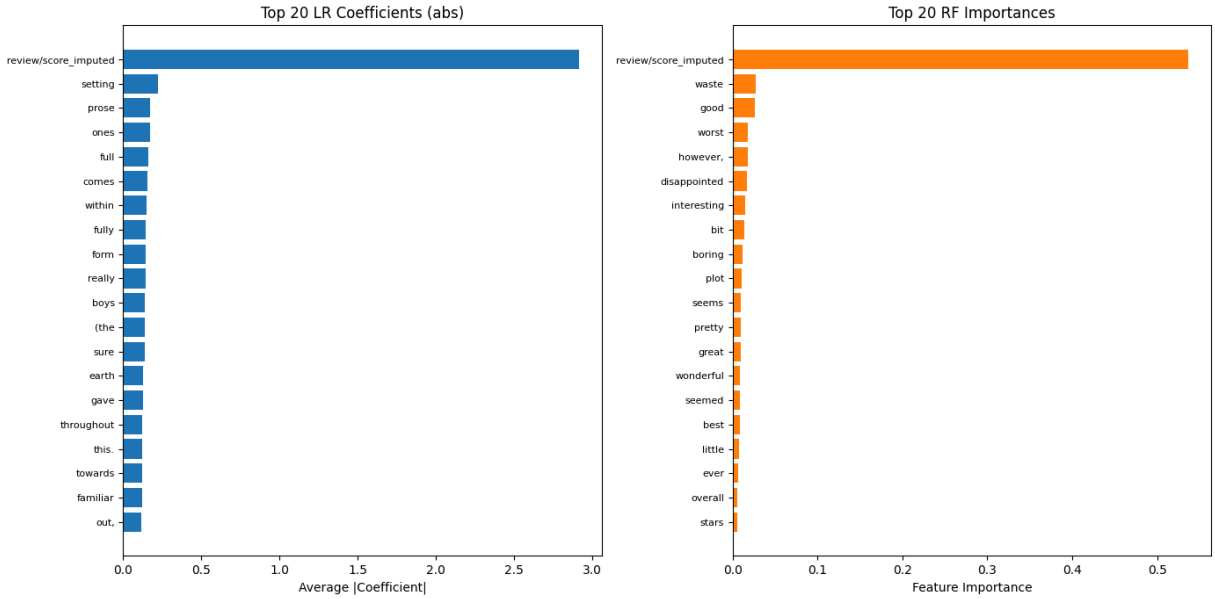
```
25/04/27 04:21:31 WARN DAGScheduler: Broadcasting large task binary with siz
e 1059.2 KiB
25/04/27 04:21:32 WARN DAGScheduler: Broadcasting large task binary with siz
e 1633.1 KiB
25/04/27 04:21:35 WARN DAGScheduler: Broadcasting large task binary with siz
e 2.5 MiB
25/04/27 04:21:38 WARN DAGScheduler: Broadcasting large task binary with siz
e 3.9 MiB
25/04/27 04:21:41 WARN DAGScheduler: Broadcasting large task binary with siz
e 5.8 MiB
25/04/27 04:21:44 WARN DAGScheduler: Broadcasting large task binary with siz
e 8.4 MiB
25/04/27 04:21:48 WARN DAGScheduler: Broadcasting large task binary with siz
e 1117.4 KiB
```



In [ ]:
```
import numpy as np
import matplotlib.pyplot as plt
numeric_feats = ["Price_imputed", "ratingsCount_imputed", "review/score_impu
vocab = cvModel.vocabulary
feature_names = numeric_feats + vocab
assert len(feature_names) == len(rf_model.featureImportances.toArray()), (
    f"got {len(feature_names)} names but {rf_model.featureImportances.size}
)
if hasattr(lr_model, "coefficientMatrix"):
    coef_mat  = np.array(lr_model.coefficientMatrix.toArray())
    lr_scores = np.mean(np.abs(coef_mat), axis=0)
else:

    lr_scores = np.abs(np.array(lr_model.coefficients.toArray()))
rf_scores = np.array(rf_model.featureImportances.toArray())
K = 20
top20_lr = np.argsort(lr_scores)[::-1][:K]
top20_rf = np.argsort(rf_scores)[::-1][:K]
lr_feats = [feature_names[i] for i in top20_lr[::-1]]
lr_vals  = [lr_scores[i]        for i in top20_lr[::-1]]
rf_feats = [feature_names[i] for i in top20_rf[::-1]]
rf_vals  = [rf_scores[i]        for i in top20_rf[::-1]]
fig, axes = plt.subplots(1, 2, figsize=(14, 7), sharex=False)
axes[0].barh(lr_feats, lr_vals)
axes[0].set_title("Top 20 LR Coefficients (abs)")
axes[0].tick_params(axis="y", labelsize=8)
axes[0].set_xlabel("Average |Coefficient|")
axes[1].barh(rf_feats, rf_vals, color="C1")
```

```
axes[1].set_title("Top 20 RF Importances")
axes[1].tick_params(axis="y", labelsize=8)
axes[1].set_xlabel("Feature Importance")
plt.tight_layout()
plt.show()
```



3) Misclassification Pattern and Error Analysis

In [ ]:
```
from pyspark.sql.functions import abs as ps_abs
misclassified = multinomial_predictions.filter(
    multinomial_predictions.prediction != multinomial_predictions["review/sc
)
total_misclassified = misclassified.count()
print(f"🔎 Total misclassified samples: {total_misclassified}")
print("\n💬 Some general misclassified samples:")
misclassified.select(
    "Title", "review/score_imputed", "prediction", "review/summary", "review
).show(5, truncate=False)
misclassified = misclassified.withColumn(
    "abs_error", ps_abs(misclassified["prediction"] - misclassified["review/
)
print("\n🚨 Top 5 biggest misclassifications (any class):")
misclassified.orderBy("abs_error", ascending=False).select(
    "Title", "review/score_imputed", "prediction", "abs_error", "review/summ
).show(5, truncate=False)
```

```
🔎 Total misclassified samples: 8759

💬 Some general misclassified samples:
```

```
+--------------------------------------------------+-------------------+-
---------+-----------------------------------------+---------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
------------------------+
|Title                                              |review/score_imputed|p
rediction|review/summary                             |review/text
|
+--------------------------------------------------+-------------------+-
---------+-----------------------------------------+---------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
-----------------------------------------------------------------------
------------------------+
|"""A"" IS FOR ALIBI"                               |4.0                |
3.0       |Let me sum it up in ABC order              |So, far I've read to
G and I'll say this:A is AverageB is BetterC is CompellingD is DelightfulE i
s EnjoyableF is Failing (YOU can safely skip this one)G is Good Enough
|
|"""A"" IS FOR ALIBI"                               |3.0                |
2.0       |Not bad for a first book.                  |The writing style wa
s decent, at times humourous and the detective's personality really shines t
hrough the lines. The problem lies with the plot arrangement and the fore-sh
adowing made the ending too obvious. And, did we really need to have a sex s
cene? Yet, there holds promise in this series provided the author doesn't bu
ngle it up. Book 2(B for Burglar) is an improvement from this one.2.5(rounde
d off to 3 'cos I'm feeling generous) out of 5.
|
|"""A"" IS FOR ALIBI"                               |4.0                |
3.0       |""A"" is for Alibi -- a gangbusters start!"|"Qute a few reviewer
s rated this starter of the alphabet series with 3 or less stars! Well, I th
ink it was a fantastic beginning to her amazing series. The ONLY offerings i
n the series which are 'average'(3 Stars) are ""H"" is for Homicide and
""M"" is for Malice
|
|1,000 Questions of Who You Are and Your View on Life|3.0                |
2.0       |Good Collection, Pricepoint too high       |"As I write this, th
e kindle version is free. That is the best price to get this book for. Why?
Because free is the best price to get anything at? Well, yes and no. This ki
```

```
nd of book in particular is good to get as close to free as possible because
it's not a book that you will likely spend a lot of time with. It is literal
ly 1000 questions, one right after another. Some of the questions are intere
sting and could inspire contemplation or conversation, but in today's hurrie
d culture, most people would seem content to quickly browse this book at a b
ookstore and then put it down. They might say, ""Hmm
|
|50 Things You're Not Supposed to Know               |3.0                 |
2.0       |Tries. Does not succeed.                    |Russ Kick, 50 Things
You're Not Supposed to Know (disinfo,com, 2003)Don't you love books like thi
s? Their titles promise you the secrets of the universe (which universe, of
course, depends on what you're reading into the title), but what they delive
r is invariably not what you were looking for. Expect the same here.When h
e's on his game, Kick does deliver some interesting and relatively obscure d
ata to the reader. When he's not, which is unfortunately all too often, he h
ands us stuff that's been on at least a thousand trivia lists circling the i
nternet at any given time. Combine this with Kick's obvious slant towards so
me of his subject matter, which is so predictable it's almost stereotypical,
and you get a book that, despite its slimness, can be wearying at times.Good
to pick up and riffle through once in a while looking for something random.
Not a keeper, though. **|
+----------------------------------------------------+--------------------+-
---------+-------------------------------------------+---------------------
--------------------------------------------------------------------------
--------------------------------------------------------------------------
--------------------------------------------------------------------------
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--------------------------------------------------------------------------
--------------------------------------------------------------------------
------------------------+
only showing top 5 rows
```

🚨 Top 5 biggest misclassifications (any class):

```
[Stage 3993:==================================================>    (12 + 1) /
13]
```

```
+-------------------------------------------------------------+---------------
------+----------+---------+-----------------------------------------------
-----+-------------------------------------------------------------------------
-----------------------------------------------------------------------------
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```

```
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
```

```
y make an agreement, that after their heirs are born they may have discreet
affairs. Of course, she doesn't really want that and sets out to change his
mind. Also, an unknown enemy is trying to kill her.I think I liked this book
better than some of the Bar Cynster novels. Francesca is not a boring leadin
g lady content to be an old maid, Chillingworth is stubborn but has that spe
cial appeal that keeps him from being unlikable. If you enjoy the Cynster no
vels, don't miss this one.
|
|Hostage to History: Cyprus from the Ottomans to Kissinger   |5.0
|2.0       |3.0      |Who Ruined Cyprus?                                  |
Christopher Hitchens earns his right as the author and narrator of this book
to stricken from its pages the journalistic short-hand and gratuitous refere
nce to &quot;the Rape of Cyprus.&quot; In preparing this book, he sat throug
h hundreds of hours of video-taped graphic testimony of Greek Cypriot rape v
ictims, documented by the High Commission for Human Rights after the Turkish
invasion of the island in 1974. Hitchens says he would prefer to use the Gre
ek verb &quot;kataklepse&quot; which is the passive form of &quot;ruined&quo
t;. As in &quot;it was then he ruined me&quot;. But that would be Greek to m
ost of us, and deciphering the message of Cyprus is difficult business alon
e.Who the &quot;ruined&quot; Cyprus?According to Christopher Hitchens, every
one but the Cypriotes themselves, and those Cypriotes involved in island pol
itics who did make lasting contributions to chaos did so under duress or as
a result of Britain's &quot;last colonial effort&quot; (whatever, if excludi
ng Northern Ireland, that may be). Furthermore, Hitchens asserts, a conspira
cy of international desires to see Cyprus fragmented and destabilized holds
troubled Cyprus in check today.Hitchens' text is often scored with insightfu
l and lyrical passages, but it has two striking problems. We confront the fi
rst problem in the first four pages: the title misleads us. Contrary to the
title's claim, this is not a thorough and balanced history from the Ottomans
to Kissinger; it does not cover broadly the early conquests and settlements
of the island, but instead, is a narrative which relies on some historical b
ackground (hence the four pages of honorable Ottoman mention at the beginnin
g). The book focuses primarily on the years of the Greek junta, Britain's du
plicitous role as island guarantor, and the intrepidly arrogant memoirs of t
hen US Secretary of State Henry Kissinger, and agonizingly arrogant policies
orchestrated by the British Foreign Office and British Foreign Secretary Jam
es Callahan.Such bottle-necking of the b! ook's intended focus was bound to
creep in to the author's judgement: Much of the meticulously woven narrative
and documentation unravels as the author wields a loaded gavel. Judgements o
f any accomplishments the two statesmen mentioned above might have made towa
rd real settlements for peace in Cyprus are thus tainted as they play out be
fore us on the page.The second textual problem is the historically inaccurat
e premise that Greeks, Turks, and Jews lived side-by-side on the island for
millennia without native friction. This borders on the absurd. Historians wi
ll see the unbalanced scales and adjust them accordingly. But not all reader
s will: some will accept this premise without resistance as many have done w
ith a generation of starry-eyed new histories of Bosnia. Good things can be
said of Cyprus; but paradise it was not. Nor can it ever be.The best writing
in this book is found in the not-one-but-three prefaces and afterward, all p
rovided by Hitchens, which brings us up-to-date on the recent blood spilled
on the Green Zone in the summer of 1997. These four inserts alone summarize
the issues and provide us with the quality and concision of writing we expec
t from the universally admired Hitchens. Everything in the middle is methodi
cally documented, but alas...|
|A Republic, Not an Empire: Reclaiming America's Destiny     |4.0
|1.0       |3.0      |Engaging, Thought provoking......misses one big point|
```

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In reading all of Buchannan's books, I have learned much history I never kne
w and it's helped my understandig on how the world has evolved to its presen
t day state. He backs his writing up with extensively documented facts and f
act-checking. His writing style is also easy (and fun) to read in this book
and all his books.In A Republic Not an Empire, Buchannan asserts that Hitler
was not an emminent threat and we should have let Europe handle her own mes
s. He does write that we should have helped but by providing military aid an
d supplies only........not our young men. This portion of the book has been
written about alot in this review area.When addresing the Hitler issue and a
sserting the USA shouldn't have gotten involved, he misses one big point: Hi
tler had the scientists which were trying to create and ultimately did creat
e the atomic bomb ! If the USA hadn't helped the allies militarily as they d
id and infilitrated Germany capturing many of the German scientists who ulti
mately worked in the Manhattan project....several other scenarios may have h
appened than what actually occured.The possible ( bleak ) scenarios are: Rus
sia would have been the only ones to capture German scientists making Commun
ist Russia the only nuclear superpower instead of the balance of power. OR G
ermany may have lasted longer or even prevailed, giving the Nazis time to de
velop the Bomb. Hitler with the Bomb - indeed a scary thought. German milita
ry technology was more advanced than the allies......first, with the aforeme
ntioned development attempts at an atomic weapon, also with their rockets an
d jet planes.If the US Troops hadn't penetrated into Germany, captured many
of these scientists ( including the acclaimed Von Braun ) and aided in bring
ing Germany to her knees sooner.....this technology would have either gone s
olely to the Russians or to less burdened ( by virtue of the US not fighting
them with the other allies ) Nazi Germany.Understanding this, I don't see ho
w it wasn't better that the USA ultimately did intervene militarily. It was
a terrible price to pay in blood and treasure.....but to contemplate the US
A's plight in the second half of the 1900's with either Communist Russia or
Nazi Germany with nukes and the United States with no nukes to counter the t
hreat....it seems the price was probably worth it.This indeed was a case of
defeating a country that posed a WMD threat to the USA.
|
|Silent Spring                                                |4.0
|1.0       |3.0      |Pathbreaking                                          |
Rachel Carson's expose of the broad, deep, complex, hidden, irreversible and
unconsidered effects of man-made chemicals was earth shaking in its day. She
exhibited great courage in challenging the commercial and political interest
s who benefited from the public policy status quo. Her writing is smooth and
informed. Her argument could truly be understood by John Q. Public. She care
fully and accurately explained scientific concepts and findings that were ne
w to most people. She is rightly hailed as a founder of the environmental mo
vement and those positive steps that have been taken to protect the environm
ent, inform public debate and raise substantive question's about man's role
in modifying nature.On the other hand, her book overstates its case, demoniz
es opponents and discredits other more balanced approaches to public policy
positions. On balance, I am thankful for her writing this book at a time whe
n environmental issues were not on the radar screen. The basic improvements
made since that time depended upon the awareness she generated. The real dis
appointment is with our lack of progress as a nation and a world in evaluati
ng environmental issues such as global warming. We generate much more heat t
han light.
|
|Alice's Adventures in Wonderland & Through the Looking-Glass|4.0
|1.0       |3.0      |Amy's Review of Alice and Wonderland                  |
The book Alice in wonderland starts when Alice finds a white rabbit. Alice f
```

```
ollows the rabbit down a hole with millions of cupboards. Alice struggles wi
th her size and finds her self talking to mice and birds. Alice meets many c
harecters in Wonderland but the worst is the Queen Of Hearts. The queens fav
orite line in &quot;... of with thier heads ...&quot;. When Alice wakes up f
rom her nap she takes Dinahs kitten through the looking glass into Looking G
lass World where everything is backwards. In Looking Glass World everyone te
lls Alice poems and she doesn't know what any mean until she finds Humpty Du
mpty. In Looking Glass World Alice moves to the 8th square and becomes a que
en. This is a book I recomend to every person that likes adventure.
|
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------+----------+---------+-------------------------------------------------
-----+-------------------------------------------------------------------------
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
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```

```
only showing top 5 rows
```

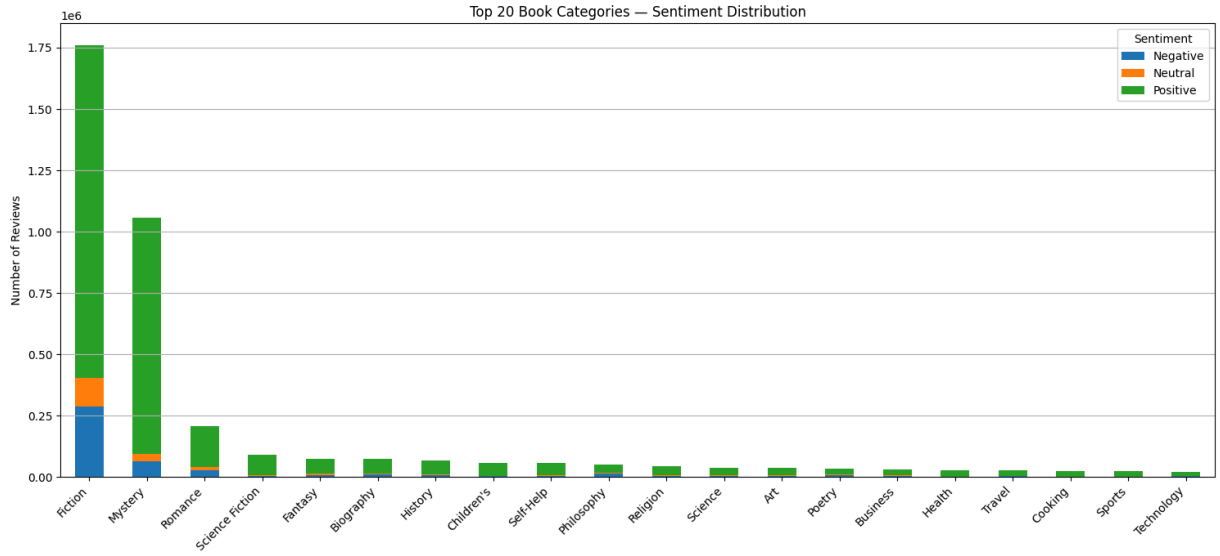
WordCloud of the Misclassified Reviews

In [ ]:
```
from wordcloud import WordCloud
import matplotlib.pyplot as plt
misclassified_texts = misclassified.select("review/text").rdd.flatMap(lambda
all_text = " ".join(misclassified_texts)
wordcloud = WordCloud(
    width=1600,
    height=800,
    background_color='white',
    max_words=200,
    collocations=False
).generate(all_text)
plt.figure(figsize=(15, 7))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis("off")
plt.title("WordCloud of Misclassified Reviews", fontsize=20)
plt.show()
```

4) Sentiment Distribution across Book Categories

In [ ]:
```
from pyspark.sql.functions import col, regexp_replace, split, explode, trim,
import matplotlib.pyplot as plt
pred = multinomial_predictions.withColumnRenamed("categories", "orig_categor
cats = processed_df.select(
    "Id",
    col("categories").alias("joined_categories")
)
full = pred.join(cats, on="Id", how="left")
```

```
full2 = full.withColumn(
    "category_arr",
    split(
        regexp_replace(col("joined_categories"), r"[\[\]']", ""),
        r",\s*"
    )
)
exploded0 = full2.withColumn("cat0", explode(col("category_arr")))
exploded = exploded0.withColumn("category", trim(col("cat0"))) \
                    .drop("cat0", "joined_categories", "category_arr")
sentimented = exploded.withColumn(
    "sentiment",
    when(col("prediction") >= 4.0, "Positive")
      .when(col("prediction") == 3.0, "Neutral")
      .otherwise("Negative")
)
counts = sentimented.groupBy("category", "sentiment").count()
pivoted = counts.groupBy("category") \
                .pivot("sentiment", ["Negative", "Neutral", "Positive"]) \
                .sum("count") \
                .fillna(0)
pdf = pivoted.toPandas()
pdf["Total"] = pdf[["Negative", "Neutral", "Positive"]].sum(axis=1)
pdf = pdf.sort_values("Total", ascending=False).drop(columns=["Total"])
top20 = pdf.head(20)
category_names = [
    "Fiction", "Mystery", "Romance", "Science Fiction", "Fantasy",
    "Biography", "History", "Children's", "Self-Help", "Philosophy",
    "Religion", "Science", "Art", "Poetry", "Business",
    "Health", "Travel", "Cooking", "Sports", "Technology",
]
mapping = dict(zip(top20.index, category_names))
top20_renamed = top20.rename(index=mapping)
ax = top20_renamed.plot(
    kind="bar",
    stacked=True,
    figsize=(15,7)
)
ax.set_ylabel("Number of Reviews")
ax.set_title("Top 20 Book Categories — Sentiment Distribution")
plt.xticks(rotation=45, ha="right")
plt.legend(title="Sentiment")
plt.grid(axis="y")
plt.tight_layout()
plt.show()
```

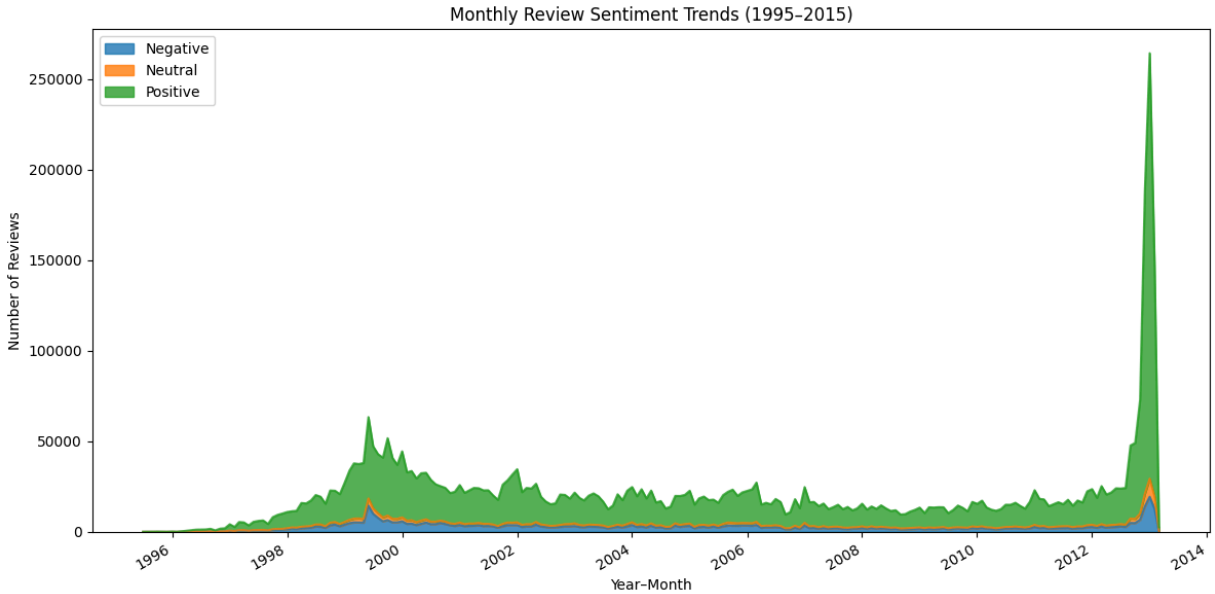


5) Temporal Analysis of Review Trends

In [ ]:

```
from pyspark.sql.functions import date_format, when, col
import pandas as pd
import matplotlib.pyplot as plt
time_df = processed_df.select(
    "Id",
    col("review/time").alias("orig_time")
)
pred_time = multinomial_predictions.join(time_df, on="Id", how="left")
pred_time = pred_time.withColumn(
    "year_month",
    date_format(col("orig_time"), "yyyy-MM")
)
pred_time = pred_time.filter(
    (col("year_month") >= "1995-01") &
    (col("year_month") <= "2015-12")
)
pred_time = pred_time.withColumn(
    "sentiment",
    when(col("prediction") >= 4.0, "Positive")
     .when(col("prediction") == 3.0, "Neutral")
     .otherwise("Negative")
)
trend = pred_time.groupBy("year_month", "sentiment").count()
trend_pivot = (
    trend
    .groupBy("year_month")
    .pivot("sentiment", ["Negative","Neutral","Positive"])
    .sum("count")
    .fillna(0)
    .orderBy("year_month")
)
trend_pd = trend_pivot.toPandas()
trend_pd["year_month"] = pd.to_datetime(trend_pd["year_month"])
trend_pd = trend_pd.set_index("year_month").sort_index()
ax = trend_pd.plot.area(figsize=(12,6), alpha=0.8)
```

```
ax.set_xlabel("Year-Month")
ax.set_ylabel("Number of Reviews")
ax.set_title("Monthly Review Sentiment Trends (1995-2015)")
plt.tight_layout()
plt.show()
```



Writing all the prediction results

In [ ]:
```
from pyspark.ml import PipelineModel
from pyspark.sql import SparkSession
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
pipelineModel = PipelineModel(stages=[cvModel, idfModel, assembler, rf_model
pipeline_dir   = "hdfs://localhost:9000/user/sathw/output/model_pipeline"
pipelineModel.write().overwrite().save(pipeline_dir)
print(f" PipelineModel saved to {pipeline_dir}")
preds = pipelineModel.transform(test_df)
preds_to_save = preds.select(
    "Id",
    "Title",
    "review/score_imputed",
    "prediction",
    "probability"
)
preds_dir = "hdfs://localhost:9000/user/sathw/output/predictions"
preds_to_save.write.mode("overwrite").parquet(preds_dir)
print(f"Predictions saved to {preds_dir}")
evaluator = MulticlassClassificationEvaluator(
    labelCol="review/score_imputed",
    predictionCol="prediction"
)
accuracy  = evaluator.setMetricName("accuracy").evaluate(preds)
precision = evaluator.setMetricName("weightedPrecision").evaluate(preds)
recall    = evaluator.setMetricName("weightedRecall").evaluate(preds)
f1_score  = evaluator.setMetricName("f1").evaluate(preds)
metrics = spark.createDataFrame(
```

```
    [(accuracy, precision, recall, f1_score)],
    ["accuracy", "precision", "recall", "f1_score"]
)
metrics_dir = "hdfs://localhost:9000/user/sathw/output/metrics"
metrics.write.mode("overwrite").json(metrics_dir)
print(f" Metrics saved to {metrics_dir}")
```

```
✅ PipelineModel saved to hdfs://localhost:9000/user/sathw/output/model_pipe
line
25/04/27 04:38:19 WARN DAGScheduler: Broadcasting large task binary with siz
e 6.3 MiB

✅ Predictions saved to hdfs://localhost:9000/user/sathw/output/predictions
25/04/27 04:38:26 WARN DAGScheduler: Broadcasting large task binary with siz
e 6.1 MiB
25/04/27 04:38:33 WARN DAGScheduler: Broadcasting large task binary with siz
e 6.1 MiB
25/04/27 04:38:40 WARN DAGScheduler: Broadcasting large task binary with siz
e 6.1 MiB
25/04/27 04:38:50 WARN DAGScheduler: Broadcasting large task binary with siz
e 6.1 MiB
[Stage 4170:==================================================>    (10 + 1) /
11]
✅ Metrics saved to hdfs://localhost:9000/user/sathw/output/metrics
```